



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 11406A-001.0
Date: 10/1/2025

Client: County of Hawai'i Carollo Project No.: 203975
Contractor: Nan, Inc.
Submittal Name: WASTE GAS BURNER
Reviewed By: Matheus Tolentino Lauar

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

1. This submittal includes components that were already reviewed under a different submittal. The submittal herein was reviewed for the waste gas burner, flare assembly, pilot system, and vendor control panel (components on tab 1 of submittal). See response to submittal 11381-001.0 - DIGESTER APPURTENANCES for the other components included in this submittal.
2. Submittal indicates that vendor does not supply the hardware for pedestal/base mounting and anchorage. Contractor responsible for anchor design for flare stack so that it meets the requirements of paragraph 11406A-2.03.E. Contractor to submit design calculations stamped by a Hawaii-registered engineer showing wind, seismic, and snow resistance as required per paragraph 11406A-1.03.B.6.
3. All vertical piping for pilot system shall have liquid collection and drain legs for liquid removal, as specified in paragraph 11406A-2.04.C. Pilot control system drawings do not show drain legs at the end of vertical pipe segments for 2" propane and 1/2" retention piping. Piping layout for pilot system shall be adjusted to include drain legs as specified.
4. Contractor to provide documentation indicating that solenoid valve on pilot system is explosion-proof, as specified in paragraph 11406A-2.04.D.
5. Contractor to confirm range of temperatures for propane used in pilot system (pilot gas). Paragraph 11406A-2.04.A specifies 60 to 130 degrees Fahrenheit. Information was not found in the submittal.
6. Provide threaded union fittings or other mechanism to allow spark plug to be easily removed, as specified in paragraph 11406A-2.04.F. Per pilot control system drawing, spark plug appears to be installed tightly between flashback

preventer and a metal bracket bolted to mounting panel. Revise layout of pilot system as needed.

7. Contractor to provide documentation indicating that all pilot system components are FM, UL, or CSA-approved, as indicated in paragraph 11406A-2.04.I.
8. Paragraph 11406A-2.06.B indicates that "NEMA 4X is acceptable if the control panel is located 10' away from DG piping". A NEMA 4X enclosure was provided for the vendor control panel, as shown in submittal. However, drawing 11-M-01-109 shows 11-VCP-2370 at less than 10' from 6" DG line routed towards flare. Explosion-proof enclosure shall be provided.
9. Contractor to provide Process and instrumentation diagrams (P&IDs) for pilot gas system and components connected to VCP, as required in paragraph 11406A-1.03.B.1.
10. Contractor to provide wiring and control diagrams, as specified in 11406A-1.03.B.2.
11. Contractor to provide complete control descriptions including sequence of operations, safety features, and alarms, as specified in 11406A-1.03.B.4.
12. Contractor to provide installation procedures and instructions on storage and protection prior to installation, as specified in 11406A-1.03.B.7 & 1.03.B.8
13. Contractor to provide documentation indicating that pressure switches supplied are explosion-proof as specified in 11406A-2.05.C

☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number: 11406A-001.0
Submittal Title: WASTE GAS BURNER ☐ For Information Only
TO: County
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec: 11406A	Paragraph: 2.01 2.02 2.03 2.04 2.05 2.06 2.07
Authorized By: Shand and Jurs	Date Submitted: 09/04/2025

Submittal Certification	
Check Either (A) or (B):	
☒ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title: Makoa Ng, QC, Project Engineer	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/4/2025
SPECIFICATION SECTION # 11406A
SPECIFICATION WASTE GAS BURNER
PARAGRAPH 2.01 2.02 2.03 2.04 2.05 2.06
DRAWING Shand and Jurs
SUBCONTRACTOR na
SUPPLIER Shand and Jurs
MANUFACTURER Shand and Jurs

CERTIFIED BY CQCM or Designee: 

SECTION 11406A

WASTE GAS BURNER

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes:
 - 1. One flare, also called waste gas burner, or burner for short, as indicated on the Drawings, and as specified herein, together with all accessories and appurtenances, to make the unit complete and operable.
- B. The burner manufacturer shall be the primary supplier for the entire waste gas burner system and shall be solely responsible for ensuring the overall waste gas burner system meets all Specification requirements.
- C. The Contractor shall provide a flare foundation similar to that indicated on the Drawings, but modified as necessary for the specific equipment furnished. The Contractor shall submit any required foundation modifications to the Engineer for review.
- D. Contractor shall obtain a Permit to Construct from the State of Hawaii Department of Health, Clean Air Branch.

1.02 REFERENCES

- A. American Gas Association (AGA).
- B. American National Standards Institute (ANSI).
- C. American Society of Mechanical Engineers (ASME):
 - 1. B16.5 - Pipe Flanges and Flanged Fittings: NPS 1/2 through 24.
- D. ASTM International (ASTM):
 - 1. A216 - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service.
 - 2. A312 - Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes.
 - 3. A351 - Standard Specification for Castings, Austenitic, for Pressure-Containing Parts.
- E. International Society of Automation (ISA).
- F. National Electrical Manufacturers Association (NEMA).
- G. National Fire Protection Association (NFPA):
 - 1. 820 - Standard for Fire Protection in Wastewater Treatment and Collection Facilities.

- H. State of Hawaii Department of Health, Clean Air Branch.
- I. Underwriters Laboratories, Inc. (UL).

1.03 SUBMITTALS

- A. Submit in Section 01330 - Submittal Procedures.
- B. The manufacturer shall submit the following:
 - 1. The make and model and catalog cut sheets for all waste gas burner system components showing materials of construction, layout plans and section drawings for all components, anchor bolt sizing calculations and drawings (if anchor bolts used), a process and instrumentation diagram meeting ISA standards, and identification of all materials and metallurgy used in the system.
 - 2. Wiring and control diagrams of the system.
 - 3. Complete piping layout drawings for the pilot gas train.
 - 4. Complete control descriptions including sequence of operations, safety features, and alarms.
 - 5. Local control panel arrangement, elevations, control system logic and control system wiring diagrams, and control schematics.
 - 6. Calculations demonstrating resistance of the equipment to loadings/forces from snow, wind, and seismic conditions specified herein. Calculations to be signed by an engineer registered in the state of Hawaii.
 - 7. Installation procedures.
 - 8. Requirements for storage and protection prior to installation.
 - 9. Complete drawings on control panel mounting frame.
- C. The following shall be included in the Operations and Maintenance Manual:
 - 1. As-built control system logic and wiring diagrams.
 - 2. Maintenance schedule.

1.04 QUALITY ASSURANCE

- A. As specified in Section 01450 - Quality Control.

1.05 DELIVERY, STORAGE AND HANDLING

- A. As specified in Section 01600 - Product Requirements - Plant.

1.06 WARRANTY

- A. Provide warranty as specified in Section 01783 - Warranties and Bonds.

1.07 SPARE PARTS AND SPECIAL TOOLS

- A. Provide the following:
 - 1. If special tools are required for the service and maintenance of the unit, provide 1 complete set of tools.
 - 2. If critical spare parts are required to keep the burner always in operation, provide 1 set of spare parts for all critical components of the burner system.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. One of the following or equal:
 - 1. Varec, 244 WL.
 - 2. Shand and Jurs, Model 97301.
- B. The burner manufacturer shall have at least five years of experience in the manufacture of similar waste gas burners meeting 40 CFR Part 60.18 control device requirements for candlestick burner.

2.02 WASTE GAS BURNER SYSTEM DESCRIPTION

- A. General:
 - 1. The waste gas burner shall be the open flame “candlestick” type with a stainless steel stack.
 - 2. Normally, parts of the waste gas burner system could be in a standby shutdown condition for an extended period of time. When required for operation a waste gas burner may be required to cycle on and off frequently to maintain the gas production and use in balance. The waste gas burner shall be designed with components and controls to meet these operating requirements.
 - 3. All gas train components exterior to the waste gas burner stack shall meet the material, design pressure, and test pressure requirements for digester gas handling equipment.
- B. Digested gas inlet conditions:
 - 1. Maximum design flow rate per waste gas burner: 200 cfm.
 - 2. Minimum design flow rate per waste gas burner: 20 cfm.
 - 3. Inlet fuel gas temperature: 60 to 120 degrees Fahrenheit.
 - 4. Minimum turndown ratio: 10 to 1.
 - 5. Inlet gas pressure to flare (minimum upstream of backpressure control valve): 4 inches water column.
 - 6. Maximum velocity at all points in the system: 17 feet per second.
- C. Service conditions:
 - 1. The gas to be burned shall be wastewater treatment anaerobic digester gas:

Component	Methane Phase
Methane, %	55 to 65
Carbon Dioxide, %	35 to 45
Hydrogen Sulfide, ppm	500 to 1,500
Moisture, saturated at	60 to 125 degrees Fahrenheit
Estimated net (lower) heating value, BTU/scf	500 to 600

- 2. The digester gas may contain trace quantities of other gases including other hydrocarbons, chlorinated compounds, and siloxanes.

- D. Regulatory requirements:
1. The burner shall be designed to reduce the possibility of flashback. The burner shall not exhibit any flame lift off or instability over the entire range of design.
 2. Assist gas shall not be required or used to maintain flame stability.
 3. The waste gas burner stack, other components and anchorage system shall be designed for the conditions specified in Section 01850 – Design Criteria.
 4. All components of the waste gas burner system shall be in accordance with NFPA 820.

2.03 FLARE ASSEMBLY

- A. Features:
1. The flare shall have a round cross section.
 2. The flare shall be freestanding, mounted to a flange, and have lifting lugs.
- B. Materials:
1. Flare stack: Type 316/316L stainless steel.
 2. Pilot gas piping: per schedule (LPG).
 3. Nuts, and washers: Type 316 stainless steel.
- C. The burner size shall be no less than 6 inches.
- D. The waste gas burner shall meet the specified pressure range. Any additional control valves, isolation valves, piping, fittings, and connections to meet this requirement shall be supplied at no additional cost.
- E. Design waste gas burner and stack.
1. Design for loadings and conditions as noted in Section 01850 - Design Criteria.
 2. Additional design requirements:
 - a. Lateral loads (wind and seismic):
 - 1) Resist forces through direct bearing against anchors and fasteners. Do not rely on friction to resist loads.
 - 2) Do not use more than 60 percent of the self-weight of equipment for resisting overturning due to seismic forces.

2.04 PILOT SYSTEM

- A. The waste gas burner pilot system shall be designed to operate using propane at temperatures between 60 to 130 degrees Fahrenheit.
- B. The supplied pilot gas shall be propane. Pilot gas train piping shall be sized for required flow at minimum supply pressure of 10 psig (measured upstream of a regulator), as field verified by the Contractor, and as required for the regulator and for connections to pressure switches, gauges, and regulators. Pilot piping shall have liquid knock out and flashback protection. All vertical piping must have liquid collection and drain legs for liquid removal.
- C. The pilot ignition system shall be an inspirating venturi type that provides a stoichiometric air/fuel mixture and senses the pilot flame by use of a Type K thermocouple installed in the continuous pilot nozzle.

- D. Pilot train shall include pilot explosion-proof solenoid, regulator, manual shutoff valves, and pressure gauges.
- E. Provide all necessary ASTM A312, type 316 stainless steel, with 150-pound screwed type 316 fittings, per pipe schedule, as required for the pilot gas train.
- F. The spark plug provided for the pilot ignition system shall be designed so that it may be removed for inspection and servicing from the exterior of the waste gas burner stack. The spark plug shall be located a maximum 70 feet away from the burner and greater than 15 feet from the burner stack for NFPA 820 compliance. The pilot shall have its own flame detection system and alarm.
- G. The pilot burner nozzles and thermocouple shall be type 316 stainless steel with thermowell.
- H. The ignition system shall include a controller to close the solenoid valve if the flame is out for 30 seconds as determined by the Type K thermocouple, adjustable time delay relay, and a pressure switch bypass mounted in a NEMA 7 enclosure.
- I. Components shall be FM, UL, or CSA-approved.
- J. Provide 316 stainless steel mounting stand and 316 stainless steel weather hood for all pilot components including the control panel.
- K. Propane piping routed along the vertical portion of the flare stack shall be provided and supported by the manufacturer.

Added as per Bid Addendum 06

2.05 ACCESSORIES

- A. See Section 11381 - Digester Appurtenances for details regarding the following waste gas burner accessories:
 - 1. Flame trap assemblies.
 - 2. Digester gas drip traps.
 - 3. Pressure/vacuum relief valves.
- B. Combination back pressure control valve and flame arrester:
 - 1. Manufacturers:
 - a. Varec, Series 440.
 - b. Shand & Jurs, Model 97160.
 - 2. Components:
 - a. Single port regulator: See Section 11381 - Digester Appurtenances.
 - b. Flame arrester: See Section 11381 - Digester Appurtenances.
 - c. 3-way solenoid valve option.
- C. Pressure switch:
 - 1. Refer to Section 17401 - Pressure/Vacuum Measurement: Diaphragm and Annular Seals.
 - 2. The pressure switch shall be explosion-proof and shall have an operating range of 4 inches to 20 inches W.C., adjustable setpoint, with a 0.4-inch to 0.6-inch W.C. deadband.
- D. Automatic drip traps:
 - 1. See Section 11381 - Digester Appurtenances.

- E. Pilot gas panel:
1. Panel shall include all necessary components for a fully functioning system. This includes but is not limited to pressure gauges, 0 to 30 inches water column, type 316 stainless steel isolation ball valve, globe valves, spark plug, 2-inch Type 347 or 316 stainless steel inspiring venturi, and an air blower.
 2. Pilot solenoid valves:
 - a. Materials:
 - 1) Body: Aluminum.
 - 2) Seals: Buna N.
 - b. Class F coil.
 - c. Explosion proof NEMA 7 enclosure, suitable for outdoor, corrosive areas.
 - d. 120 Volt AC operation.
 3. Flame check:
 - a. Materials:
 - 1) Housing: Low copper cast aluminum.
 - 2) Element: Type 316 stainless steel.
 - b. Working pressure:
 - 1) 25 psig (172 kPa) maximum.
 - c. NPT connections provided on all sizes.
 4. Panel shall be constructed of type 316 stainless steel weatherhood and stand.
 5. All tubing, piping, and threaded fasteners shall be provided in type 316 stainless steel.

✓ 2.06 **VENDOR CONTROL PANEL (VCP)**

- A. Panel shall be tagged 11-VCP-2370 with flare 11-FLR-2370.
- B. Provide explosion proof control panel in accordance with Section 16134 – Boxes. NEMA 4X is acceptable if the control panel is located 10' away from DG piping. Provide control panel with type 316 stainless steel, self standing mounting frame with a sun shield, and with UL508A label. The control panel, components and instrumentation shall meet the requirements of Section 16075 – Identification for Electrical Systems and Section 17050 – Common Work Results for Process Control and Instrumentation Systems of the specifications.
- C. The VCP shall house the waste gas burner electronic control unit. The waste gas burner control unit shall include the automatic ignition pilot system, a waste gas burner system HAND/OFF/AUTO selector switch, any necessary control relays and timer relays, status indicator lights (on/off/alarm) for all equipment, terminal strips, and internal wiring as necessary to provide a complete system.
1. Where possible, all electronic circuit boards shall be conformal coated.
- D. Panel mounted devices. As a minimum the following devices shall be provided on the VCP.
1. White POWER ON status light.
 2. SYSTEM HAND/OFF/AUTO selector switch.
 3. Red PILOT FLAME ON status light.
 4. Red SYSTEM RUN status light.
 5. Amber SYSTEM TROUBLE alarm light.
 6. MAIN POWER DISCONNECT hand switch.
 7. ALARM RESET push button.

- E. Plant PLC communications. Provide the following via 120 VAC contacts:
 - 1. SYSTEM ALARM status.
- F. Power supplied to the vendor control panel shall be 120 VAC, single-phase, 60 hertz. All other required voltages and control power requirements within the VCP shall be provided through devices mounted in the VCP by the manufacturer.
 - 1. Power Disconnect:
 - a. Flange mounted thermal magnetic circuit breaker:
 - 1) Lockable in the OFF position.

2.07 CONTROL OF THE WASTE GAS BURNER

- A. VCP system auto control:
 - 1. When the SYSTEM HAND/OFF/AUTO selector switch is in the SYSTEM AUTO mode the burner start up is automated by the VCP.
 - a. The waste gas pressure shall be monitored via vendor-furnished pressure switch. The pressure switch shall be wired directly to the VCP to allow for VCP automatic ignition control based on high pressure at the switch. The VCP also controls the pilot gas solenoid valve.
 - b. The pilot valve shall remain open at all times when the burner is in the AUTO position, unless the pilot flame is not detected. The system will then activate an alarm. In hand position, the solenoid valve shall remain closed.
- B. VCP system manual control:
 - 1. When the SYSTEM HAND/OFF/AUTO selector switch is in the HAND mode, the burner start up is manually started by the operator. The VCP shall send a signal to open the pilot gas supply solenoid valve.
- C. Protective interlocks:
 - 1. The following characteristics shall prevail in AUTO modes as a minimum. Vendor to provide all required protective interlocks to maintain safe system:
 - a. If the pilot light is not lit, the VCP shall activate an alarm, and send it to the Plant area PLC. The solenoid valve on the pilot gas supply line shall close after 30 seconds. The alarm shall be turned off by the operator pressing the RESET pushbutton on the VCP.
 - b. When the SYSTEM HAND/OFF/AUTO hand switch is in the OFF mode, the VCP shall close the pilot gas supply solenoid valve and initiate the SYSTEM alarm at the VCP and Plant area PLC/SCADA.
- D. The SYSTEM alarm shall be indicated at the VCP and indicated and recorded at the Plant area PLC.

PART 3 EXECUTION

3.01 GENERAL

- A. Waste gas burners shall be installed in accordance with the manufacturer's installation instructions.

- B. All electrical or controls shall be complete and conform to the requirements of these Specifications.
- C. The Contractor may be required to modify the piping as indicated on the Drawings to match the equipment supplied. All modifications shall be at no additional cost.

3.02 FIELD QUALITY CONTROL

- A. After substantial completion of the construction of the waste gas burner system, the primary supplier shall conduct field checkout, testing and adjustment of the combustion air blower, pilot ignition, and burner systems to ensure that all control systems function properly. After completing the testing and adjustment of the control systems, the primary supplier shall certify in writing that the system is ready for normal plant operations.
- B. Performance and function testing:
 - 1. As specified in Section 01756 - Commissioning.
- C. Certificate of Installation:
 - 1. As specified in Section 01756 - Commissioning.

3.03 MANUFACTURER'S SERVICES

- A. Manufacturer services:
 - 1. Provide certificates:
 - a. Manufacturer's Certificate of Installation and Functionality Compliance.
 - 2. Manufacturer's Representative onsite requirements: Require manufacturer's representative to perform the following services as described below and as specified in Section 01756 - Commissioning. The specified durations are the minimum required time on the job site. Additional services and/or longer durations shall be provided as needed at no cost to the Owner to meet the required quality of work. Work to be done in a minimum of 2 trips:
 - a. Installation assistance: As required:
 - 1) Advise/observe the Contractor on the installation of the equipment.
 - 2) Provide additional assistance as required.
 - b. Installation inspection: 1 trip, 2-day minimum.
 - c. Functional Testing: 1 trip, 2-day minimum.
 - 3. Training: As defined in Section 01756 - Commissioning and this Section (trip can be combined with functional testing):
 - a. Operations: 2-hour class, 2 sessions.
 - b. Mechanical Maintenance: 3-hour class, 1 session.
 - a. Combined Electrical and I&C Maintenance: 1-hour class, 1 session.

END OF SECTION

Nan Inc.

*Hilo WWTP Rehabilitation and Replacement Project
Phase 1
Hilo, HI*

Data Sheets & Approval Drawing Submittal

**Shand & Jurs Sales Order: IB2090
Purchase Order: 24077-00048**

July, 2025

Equipment

97301 - 6" Waste Gas Burner
97162 - 6" Pressure Relief/Flame Trap Assembly
97200 - 1/2" NPT Flame Check
97100E - 6 Qt. Electric Drip Trap
94407 - 6" Horizontal Inline Deflagration Flame Arrester
97120 - 6" Sediment Trap
97570 - 6" Combination Conservation Vent & Flame Arrester
97190 - 6" 3-Way Safety Selector Valve
97401 - 15" U-Tube Type Manometer
97150 - 4" Single Port Regulator
97100E - 6 Qt. Electric Drip Trap



L&J TECHNOLOGIES



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L&J engineering

SHAND & JURs

Data Sheet and Approval Drawings for Digester Gas Safety and Control Equipment

Project Information

Project: Hilo WWTP Rehabilitation and Replacement Project
Phase 1
Hilo, HI

Equipment: Digester Gas Safety
and Control Equipment

Contractor: Nan Inc.
636 Laumaka Street
Honolulu, Hawaii 96819

Manufacturer,
Service & Parts: Shand & Jurs
5911 Butterfield Road
Hillside, IL 60162
Ph. (708) 236-6000 Fax (708) 236-6006

Shand & Jurs Order #: !B2090
Purchase Order #: 24077-00048

L&J TECHNOLOGIES

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SHAND & JURS

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SHAND & JURS

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Data Sheet	
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Foreword

Sales Order !B2090



BIOGAS
SHAND & JURS



L&J TECHNOLOGIES

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GPE CONTROLS,LLC / SHAND&JURS
An L&J Technologies Company

SALES ORDER
4-MANUFACTURING

Order Number
!B2090

Cust.number: 2504600

B NAN, INC

I 363 LAUMAKA

L GPE CONTROLS,LLC / SHAND&JURS

L 5911 BUTTERFIELD RD

HONOLULU HILLSIDE, IL 60162

T HI 96819 PH (708) 236-6000

O USA FX (708) 236-6006

FAX: JCJHUO@NANHAWAII.COM

Ship To Number: 2504600

S NAN, INC

H 636 LAUMAKA ST

I

P HONOLULU

T HI 96819

O USA

Date entered	Date Printed	COLL/PP	Salesman	Customer PO number	Page#		
2/28/2025	7/07/2025	L&J Pays Frght	100 1275	24077-00048	1		
Partials		VIA	INCOTERMS	Terms			
N		BEST WAY- EST FRT \$14,800.00	FOB HILLSIDE IL	NET 30			
Item	QTY	Part Number	Description	Quantity	Estimated Date	Price Each	Extended Price
01	1	97301-153773001	6" A316 PSH316 316P CB316 LUGS	1	10/16/2025		
		INSTRUCTION 97-301-!B2090					
		TEST REPORT 853422					
02	1	122200	PRESS XMTR XP 1/2NPT 4-20 1.5#	1	10/16/2025		
03	1	97162-064210001	6"SPR+407FA ALUM X SOL 9-15"	1	10/16/2025		
		INSTRUCTION 97-162					
		TEST REPORT RA800-503					
04	1	97200-22	1/2"NPT FLAME CHECK ASY AL/316	1	10/16/2025		
		INSTRUCTION 97-200					
		TEST REPORT RA800-508					
05	6	97100E631203	6QT AL,NEOP,A/I N7-TMR,AN-C/D	6	10/16/2025		
		INSTRUCTION 97-100E-12					
		TEST REPORT RA800-450-3					
06	4	94407-632205	6"AL 316TBS 2-3/4 2-1/2	4	10/16/2025		
		INSTRUCTION 94-407					
		TEST REPORT RA800-557					
07	2	97120-061032103	6"FF ST-316 SG CR 25# AIS	2	10/16/2025		
		INSTRUCTION 97-120					
		TEST REPORT RA800-449-1					
08	4	97570=160718AD2	6"AL 316:TB+SH+TR AW XOS IW	4	10/16/2025		
		PRES 15.000 inWC VAC 2.000 inWC					
		INSTRUCTION 97-570					
		TEST REPORT RA800-543					
		6"-16" W.C. IN 1" INCREMENTS INCLUDING (1)-1/2 IN. &					
		AND (2)-1/4 IN. W.C. WTS					
09	2	97190-64331111	6"AL/FF 316/RTR 316/NEO NUT DR	2	10/16/2025		
		INSTRUCTION 97-190-1					
		TEST REPORT RA800-509-1 & RA800-509-2					
10	1	122204	30MM IND 120V 4X/PTT/LED/WHITE	1	10/16/2025		
11	1	122149	30MM IND 120V 4X/PTT/LED/GREEN	1	10/16/2025		
12	1	122147	30MM IND 120V 4X/PTT/LED/RED	1	10/16/2025		
13	1	122088	CKT BKR 240V 10A 1P DIN BRANCH	1	10/16/2025		
14	1	649021	FUSE 2A SLOWBLOW 250VAC 5X20MM	1	10/16/2025		
15	1	649022	FUSE 7A SLOWBLOW 125VAC 5X20MM	1	10/16/2025		
16	1	9730-10020	FLARE SPARK PLUG I-31-1 P5022	1	10/16/2025		
17	1	9730-13311	THMCPL 48"P/192"L 2200F 600 HT	1	10/16/2025		
18	1	121258	PLC 4POS RLY OUT SPDT 7A 240VA	1	10/16/2025		
19	2	97401-15112	15" U-TUBE ENG ALUM/PNL FCV	2	10/16/2025		
20	1	97150-042300003	4" SPR,EXT,SOL,SS/HDW,7-13"WC	1	10/16/2025		
		PRES 10.000 inWC					

SERVICE CHARGE OF 1.5% PER MONTH(18% PER YEAR) ON ALL INVOICES OVER 30 DAYS

FORM# 853304

GPE CONTROLS,LLC / SHAND&JURS
An L&J Technologies Company

SALES ORDER
4-MANUFACTURING

Order Number
!B2090

Cust.number: 2504600

B NAN, INC

I 363 LAUMAKA

L

L

HONOLULU

T HI 96819

O USA

FAX: JCJHUO@NANHAWAII.COM

GPE CONTROLS,LLC / SHAND&JURS

5911 BUTTERFIELD RD

HILLSIDE, IL 60162

PH (708) 236-6000

FX (708) 236-6006

Ship To Number: 2504600

S NAN, INC

H 636 LAUMAKA ST

I

P

HONOLULU

T HI 96819

O USA

Date entered	Date Printed	COLL/PP	Salesman	Customer PO number	Page#
2/28/2025	7/07/2025	L&J Pays Frght	100 1275	24077-00048	2
Partials	VIA	INCOTERMS	Terms		
N	BEST WAY- EST FRT \$14,800.00	FOB HILLSIDE IL	NET 30		
QTY	Part Number	Description	Quantity	Estimated	Price Each
Item	Ordered				Extended Price

		INSTRUCTION 97-150-10			
		TEST REPORT RA800-503			
21	1	97100E631203 6QT AL,NEOP,A/I N7-TMR,AN-C/D	1	10/16/2025	
		INSTRUCTION 97-100E-12			
		TEST REPORT RA800-450-3			
22	13	3001 APPROVAL DRAWINGS	13	10/16/2025	
23	1	5983 FLD. SERVICE-WW	1	10/16/2025	
		SERVICE TRIP: 2 TRIPS, 3 DAYS			
24	1	5000 OPERATION & MAINTENANCE BOOKS	1	10/16/2025	
25	1	5989 MISCELLANEOUS	1	10/16/2025	
		EXTENDED WARRANTY: 2 YEARS			

MARK SHIPMENT PO# 24077-00048

JOB# WW-4705R

#WW-4705R - HILO WASTEWATER TREATMENT PLANT (WWTP)

LITATION AND

REPLACEMENT PROJECT PHASE 1, WAIAKEA, HILO, HAWAII

Order Value Taxable:

Total Taxes:

Order Value Non-Taxable:

Total Order Value:

Tab 1

(97301-153773001)

6" Waste Gas Burner

- Data Sheet

- Drawing



BIOGAS
SHAND & JURS



L&J TECHNOLOGIES

A Cognesense Company

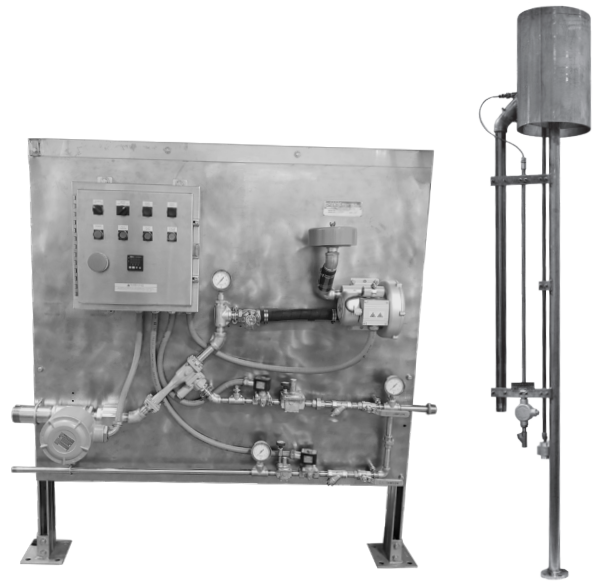
97301 Waste Gas Burner

Designed for complete automatic operation of the entire Flare System, The S&J 97301 Waste Gas Burner and TouchScreen Control Box are designed to combust biogases generated in fermentation processes. It efficiently incinerates low BTU gases from anaerobic digesters, lagoons, and municipal landfills, minimizing odors and VOC's. The stoichiometric pilot ensures that a proper air to fuel mixture is maintained throughout the wide range of pressure and BTU fluctuations. A continuous burning pilot in the flame area provides stable, controlled, nonsmoking, combustion.

The 97301 is designed to withstand wind speeds up to 150mph as well as seismic zone 4 loads. Its stainless steel components endure the severest of process environments. The burner tip is designed with swirl inducers that create a cyclonic effect producing an efficient air/fuel mixture and maximizes flame retention. The wind shroud induces sufficient air to the flare tip for proper mixing and combustion throughout the operating range.

The Pilot Control Box utilizes state of the art electronics and instrumentation for safe operation. Pilot controls are enclosed in a NEMA 4, carbon steel, electrical enclosure (NEMA 4X, 304 or 316 optional). For automatic operation, the flare's pilot can be designed to operate only during initial startup or continuously. The Pilot Control Box includes a dry contact input for Remote Start from a SCADA system or a pressure switch. A standard system uses a PLC which can be wired to the SCADA system. The control box also includes Contact Status outputs for Alarm and Flame Proven.

The auto-ignition sequence is started by the closing of the remote start contact or pressure switch contact. This will indicate that the gas pressure limit has been reached and flaring of excess gas should begin. The operation of the burner will continue until the contact opens.



Features

- High Performance Stoichiometric Pilot
- Sizes ~~2" Through 12"~~ 6"
- Burns High Flow, Low BTU "Wet" Methane
- Stainless Steel Flame Area
- Superior Pilot Wind Protection
- Solid State Controls
- Fully Automated Pilot: Continuous ~~or Intermittent~~ While Flaring
- Ground Level Venturi Pilot Ignition
- Provides Alarm Outputs

Specifications:

Sizes:

~~2", 3", 4", 6", 8", 10" & 12"~~

Stack Burner Connection:

ANSI 150 lb. Raised Face Flange

Contact Outputs:

Pilot On/Off: SPDT, 120 VAC 1 Amp

Pilot Fail: SPDT, 120 VAC 1 Amp

Other Status Outputs Available

Power Requirements:

120 VAC, 10 Amp

~~220 VAC, 5 amp (Optional)~~

Controller:

Temperature Range: -20 to 150 degrees F

Enclosure: ~~NEMA 4, Steel~~; NEMA 4X,
Stainless Steel; ~~NEMA 7,~~
~~Aluminum~~

Functions: Manual Start, Remote Start
Automatic Sequencing
Continuous Pilot ~~or~~
~~Intermittent Pilot~~

Stack Materials:

Stack with Shroud: ~~304L Stainless Steel~~
316L Stainless Steel (Optional)

Piping: ~~304L Stainless Steel~~
316L Stainless Steel (Optional)

Biogas Criteria Composition:

50%-70% CH₄, 50%-30% CO₂, with trace amounts of
H₂S, Inert Gases and Air

Moisture Content:

Saturated (100% Humidity)

Pilot Gas:

~~Natural Gas~~

Propane (LPG)

~~Waste Gas (500 BTU/ Cubic Foot Minimum)~~

Pilot/Ignition Gas Pressure:

4"-30" W.C. Low Pressure - Standard

1-30 PSIG (5 PSIG Minimum Required)

Functions:

Manual Start:

The operator puts selector to manual and initiates ignition by depressing the start push-button on the control panel.

Remote Start:

Remote ignition is performed by placing selector switch in the auto position and closing the remote location dry contact to initiate the operation of the waste gas burner.

Auto Start:

Automatic Start is performed by the sensing of a pressure permissive in the system. The pilot control panel must be set to "Auto" position for this to be controlled by the pressure switch. When the pressure switch contacts close, the auto flaring sequence will begin. Once the pressure drops below the pressure switch set point the contacts will open, halting operation.

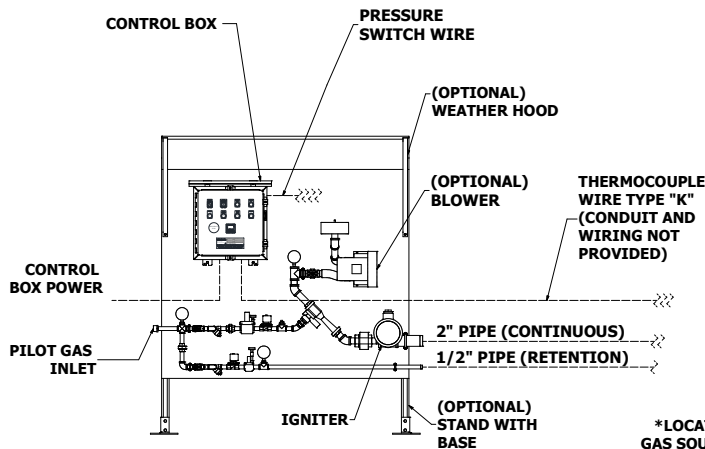
Accessories:

A back pressure regulator / deflagration arrester should be installed in the digester line just upstream of the flare. For automatic operation, the solenoid option must be selected on the back pressure regulator.

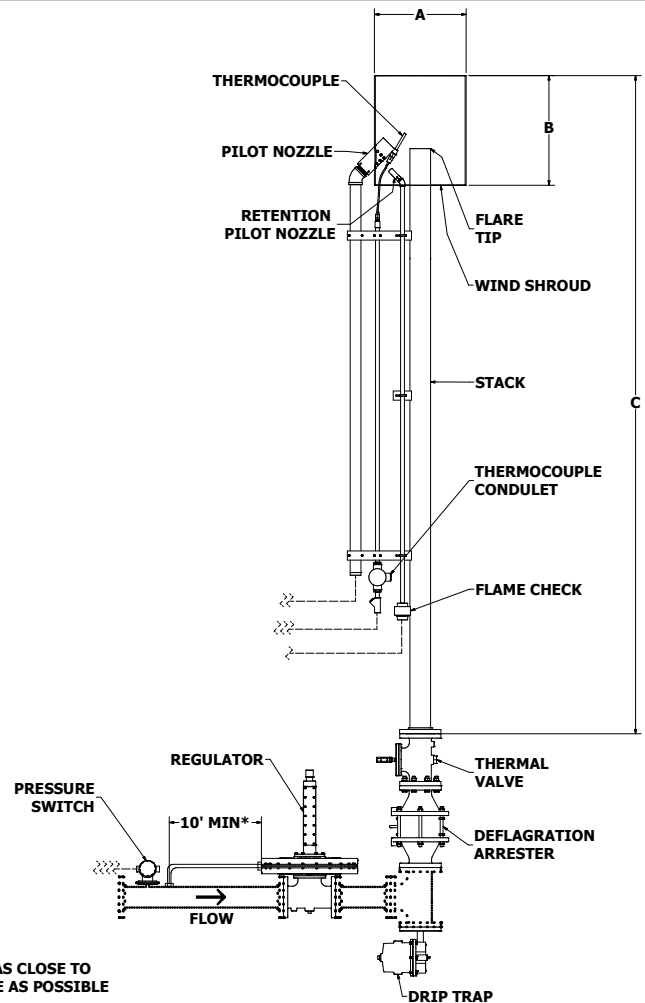
Dimensions:

Additional Options

- Weatherhood
- Stand with Base
- Pressure Switch (Remote/Automatic Control)
- 316 Stainless Steel Piping and Fittings
- NEMA 7, Explosion Proof Configuration



*LOCATE AS CLOSE TO GAS SOURCE AS POSSIBLE



Stack Dimensions

Dimensions (Inches [mm])			
Size	A	B	C
2 [50]	16 [406]	24 [610]	88 [2235]
3 [75]	18 [457]	24 [610]	92 [2337]
4 [100]	20 [508]	24 [610]	92 [2337]
6 [150]	24 [610]	36 [914]	128 [3251]
8 [200]	24 [610]	48 [1219]	144 [3658]
10 [250]	30 [762]	48 [1219]	176 [4470]
12 [300]	36 [914]	60 [1524]	188 [4775]

Flow specified for gas with 0.8 specific gravity, air at 60°F, and .5" WC pressure drop

Capacity

Size (Inches mm)	Capacity (FT ³ /Hr.)
2 [50]	4000
3 [75]	9970
4 [100]	22250
6 [150]	44200
8 [200]	76800
10 [250]	129000
12 [300]	218600

All designs subject to change. Certified dimensions and specifications available upon request.

L&J Technologies or any of its subsidiaries assume no responsibility and shall not be liable for any damage, injury or death caused by the mis-application or improper installation of the products that it provides. Installation shall be per manufacturer's instructions in accordance to any applicable local, state or federal regulations. It is the responsibility of the purchaser to ensure these guidelines are followed and that the products are applied properly.

97301 Ordering Guide

Model Number Selection

The model number will have a base number **97301** followed by 9 digit numbers. These digits will represent 9 sets of option tables.

97301 - AB - CD - EF - ~~GH~~ - ~~X~~001 = Special: Lifting Lugs; PSH316

Table A - Pilot Gas

Option A	Pilot Gas
0	Natural
1	Propane
2	Bio
3	Auto Dual*
4	Manual Dual*

* Biogas or other gas continuous pilot

Table B - Unit Size

Option B	Unit Size
2	2"
3	3"
4	4"
5	6"
7	8"
8	10"
9	12"
A	2" Beveled Shroud
B	3" Beveled Shroud
C	4" Beveled Shroud
D	6" Beveled Shroud
E	8" Beveled Shroud
F	10" Beveled Shroud
G	12" Beveled Shroud

Table C - Power Source / P.E. Certificate

Option C	Description
1	120 VAC, 60HZ
2	220/240 VAC, 60HZ
3	120 VAC, 60HZ with P.E. Cert
4	220/240 VAC, 60HZ with P.E. Cert

Table D - Enclosure Rating / UL

Option D	Description
0	NEMA 4 - Carbon Steel
1	NEMA 7 - Cast Aluminum
2	NEMA 4X - 304 Stainless Steel
3	NEMA 4X - 316 Stainless Steel
4	NEMA 4 - Carbon Steel (UL 508A)
5	NEMA 7 - Cast Aluminum (UL 508A)
6	NEMA 4X - 304 SS (UL 508A)
7	NEMA 4X - 316 SS (UL 508A)

Table E - Control Inputs

Option E	Description
1	1 Dry Contact Input
2 - 4	Reserved
5	2 Dry Contact Inputs
6	1 4-20MA Analog Input
7	1 4-20MA Analog + 1 Dry Contact Input

Typically Dry Contacts are used for SCADA or pressure switch. Analog for pressure transmitter.

Table F - Pilot

Option F	Description
1	Continuous (ON While Flaring)
2	Intermittent (OFF While Flaring)
3	Continuous (ON While Flaring; 347 SS*)
4	Intermittent (OFF While Flaring; 347 SS*)
5	Continuous (ON While Flaring) + 4-20MA Temp Out
6	Intermittent (OFF While Flaring) + 4-20MA Temp Out
7	Continuous (ON While Flaring; 347 SS*) + 4-20MA Temp Out
8	Intermittent (OFF While Flaring; 347 SS*) + 4-20MA Temp Out

* 347 Stainless Steel Venturi

Table G - Blower/Control Box Location

Option G	Description
1	Blower Motor - General Purpose / Local C.B.
2	Blower Motor - NEMA 7 / Local C.B.
3	Blower Motor - General Purpose / Remote C.B.
4	Blower Motor - NEMA 7 / Remote C.B.
5	No Blower / Local C.B.*
6	No Blower / Remote C.B.*

*5 PSIG minimum pressure required

Table H - Control Panel Construction*

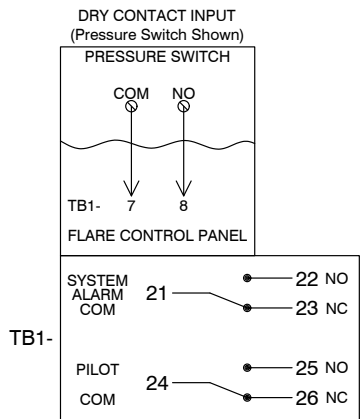
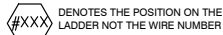
Option H	Description
2	304 SS Panel, Pipe & Fittings
3	304 SS Panel, Pipe, Fittings & Base
4	304 SS Panel, Base & Weatherhood
5	304 SS Panel w/ 316 SS Pipe & Fittings
6	304 SS Panel & Base w/ 316 SS Pipe & Fittings
7	304 SS Panel, Base & Weatherhood with 316 SS Pipe & Fittings

NOTE: *Other materials available upon request.

Table I - Materials of Construction

Option I	Stack / Piping
2	304(L) Stainless Steel / 304(L) Stainless Steel
4	316(L) Stainless Steel / 316(L) Stainless Steel
5	304(L) Stainless Steel / 316(L) Stainless Steel
6	316(L) Stainless Steel / 304(L) Stainless Steel

NOTE: Pilot Material 316 Stainless Steel



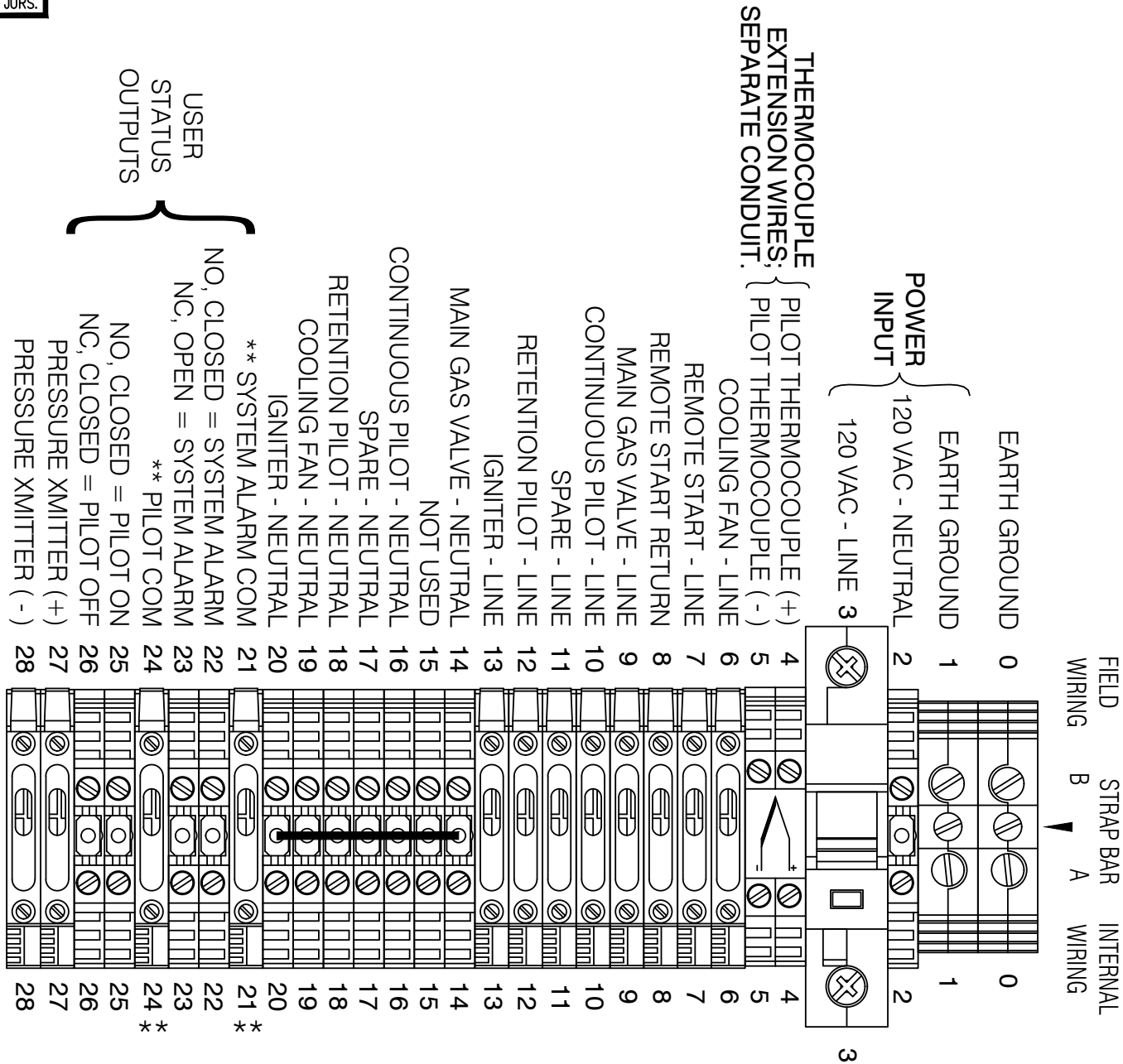
A	Submittal	7322	CWS	Jun 13 25	RT
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND & JURS an L&J TECHNOLOGIES Company HILLSIDE, ILLINOIS USA 97301 Candlestick Flare Ladder Logic					
APRV CNO D DATE DRAW	CWS	Type:	Elec.	Size: C: 22 x 17	
	Jun 13 25	Scale:	None	Designed: CWS	
	CWS	T:\Flares\97301\B2090\97301-A-B2090.dwg			
	RT	97301-A-B2090			

97301-B-!B2090

T:\Flares\97301\B2090\97301-B-!B2090.dwg

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- * ; FACTORY WIRED
** ; 6-240VAC/6-27VDC, 2 AMP MAX.



PROJECT: **HILD WWTP PROJECT**
CUSTOMER: **NAN, INC.**
CUSTOMER P.O. NO: **24077-00048**
DESCRIPTION: **Waste Gas Burner**
MODEL NO: **97301-153773001**
SPECIFICATION: **11406A** SECTION: **2.02-2.07**
S&J SALES ORDER NO: **!B2090** DATE: **June 13, 2025**

APPROVED ☐ APPROVED w/CHANGES ☐ NOT APPR.-RESUB. PER COMMENT ☐
CUST. SIGNATURE: _____ DATE: _____

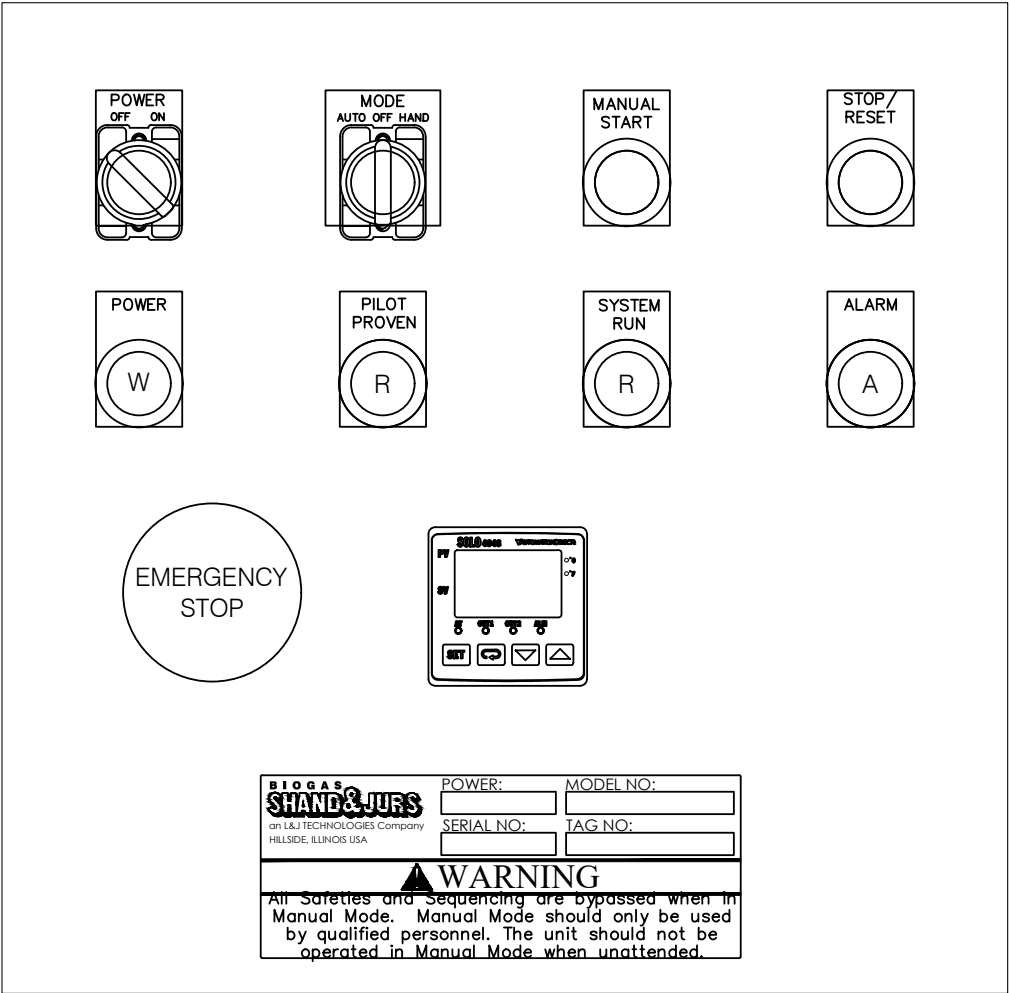
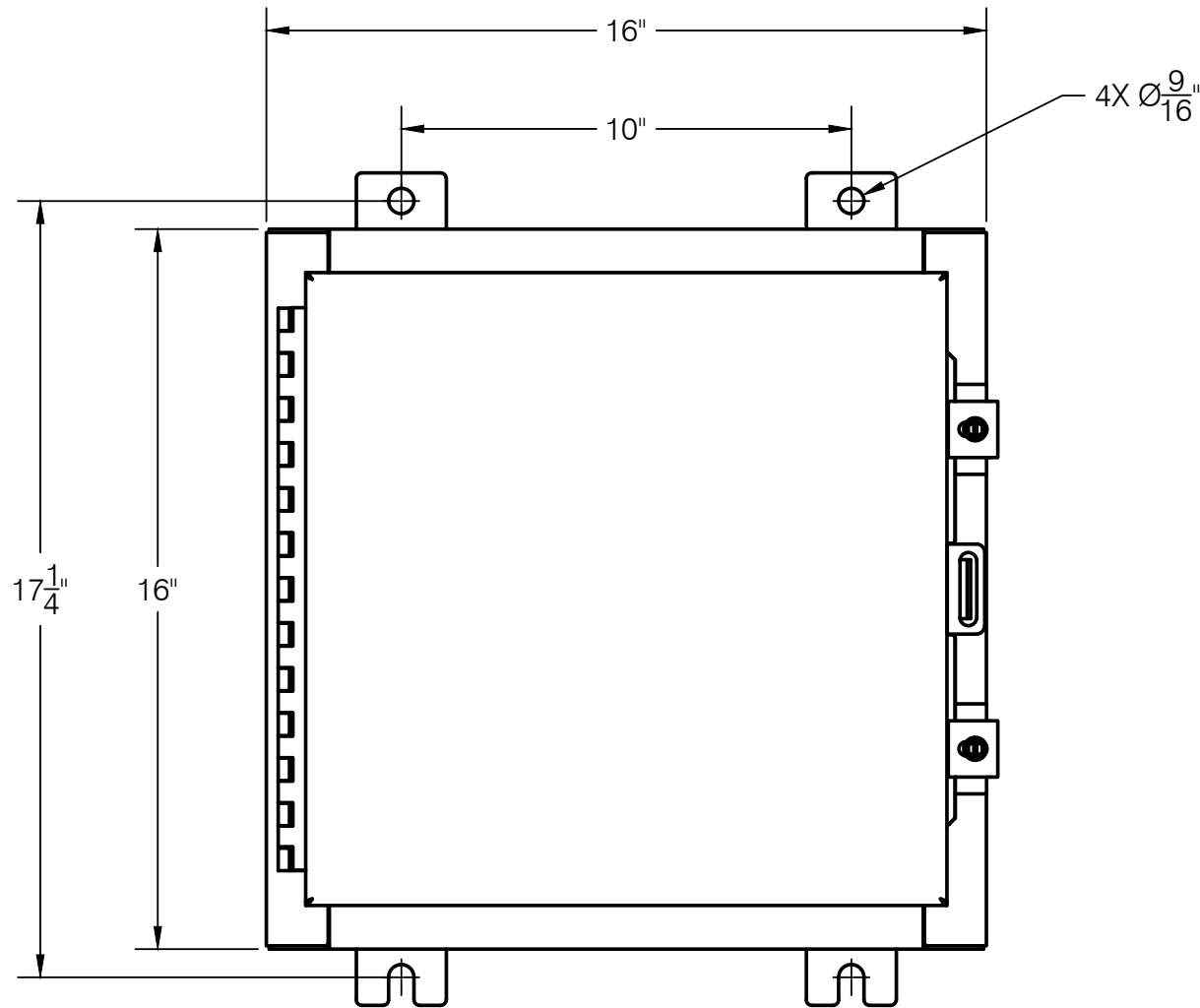
Note:

- SEE INSTRUCTION MANUAL **97-301-!B2090** FOR WIRE TYPES.
- TB1-4 AND TB1-5 USE TYPE K THERMOCOUPLE EXTENSION WIRE (USER SUPPLIED).

A	Submittal	7322	CWS	Jun 13 25	RT
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND&JURS an Lej TECHNOLOGIES Company HILLSIDE, ILLINOIS USA					
97301 Candlestick Flare Field Wiring Termination					
DRWN	CWS	Type:	Dim's	Size:	B: 17 x 11
DATE	Jun 13 25	Scale:	None	Designed:	CWS
APRV	CWS	T:\Flares\97301\B2090\97301-B-!B2090.dwg			
	RT	97301-B-!B2090			

97301-C-!B2090

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Notes:

- 1) NEMA 4X ENCLOSURE - 316 STAINLESS STEEL. UL508A CERTIFIED
- 2) DEPTH: 6 INCHES NOMINAL.
- 3) POWER: 120VAC @ 10 AMPS.
- 4) FIELD WIRING: SEE 97301-B-!B2090.
- 5) CONDUIT OPENING: BY INSTALLER.
- 6) WEIGHT: APPROXIMATELY 35 LBS.
- 7) INCLUDES HEATER AND THERMOSTAT.
- 8) PANEL MUST BE MOUNTED AT LEAST 15' FROM FLARE STACK IN NON-HAZARDOUS LOCATION.

PROJECT: HILD WWTP PROJECT		
CUSTOMER: NAN, INC.		
CUSTOMER P.O. NO: 24077-00048		
DESCRIPTION: Waste Gas Burner		
MODEL NO: 97301-153773001		
SPECIFICATION: 11406A SECTION: 2.02-2.07		
S&J SALES ORDER NO: !B2090 DATE: June 13, 2025		
APPROVED ☐	APPROVED w/CHANGES ☐	NOT APPR.-RESUB. PER COMMENT ☐
CUST. SIGNATURE: _____		DATE: _____

Quantity: 1
Cust. Tag No(s): 11-VCP-2370

A	Submittal	7322	CWS	Jun 13 25	SB
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND&JURS an L&J TECHNOLOGIES Company HILLSIDE, ILLINOIS USA					
97301 Candlestick Flare Control Box					
APRV/CHKD DATE	CWS	Type:	Dim's	Size:	B: 17 x 11
	Jun 13 25	Scale:	1 : 4	Designed:	CWS
	CWS	T:\Flares\97301\B2090\97301-C-!B2090.dwg			
	SB	97301-C-!B2090			

97301-J-!B2090

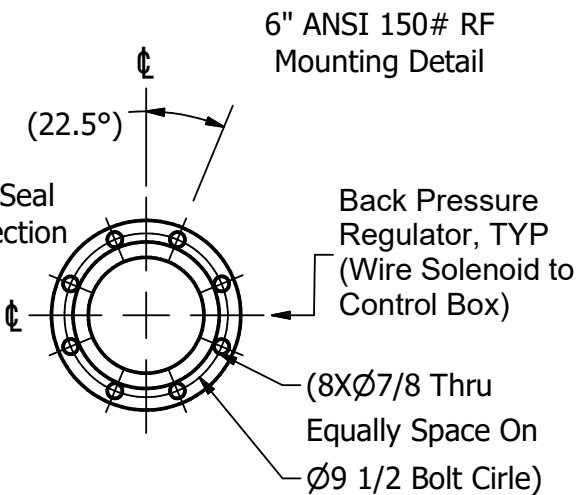
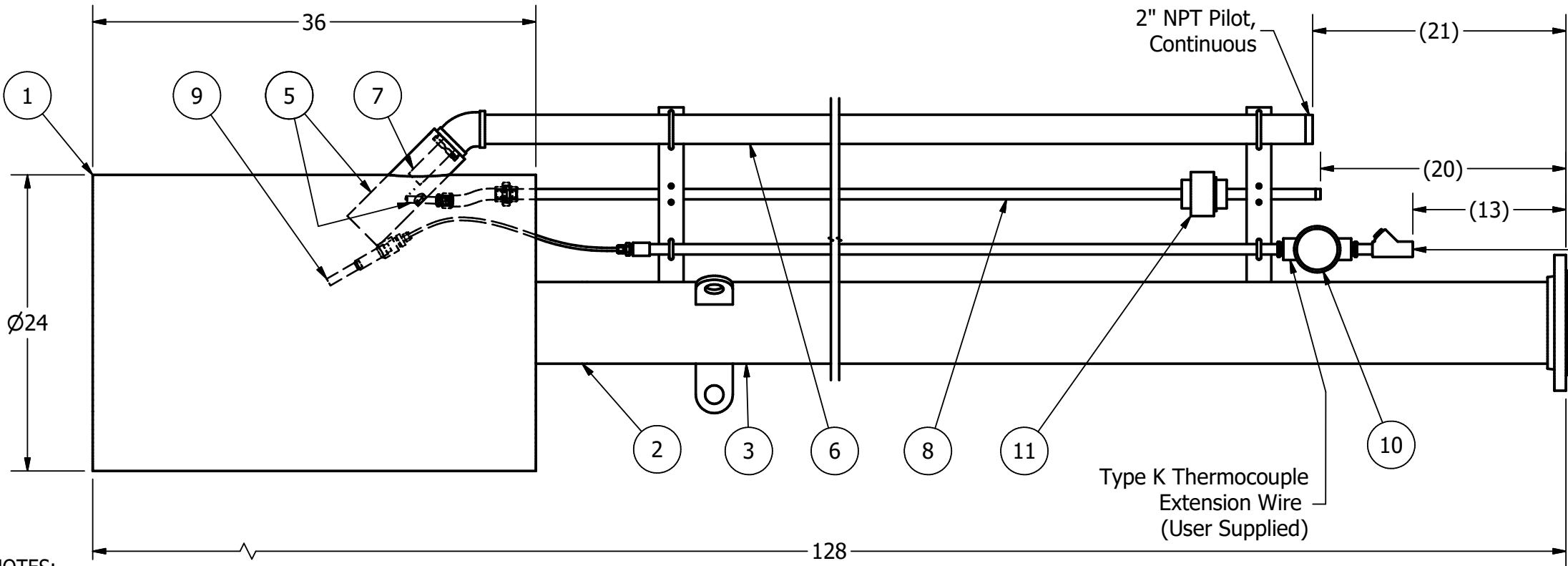
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Design Parameters

Inlet Connection: 6" RF ANSI 150# Flange
Burning (Flow) Capacity: 12,000 SCFH MAX
Fuel Gas Temperature: 60 to 120°F
Min Turndown Ratio: 10 to 1
Gas Composition: 55-65% CH₄, 35-45% CO₂, 500 to 1,500 PPM H₂S + Trace
Elements (TYP)
Net (Lower) Heating Value: 500 to 600 BTU/SCF
Pilot Gas: Propane Gas, 10 PSIG MIN
Pilot System: Automatic/Continuous (Flaring)

PROJECT: *HILO WWTP PROJECT*
CUSTOMER: *Nan, Inc.*
CUSTOMER P.O. NO: *24077-00048*
DESCRIPTION: *Waste Gas Burner*
MODEL NO: *97301-153773001*
SPECIFICATION: *11406A* Section: *2.02 - 2.07*
S&J SALES ORDER NO: *!B2090* DATE: *Jun 13, 2025*

APPROVED ☐ APPROVED ☐ NOT APPR.-RESUB. ☐
w/CHANGES PER COMMENT
CUST. SIGNATURE: _____ DATE: _____



- NOTES:
- 1. All Dimensions in Inches Unless Otherwise Stated.
 - 2. Wiring Per 97301-B-!B2090.
 - 3. 6" 150# RF Mounting Hardware and Support Structure; Not Supplied.
 - 4. Not Used.
 - 5. Not Used.
 - 6. Flare Assembly With All Associated Loads Must Be Fully Supported By Mounting Structure.
 - 7. Clean, Dry, Propane Gas Required; Regulated by Customer. Coalescing Filter Recommended; Not Supplied.
 - 8. Interconnecting Piping; Not Supplied. Piping Arrangement Shall be Straight Piping From Control Panel and May Include (2) 45° Bends at Flare.
 - 9. Continuous Pilot and Retention Lines Must Have Drip Legs; Not Supplied.
 - 10. Pedestal/Base Mounting and Anchorage Hardware; Not Supplied.
 - 11. Heat Tracing and Insulation Recommended/ As Required; Not Supplied.

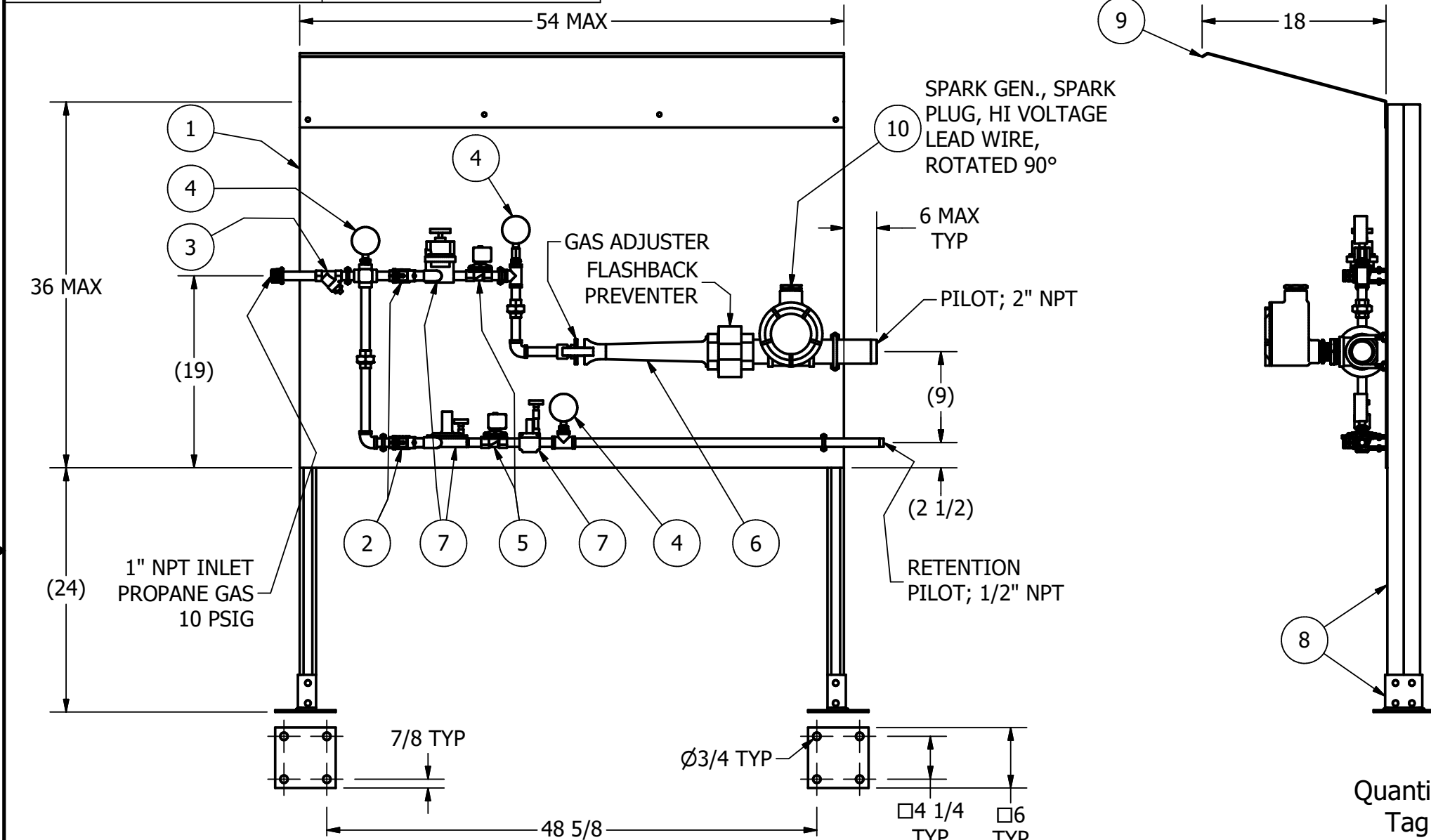
ITEM	DESCRIPTION
1	Wind Shroud; 316 SS
2	Flare Stack Tip; 316 SS
3	Flare Stack; 316 SS
4	ANSI, 150#, RFSO Flange; 316 SS
5	Continuous and Retention Nozzles; 316 SS
6	2" Continuous Pilot Pipe; 316 SS
7	2" Continuous Pilot; 316 SS
8	1/2" Retention Pilot Pipe; 316 SS
9	Thermocouple (Type K)/Thermowell; Inconel
10	Junction Box, NEMA 7; Aluminum
11	Flame Check; Aluminum/316 SS

Quantity: 1
Tag No.: 11-FLR-2370

A	Submittal	7322	BG	06/13/2025	SB
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND&JURS an L&J TECHNOLOGIES Company HILLSIDE, ILLINOIS USA					
97301 Waste Gas Burner General Arrangement					
APRV CHKD DATE DWG	BG	Type: DIM	Size: B: 17 X 11		
	06/13/2025	Scale: NONE	Designed: <i>CWS</i>		
	CWS SB	97301-J-!B2090			

97301-K-!B2090

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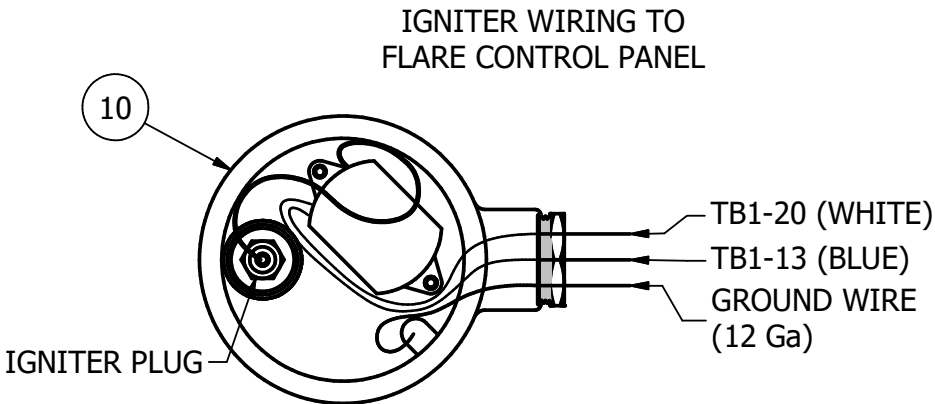


ITEM	QTY	DESCRIPTION
1	1	54" W X 36" H Mounting Panel; 316 SS
2	2	Isolation Valve; 316 SS/Teflon
3	1	Wye Strainer; 316 SS/304 SS
4	3	Pressure Gauge; Brass
5	2	Solenoid Valve, NEMA 4X; Aluminum/SS
6	1	Venturi Inspirator; 347 SS
7	3	Gas Regulator; Aluminum
8	2	Support/Base; 316 SS
9	1	Weather Hood; 316L SS
10	1	Ignitor Junction Box; NEMA 4X, Aluminum

PROJECT: **HILO WWTP PROJECT**
CUSTOMER: **Nan, Inc.**
CUSTOMER P.O. NO: **24077-00048**
DESCRIPTION: **Waste Gas Burner**
MODEL NO: **97301-153773001**
SPECIFICATION: **11406A** Section: **2.02 - 2.07**
S&J SALES ORDER NO: **!B2090** DATE: **Jun 13, 2025**

APPROVED ☐ APPROVED ☐ NOT APPR.-RESUB. ☐
w/CHANGES PER COMMENT
CUST. SIGNATURE: _____ DATE: _____

- NOTE:
- ALL UNITS IN INCHES UNLESS OTHERWISE SPECIFIED
 - CONTROL BOX: REMOTE TO THE GAS LINE AND LOCATED OUTSIDE THE CLASSIFIED ENVELOPE. SEE DRAWING 97301-C-!B2090.
 - CONTROL BOX INCLUDES HEATER AND THERMOSTAT.
 - SOLENOIDS AND IGNITER ARE TO BE FIELD WIRED (NOT SHOWN).
 - REGULATED PROPANE GAS SUPPLY. CLEAN DRY GAS REQUIRED. IF NECESSARY, COALESCING FILTER RECOMMENDED, NOT SUPPLIED.
 - ALL PIPING, TUBING AND THREADED FITTINGS ARE 316 STAINLESS STEEL, UNLESS OTHERWISE SPECIFIED.
 - NOT USED.
 - INTERCONNECTING PIPING NOT PROVIDED. PILOT LINES MUST BE CONNECTED WITH STRAIGHT PIPING.
 - HEAT TRACING AND INSULATION RECOMMENDED/AS REQUIRED (NOT PROVIDED).
 - PIPING TO FLARE MAY INCLUDE UP TO (2) 45° BENDS.
 - CONTRACTOR SHALL PROVIDE ALL NECESSARY PIPING, VALVES, AND REGULATORS.
 - APPROXIMATE WEIGHT: 500LBS
 - CUSTOMER MUST CONFIRM INLET/OUTLET ORIENTATION AS SHOWN. SIGNED: _____



Quantity: 1
Tag No.: NA

A	Submittal	7322	BG	6/13/2025	SB
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND&JURS an L&J TECHNOLOGIES Company HILLSIDE, ILLINOIS USA					
97301 Waste Gas Burner Pilot Control System					
DRWN	BG	Type:	DIM	Size:	B: 17 x 11
DATE	6/13/2025	Scale:	NONE	Designed:	CWS
CHKD	CWS	97301-K-!B2090			
APRV	SB				

Tab 2

(97162-064210001)

6" Pressure Relief/Flame Trap Assembly

- Data Sheet

- Drawing



BIOGAS
SHAND & JURS



L&J TECHNOLOGIES

A Cognesense Company

97162 Pressure Relief/Flame Trap Assembly

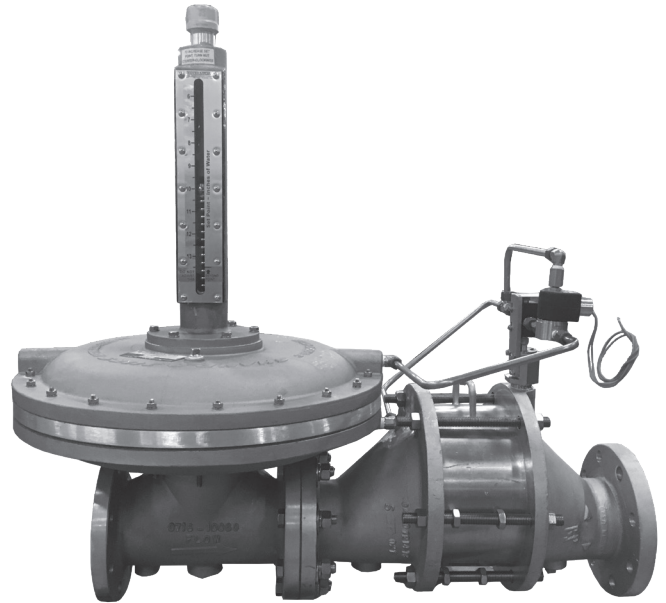
The Shand & Jurs 97162 Pressure Relief and Flame Trap Assembly combines a 97150 Back Pressure Regulator, a Thermally Deflagration Operated Shut Off Valve, and a 94407 Horizontal Flame Arrester to maintain upstream pressure and effectively inhibit flame propagation in low-pressure gas lines.

The Flame Arrester tube bank assembly absorbs heat faster than the ignited gas can produce it. This lowers the gas temperature to below the ignition point, thus quenching the flame. The back pressure regulator maintains an adjustable preset upstream pressure. When upstream pressure exceeds set point, the diaphragm will open the valve. After relieving gas to flare the pressure in the line will drop below set point closing the valve. The Pilot Valve Assembly contains a fusible material that melts at 255°F and closes the regulator valve effectively shutting off flow during emergency thermal event conditions.

The Flame Trap Assembly should be installed upstream of a waste gas flare or burner. External Sense Connection should be connected minimum 10 pipe diameters upstream.

Standard materials of construction include an Aluminum Body and Buna-N diaphragm. Its aluminum and stainless steel components withstand the severest of process environments.

The Shand & Jurs 97162 is especially designed for hydrogen sulfide and hot, wet methane which are the main components of digester gas streams in municipal waste water treatment facilities.



Features

- External Sense Connection
- Positive Emergency Shutoff
- Maintain Upstream Pressure
- 255°F Fusible Link
- Sizes ~~2" Through 12"~~
- 3-Way Solenoid Shut-off Option for Flare System
- Large Diaphragm for Sensitivity
- ~~Manual Bypass for Solenoid Option Available~~
- Horizontal Configuration Only

Standard Materials of Construction:

Body:

Cast Aluminum

Flame Arrester Housing:

Cast Aluminum

Pallet:

Low Copper Aluminum

Tube Bank:

316 Stainless Steel

Diaphragm:

Nylon Reinforced Buna-N

Specifications:

Flanged Connections:

125 lb. ANSI FF Flange

Installation:

Locate 97162 per deflagration arrester approval

Drain Connection:

1/2" NPT Connection

Maximum Operating Pressure:

20" W.C. (Sizes 2"-8")

~~15" W.C. (Sizes 10" and 12")~~

Consult Factory for other options.

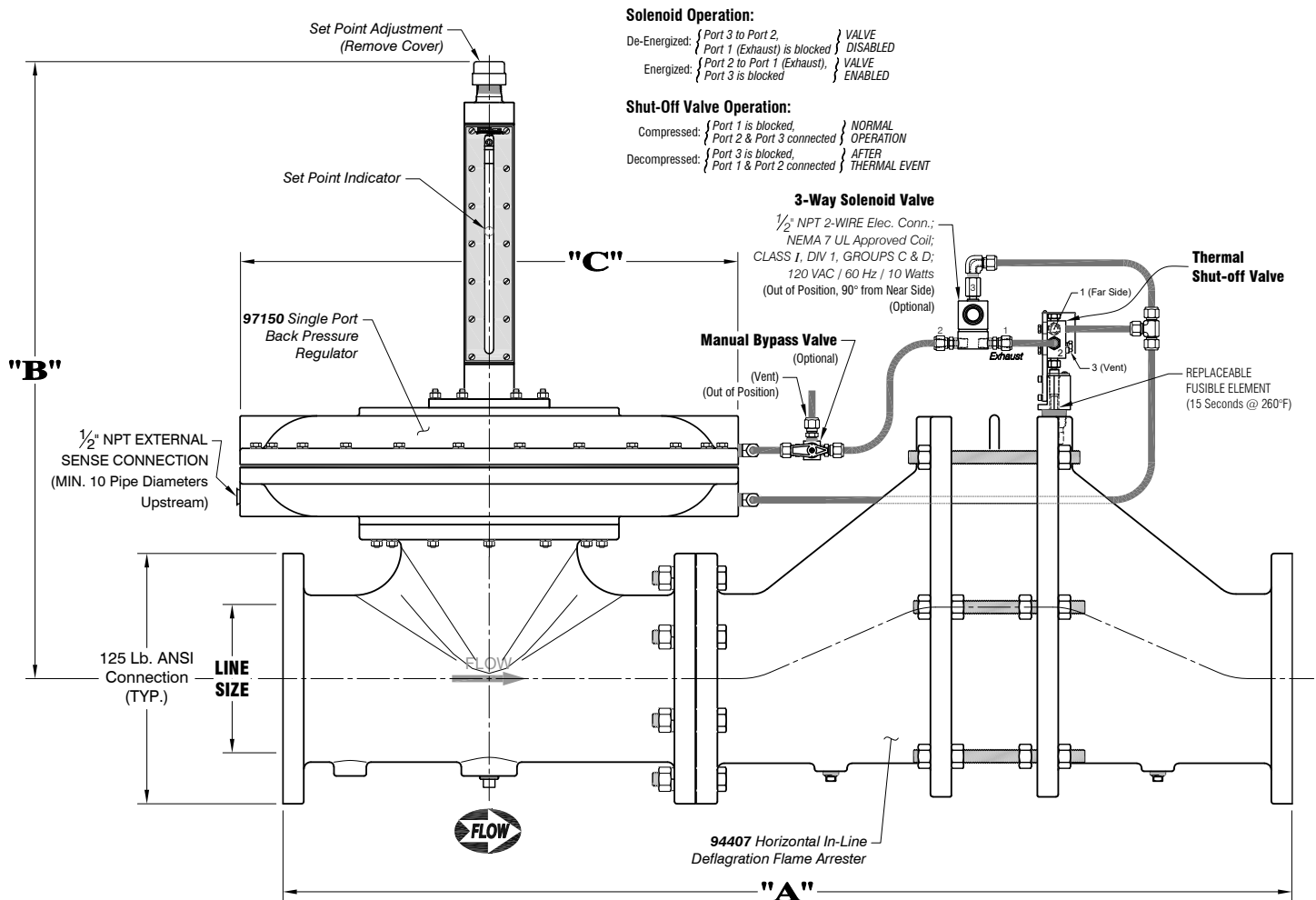
Pressure Rating:

5 psi Standard

Air Flow Capacity in Standard Cubic Feet per Hour x 1000 @ 60°F

Over Pressure Inches W.C.	Line Diameter						
	2"	3"	4"	6"	8"	10"	12"
1	0.9	2.1	4.2	9.0	12.8	21.0	34.2
2	1.4	3.4	6.4	13.1	19.0	34.8	53.4
3	1.8	4.2	8.0	16.6	24.3	45.4	69.8
4	2.0	4.8	9.3	19.4	28.5	54.6	83.2
5	2.4	5.4	10.4	21.9	32.4	62.6	96.0
6	2.6	6.1	11.6	24.3	44.5	69.8	107.2
7	2.9	6.7	12.5	26.3	48.2	76.4	115.2
8	3.1	7.2	13.5	28.2	51.8	82.4	125.6
9	3.3	7.7	14.3	30.1	55.3	88.8	133.6
10	3.5	8.2	15.3	31.7	58.6	94.4	140.8
11	3.7	8.7	16.2	33.5	61.6	100.0	148.0
12	3.9	9.0	16.8	35.1	64.5	104.8	155.2
13	4.1	9.4	17.7	36.4	67.2	109.6	161.6
14	4.3	9.7	18.3	37.9	69.9	114.4	168.0
15	4.4	10.1	19.0	39.2	72.8	118.4	173.6
16	4.6	10.5	19.6	40.6	75.0	122.4	179.2
17	4.8	10.8	20.2	41.7	77.4	126.4	184.8
18	4.9	11.1	20.8	43.0	80.0	129.6	190.4
19	5.1	11.6	21.3	44.2	81.6	132.8	195.2
20	5.2	11.9	21.9	45.4	84.0	136.0	200.0

Dimensions



Line Diameter (Inches [mm])	A (Inches [mm])	B (Inches [mm])	C (Inches [mm])
2 [50]	23 $\frac{3}{8}$ [594]	27 $\frac{7}{8}$ [708]	21 [533]
3 [75]	26 $\frac{1}{16}$ [661]	28 $\frac{1}{2}$ [724]	21 [533]
4 [100]	31 $\frac{7}{16}$ [799]	29 [736]	21 [533]
6 [150]	39 $\frac{7}{16}$ [1002]	31 $\frac{3}{8}$ [797]	26 $\frac{7}{8}$ [683]
8 [200]	54 $\frac{7}{16}$ [1383]	33 $\frac{3}{8}$ [847]	26 $\frac{7}{8}$ [683]
10 [250]	63 $\frac{5}{16}$ [1608]	37 $\frac{7}{8}$ [962]	36 $\frac{1}{2}$ [927]
12 [300]	67 $\frac{3}{8}$ [1711]	41 $\frac{1}{4}$ [1047]	36 $\frac{1}{2}$ [927]

All designs subject to change. Certified dimensions and specifications available upon request.

97162 Ordering Guide

Model Number Selection

The model number will have a base number **97162** followed by 7 digit numbers. These digits will represent 6 sets of option tables.

97162 - AB - CD - EF - G - 1 = Special: Set Point 12"
Range: 9"-15"

Table AB - Line Size**Table C - Setting Range**

Option AB	Line Size*	Option C	Setting Range
02	2"	3 - Standard / External	2" - 8" W.C.
		4 - High / External	>8" - 20" W.C.
03	3"	3 - Standard / External	2" - 8" W.C.
		4 - High / External	>8" - 20" W.C.
04	4"	3 - Standard / External	2" - 8" W.C.
		4 - High / External	>8" - 20" W.C.
06	6"	3 - Standard / External	2" - 8" W.C.
		4 - High / External	9" - 15" W.C.
08	8"	3 - Standard / External	2" - 8" W.C.
		4 - High / External	>8" - 20" W.C.
10	10"	3 - Standard / External	2" - 8" W.C.
		4 - High / External	>8" - 15" W.C.
12	12"	3 - Standard / External	2" - 8" W.C.
		4 - High / External	>8" - 15" W.C.

Specify operating set point. Valve range will be +/- 3" W.C. of set point for 2" through 8" size and +/- 2" W.C. for 10" through 12" sizes. Range will not exceed listed setting range (i.e. set point of 3" will have a range of 2"-8", set point of 19" will have a range of 14"-20"). For other ranges, please Consult Factory.

Table D - Tube Bank Material

Option D	Tube Bank Material
2	Stainless Steel

Table F - Hardware Material

Option F	Hardware
0	Stainless Steel
1	316 Stainless Steel

Table E - Configuration

Option E	Configuration
0	Horizontal
1	Horizontal with Solenoid
2	Horizontal with Solenoid and Manual Bypass

Table G - Accessories

Option G	Accessories
0	None
1	Insulation Jacket

9716-00041-06-IB2090

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Set Point: 12" W.C.

Set Point Range: 9-15" W.C.

NOTE:

OUR TYPICAL RANGE IS $\pm 3"$ W.C. OF SET POINT
CONFIRM SET POINT

Notes:

- 1) MAX. 5 PSIG (LEAKPROOF).
- 2) $\Delta P = 1.0"$ W.C. @ 9,040 SCFH
- 3) MOUNTING HOLES STRADDLE CENTERLINE.
- 4) DIMENSIONS EXPRESSED IN INCHES [mm].
- 5) ARRESTER BANK FREE AREA EQUALS OR EXCEEDS 3X LINE SIZE AREA.
- 6) THE REGULATOR SHALL PROVIDE TIGHT SHUTOFF AND MAINTAIN THE BACK PRESSURE WITHIN 10% OF SETTING.

SET POINT ADJUSTMENT
(REMOVE COVER)

SOLENOID OPERATION:

De-Energized: { Port 3 to Port 2, } VALVE
{ Port 1 (Exhaust) is blocked } DISABLED
Energized: { Port 2 to Port 1 (Exhaust), } VALVE
{ Port 3 is blocked } ENABLED

SHUT-OFF VALVE OPERATION::

Compressed: { Port 1 is blocked, } NORMAL
{ Port 2 & Port 3 connected } OPERATION
Decompressed: { Port 3 is blocked, } AFTER
{ Port 1 & Port 2 connected } THERMAL EVENT

SET POINT
INDICATOR

97150 Single Port
Regulator Assembly

3-WAY SOLENOID VALVE:
110 VAC, With $\frac{1}{2}"$ NPT
ELEC. CONNECTION

THERMAL
SHUT-OFF VALVE

1 (Far Side)

Exhaust

3 (Vent)

REPLACEABLE
FUSIBLE ELEMENT
(15 Seconds @ 260°F)

94407 Flame
Arrester Assembly

FLOW

6" Line

$\varnothing 11$ [Ø279]
6"FF 125 Lbs.
ANSI Conn.
(TYP.)

(TYP.) $\frac{1}{2}"$ NPT DRAIN PLUG

39 $\frac{7}{16}$ [1002]

94407 Horizontal Flame Arrester

Maximum Pressure: 10 PSIG

Part Description Material

Body: 356-T51 Cast Aluminum
Tube Bank Shell: 316 Stainless Steel
Tube Bank: 316 Stainless Steel
Hardware: Stainless Steel
Gasket(s): High Temp. Synthetic

97150 Single Port Regulator

Maximum Pressure: 5 PSIG

Part Description Material

Valve Body, Housings: 356-T6 Aluminum Alloy
Pallet: #6061 Aluminum
Diaphragm Plate: Coated Steel
O-Ring(s) / Gasket(s): NBR / Fiber Composite
Diaphragm: NBR
Spring: 17-7 PH S.S. (Electroless Nickel Plated)
Hardware: 316 Stainless Steel
Internal Parts: Types 304/316 S.S. + Aluminum
Fusible Element Assembly: 316 S.S. + Cerrobased Alloy
(260°F Melting Point):

PROJECT: **HILO WWTP PROJECT**
CUSTOMER: **NAN, INC.**
CUSTOMER P.O. NO: **24077-00048**
DESCRIPTION: **Pressure Relief + Flame Trap**
MODEL NO: **97162-064211001**
SPECIFICATION: **11381** SECTION: **2.02**
S&J SALES ORDER NO: **IB2090** DATE: **June 13, 2025**

APPROVED ☐ APPROVED ☐ NOT APPR.-RESUB. ☐
w/CHANGES PER COMMENT
CUST. SIGNATURE: _____ DATE: _____

~	Approval Drawing	45	SB	Jun 13 25	Slaw
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND & JURIS an L&J TECHNOLOGIES Company HILLSDALE, ILLINOIS USA					
97162 Pressure Relief & Flame Trap 6"FF Dim's w/Solenoid					
DRWN	SB	Type:	Dim's	Size:	B: 17 x 11
DATE	Jun 13 25	Scale:	None	Designed:	Standard
CHKD	SB	T:\SO#\XXXX\BXXXX\B2090\9716-00041-06-IB2090.dwg			
APRV	Slaw	9716-00041-06-IB2090			

Tab 3

(97200-22)

1/2" NPT Flame Check

- Data Sheet

- Drawing



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L&J TECHNOLOGIES

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97200 Flame Check

The S&J 97200 Flame Check provides an in-line barrier against flame flashback in small gas lines. They are typically installed in pilot lines supplying fuel burning devices used in flares and waste gas burners. They are also used in vent piping servicing manometer pressure taps and diaphragm regulators.

Optimally, the flame check should be located as close as possible to the open end of the vent pipe or to the potential source of the flame. The flame check quenches in-line fires by lowering the gas temperature to below its ignition point. The 97200 provides safe, reliable fire protection with a woven wire design.

The flame check allows maximum flow with minimal pressure drop. It is also easily serviced and installed with pipe union housing.

Its aluminum and stainless steel components withstand the severest of process environments. The S&J 97200 is especially designed for hydrogen sulfide and hot, wet methane which are the main components of digester gas streams in municipal waste water treatment facilities.

Specifications:

Body:

Aluminum ~~or Stainless Steel~~

Element:

Stainless Steel

Gasket:

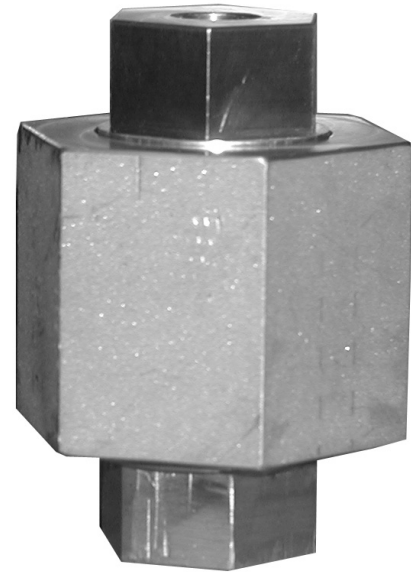
Nitrile Fiber

Pressure Rating:

25 PSIG

Gas Application:

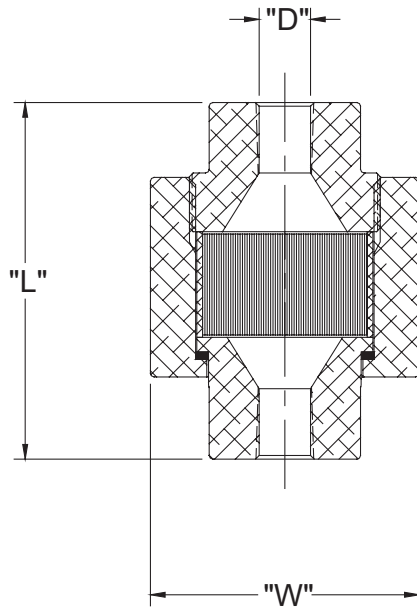
NEC Group D



Features

- Flame Flashback Prevention
- High Flow Capacity
- Standard Stainless Steel
- Sizes ~~1/4"~~ Through ~~1/2"~~
- 25 PSIG Working Pressure

Outline Dimensions



Dimensions (Inches [mm])							
Size Code		1	2	3	4	5	6
D	Nominal Pipe Size	¼ [6 m]	½ [13]	¾ [19]	1 [25]	1 ½ [38]	2 [51]
W	Width	2 [51]	3 ½ [89]	3 ½ [89]	3 ½ [89]	4 ¾ [121]	4 ¾ [121]
L	Length	3 [76]	3 ½ [89]	3 ½ [89]	3 ½ [89]	4 ½ [114]	4 ½ [114]
Shipping Weight		1	2	2	3	3	4

Air Flow Capacity in Standard Cubic Feet per Hour

Pressure Inches W.C.	Line Diameter					
	1/4"	1/2"	3/4"	1"	1-1/2"	2"
0.25	39	54	78	148	316	886
0.50	62	91	129	260	536	1,430
0.75	80	124	176	353	716	1,877
1.00	96	151	213	443	892	2,236
1.25	107	175	247	519	1,032	2,562
1.50	117	193	271	596	1,165	2,863
1.75	126	204	294	659	1,288	3,115
2.00	133	213	311	720	1,395	3,279

All designs subject to change. Certified dimensions and specifications available upon request.

97200 Ordering Guide

Model Number Selection

The model number will have a base number **97200** followed by 2 digit numbers. These digits will represent 2 sets of option tables.

97200 - AB

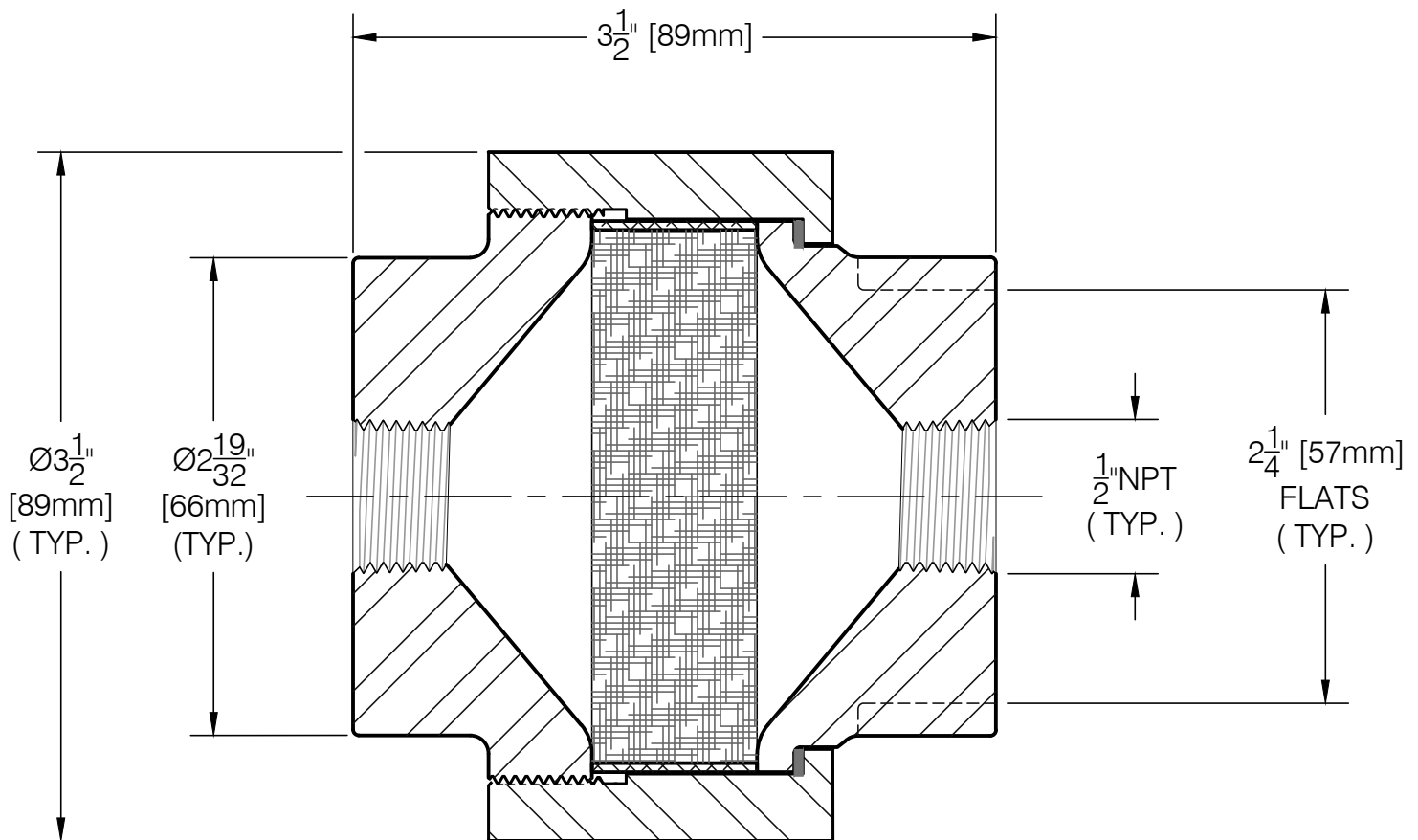
Table A - Line Size & Connection

Option A	Line Size & Connection
1	1/4" NPT
2	1/2" NPT
3	3/4" NPT
4	1" NPT
5	1-1/2" NPT
6	2" NPT

Table B - Body Material

Option B	Element / Body Material
1	316 Stainless Steel / 316 Stainless Steel
2	316 Stainless Steel / Aluminum

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97200 Flame Check Materials of Construction:

Body, Ends Cap: #6061-T6511 Aluminum Alloy
Mesh Element: Type 316 Stainless Steel
Gasket(s): High Temp. Synthetic

Max. Pressure: 25 PSIG
Quantity: 1
Cust. Tag No('s): N/A

PROJECT: **HILO WWTP PROJECT**
CUSTOMER: **NAN, INC.**
CUSTOMER P.O. NO: **24077-00048**
DESCRIPTION: **Model Number Description**
MODEL NO: **97200-22**
SPECIFICATION: **11381** SECTION: **2.08**
S&J SALES ORDER NO: **!B2090** DATE: **June 13, 2025**

APPROVED ☐ APPROVED w/CHANGES ☐ NOT APPR.-RESUB. PER COMMENT ☐
CUST. SIGNATURE: _____ DATE: _____

~	Approval Drawing	45	TKL	Jun 13 25	JJ
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND & JUR'S an L&J TECHNOLOGIES Company HILLSIDE, ILLINOIS USA					
97200 Flame Check 1/2" NPT Dimensions					
APRV	CHKD	DATE	DRWN	Type: Dim's	Size: A: 8.5 x 11
				Scale: None	Designed: Standard
				T:\SO#\XXXXX\BXXXX\B2090\9720-00010-22-!B2090.dwg	
9720-00010-22-!B2090					

Tab 4

(97100E631203)

6 Qt. Electric Drip Trap

- Data Sheet

- Drawing



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L&J TECHNOLOGIES

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97100E Electric Drip Trap

The S&J 97100E Electric Drip Trap provides for the continuous, safe removal of condensate from low points in low pressure gas piping systems. Available in either 3 or 6 quart sizes and with 1" NPT inlets and outlets, the S&J 97100E adapts to any condensation or drainage collection requirement. The 97100E can be mounted in Class 1, Division 1 & 2, Gas Group D hazardous locations and is rated for both NEMA 4 and 7.

The unit collects liquid until the ported rotating disc valve is electrically opened. The actuator seals the gas inlet port prior to opening the drain port allowing for removal of condensate without gas escaping. A visual indicator is located on the front to identify the "open" or "closed" condition. An optional timer controlled electrical actuator is available which will automatically drain the trap at a user programmable time interval.

An optional Local Control Panel, the 97101P, is available and can control up to 8 units maximum. The panel is available in NEMA 4X or NEMA 7 enclosures.

For higher pressure systems, see the Shand & Jurs model 97101E.

Specifications:

Available Sizes/Capacity

~~3 Quart~~, 6 Quart

Inlet/Outlet Connections

1" NPT

Materials of Construction

Low Copper Aluminum ~~or Stainless Steel (SS) body~~, cover plate and (Drip Trap) disc; Stainless Steel internals

Maximum Working Pressure

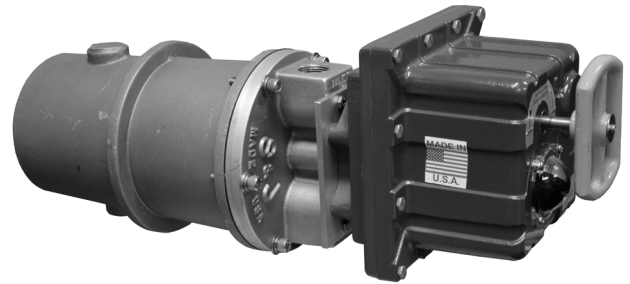
5 PSIG Standard, ~~25 PSIG Optional~~

Mounting Bracket

~~Available in Aluminum or~~ Stainless Steel

Electronic Actuator:

120 VAC 60 Hz, 1 Phase (220 VAC Model available)
Temp. Rating: -40°F to +150°F (with heater)
NEMA 4/7, Aluminum Enclosure Class 1, Div 1 & 2,
Gas Group D CSA Approved
Permanent Lubrication; High Torque Capability
Visual Indication: Open/Close
Indication Contacts: Open/Close SPDT;
Auxiliary Limit Switches (Optional): Open/Close SPDT



97100E Drip Trap Standard Model with Handwheel

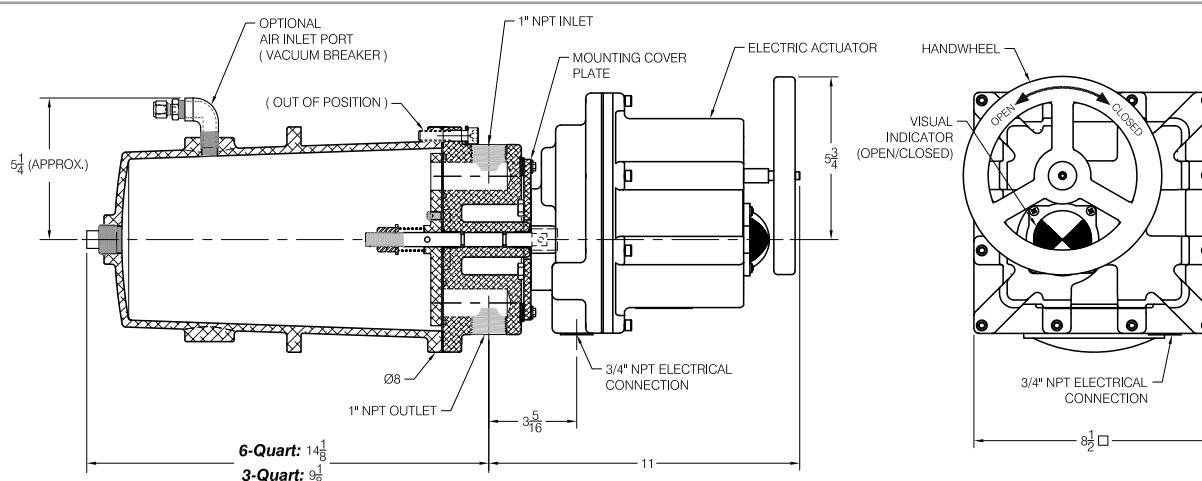
Features

- Positive Gas Shut-off While Draining
- Automatic Operation with an Electronic Timer
- Local Control Panel Available
- 5 PSI ~~or 25 PSI~~ Operating Pressures ~~Available~~
- ~~3 or~~ 6 Quart Capacities
- Air Inlet ~~and Insulation Jackets available~~
- Manual Handwheel Standard
- New - Easy to Read Visual Indicator
- NEMA 4/7 Approved
- Isolation Valve Available

97100E Electric Drip Trap

SHAND & JURS

Dimensions:



All designs subject to change. Certified dimensions and specifications available upon request.

97100E Ordering Guide

Model Number Selection

The model number will have a base number **97100E** followed by 6 digit numbers. These digits will represent 6 sets of option tables.

97100E - AB - CD - EF

Table A - Capacity

Option A	Capacity	MAWP
3	3 Quart	5 PSIG
6	6 Quart	
4	3 Quart	25 PSIG
7	6 Quart	

Table D - Options

Option D	Options
0	None
1	Insulation Jacket
2	Air Inlet
3	Air Inlet with Insulation Jacket

Isolation valve available upon special request.

Table B - Body Material

Option B	Material
2	Aluminum
3	Aluminum w/ Anodized Disc/Cover
4	Aluminum w/ Anodized Disc/Cover/Body
5	304 Stainless Steel
6	316 Stainless Steel

Table E - Hardware

Option E	Options
0	Stainless Steel Bracket / Stainless Steel Hardware
2	Aluminum Bracket / Stainless Steel Hardware
3	316 Stainless Steel Bracket / Stainless Steel Hardware

Table C - Softgoods

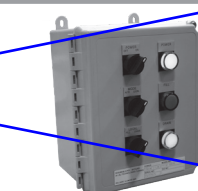
Option C	Softgoods
0	NBR (Nitrile-Butadiene)
1	CR = Chloroprene (Neoprene)

Table F - Actuator

Option F	Actuator
2	No Timer, NEMA 4/7 w/ Manual Handwheel
3	With Timer, NEMA 4/7 w/ Manual Handwheel

97100P Local Control Panel

The 97100P Local Control Panel can control up to eight 97100E units maximum. Selectors include Auto/Local Mode and Local Fill/Drain. Indicators show when the drip trap is filling or draining. An optional fail indicator is also available. Optional status and fail dry contacts are available for user control. The Local Control Panel is available for manual control only or with automatic timers. Typical enclosures available are NEMA 4X Fiberglass, 304 Stainless Steel, 316 Stainless Steel or NEMA 7 Aluminum.



NEMA 4X Fiberglass Panel



NEMA 7 Panel

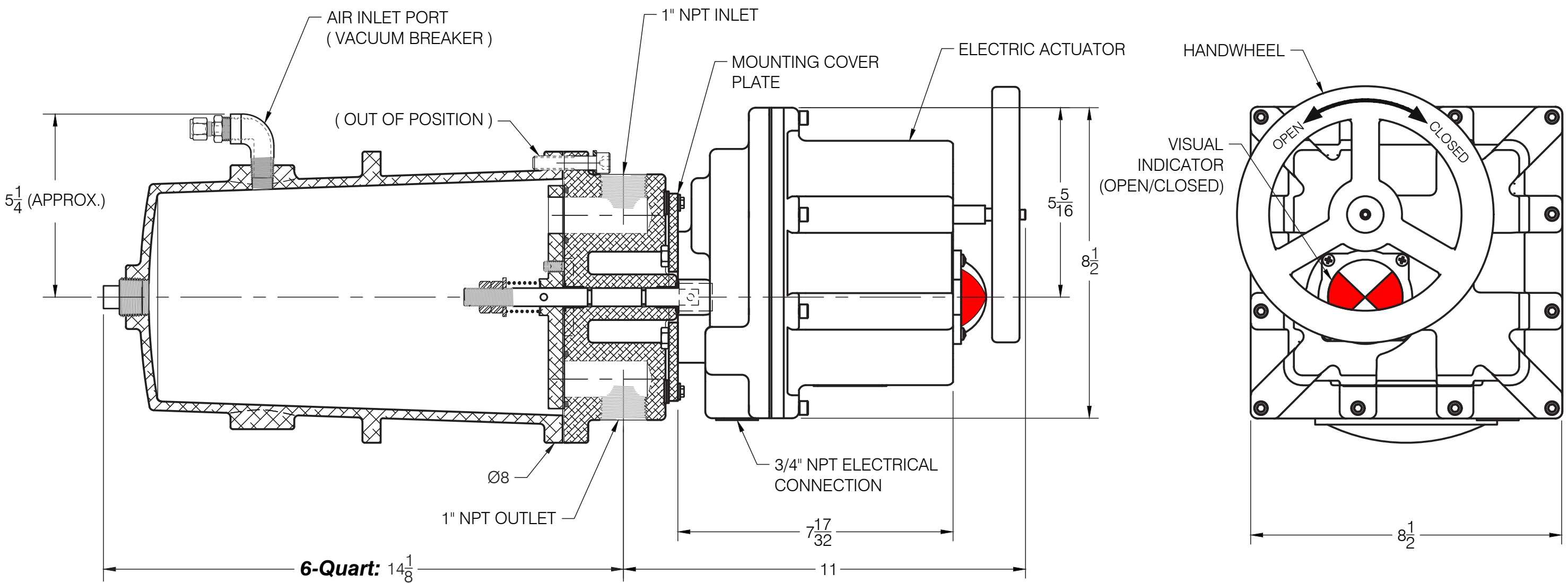
SHAND & JURS

5911 Butterfield Road Hillside, IL 60162 • tel: (708) 236-6000 • fax: (708) 236-6006
sales@ljtechnologies.com • www.ljtechnologies.com • ISO 9001:2015 Certified

Rev: 97100E-3G

9710-00046-!B2090

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ACTUATOR SPECIFICATION:

- 1) 120VAC 60HZ, 1 PHASE, 1 AMP.
- 2) TEMP.RATING -40° TO +150°F.
- 3) NEMA 4X/7 ENCLOSURE,
Class *I* Groups *C & D*;
Class *II* Groups *E, F, & G*. DIV 1 & 2.
- 4) PERMANENT LUBRICATION.
- 5) VISUAL INDICATION - OPEN/CLOSE.
- 6) HIGH TORQUE CAPABILITY, REVERSIBLE.
BUILT IN OVERLOAD PROTECTION.
- 7) MANUAL OVERRIDE DECLUTCHING HANDWHEEL.
- 8) DIMENSIONS ARE SHOWN IN INCHES.
- 9) INTERNAL PROGRAMMABLE TIMER.
5 MINUTE TO 10 HOURS (BUILT IN).

PROJECT: <i>HILD WWTP PROJECT</i>		
CUSTOMER: <i>NAN, INC.</i>		
CUSTOMER P.O. NO: <i>24077-00048</i>		
DESCRIPTION: <i>Electric Drip Trap</i>		
MODEL NO: <i>97100E631203</i>		
SPECIFICATION: <i>11381</i>		SECTION: <i>2.11</i>
S&J SALES ORDER NO: <i>!B2090</i> DATE: <i>June 13, 2025</i>		
APPROVED ☐	APPROVED ☐ w/CHANGES	NOT APPR.-RESUB. PER COMMENT ☐
CUST. SIGNATURE: _____		DATE: _____

Max. Working Pressure: 5 PSIG
Capacity: 6 Quart
Quantity: 6
Cust. Tag No(s): N/A

97100 Electric Drip Trap

Part Description: Material:

- Body: *Low Copper Aluminum Alloy 356*
Cover: *Low Copper Aluminum Alloy 356, Anodized*
Disc: *Low Copper Aluminum Alloy 6061, Anodized*
Internals: *Stainless Steel*
Shaft: *Stainless Steel*
O-Rings: *Chloroprene*
Coupling: *Stainless Steel*
Hardware: *Stainless Steel*
Bracket/ Handle: *Stainless Steel*

A	Approval Drawing	45	CWS	Jun 13 25	SB
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND&JURS an L&J TECHNOLOGIES Company HILLSIDE, ILLINOIS USA					
97100E Drip Trap Dimensions 6 Qt. Dim's w/ Vacuum Breaker; Actuator With Timer					
DRWN	CWS	Type:	Dim's	Size:	B: 17 x 11
DATE	Jun 13 25	Scale:	None	Designed:	Standard
CHKD	CWS	T:\SO#\XXXXXX\BXXXX\B2090\9710-00046-!B2090.dwg			
APRV	SB	9710-00046-!B2090			

Tab 5

(94407-632205)

6" Horizontal Inline Deflagration Flame Arrester

- Data Sheet

- Drawing



BIOGAS
SHAND & JURS



L&J TECHNOLOGIES

A Cognesense Company

94407 Horizontal Inline Deflagration Flame Arrester

The Shand & Jurs 94407 Horizontal Inline Deflagration Flame Arrester is designed to provide a positive flame stop in horizontal gas piping systems containing flammable vapors having a low flash point. The 94407 not only provides exceptional protection against propagation of flame from external source but also offers maximum flow capacity. It is specifically designed to prevent liquid accumulation in the tube bank assembly.

The tube bank design, consisting of a spiral-wound crimped ribbon around a solid core, maximizes flow capacity with minimum pressure drop. The standard flame arrester is Factory Mutual (FM) approved to meet the ATEX application requirements per EN 12874 and ISO 16852.

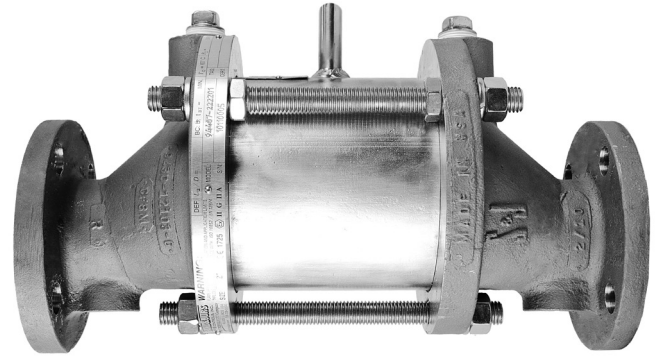
Periodic inspection, maintenance and replacement of the tube bank is easily accomplished by simply removing tie-bolts and minimally expanding the remaining jack screws. Once the upper and lower body sections are expanded, the tube bank is easily removed with the aid of a handle.

Standard body construction includes light weight cast aluminum, cast steel, 304 stainless steel and 316 stainless steel body materials suitable for most environments. Tube bank is available in 304 stainless steel or 316 stainless steel as standard. A complete range of sizes from 2" [50 mm] through 12" [300 mm] are available with flat face or raised face flanges to match ANSI, EN1092-1 or JIS connections.

For highly corrosive applications, body and tube bank construction of Duplex SS, Hastelloy C, or Alloy 20 are available.

Temperature monitor device to be installed on unprotected side of Tube Bank.

Temperature monitor devices available. Consult Factory.



Features

- ~~ATEX EN 12874 Approved~~
- ISO 16852 Approved
- Unitized tube bank design
- Maximum protection and efficiency with minimum pressure drop
- Wide range of standard construction materials
- Easy inspection and maintenance, due to simple removal of tube bank
- Complete range of flange sizes from ~~2" (50 mm) to 12" (300 mm)~~ 6" ANSI, EN1092-1, JIS. ~~Consult factory for larger sizes~~

94407

Horizontal Inline Deflagration Flame Arrester

SHAND & JURS

Specifications

Sizes:

2", 3", 4", 6", 8", 10" and 12"

Max. Static Pressure:

3.45 BARG (50 PSIG)

Max. Operating Pressure:

Sizes 2"-6" 1.1 BARA (16.20 PSIA)

8"-12" 1.09 BARA (15.95 PSIA)

Max. Operating Temp:

60°C (140°F)

Flange Rating:

To match drilling of ANSI 125/150 lb. flat face, raised face flange, EN1092 PN 10/16, and JIS 10K options.

Approval:

ATEX Directive 94/4/EC and EN 12874, ISO 16852 (Short Term Burn)

NOTE: Aluminum and Cast Iron bodies are only approved for ISO 16852.

Gas Application:

IIA (NEC Group D)

Materials of Construction

Body:

Carbon Steel, 304 Stainless Steel, 316 Stainless Steel, Aluminum, Duplex Stainless Steel, Hastelloy C or Alloy 20

Hardware:

Carbon Steel, 304 Stainless Steel, 316 Stainless Steel, Duplex Stainless Steel, Hastelloy C or Alloy 20

Tube Bank:

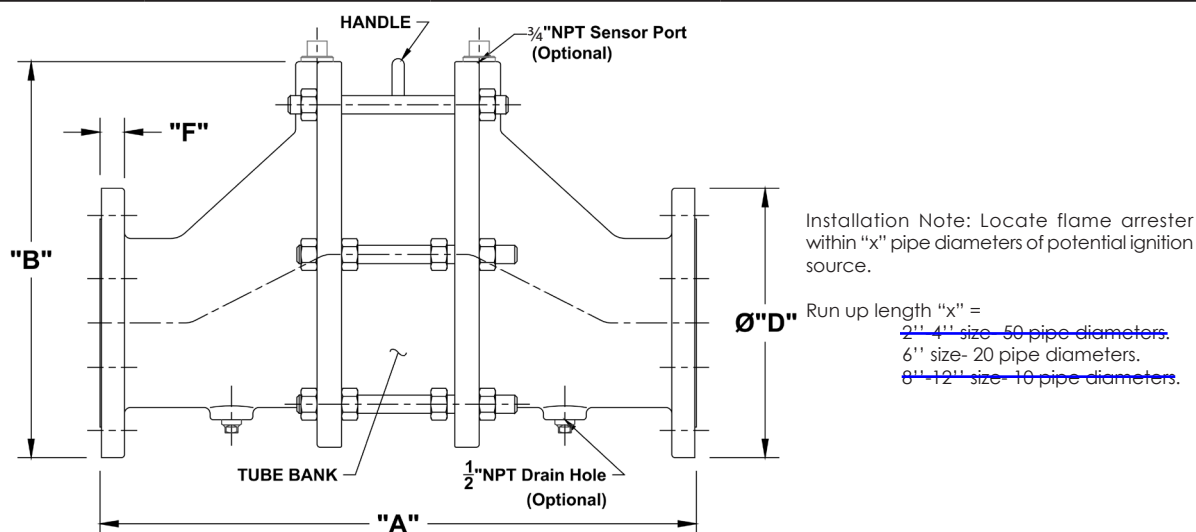
Carbon Steel, 304 Stainless Steel, 316 Stainless Steel, Duplex Stainless Steel, Hastelloy C or Alloy 20

Gaskets:

High Temperature Synthetic Composition

Outline Dimensions

Dimensions in Inches				
Vent Size	"A"	"B"	Diameter "D"	"F"
2"	14 1/2	8 5/32	6	5/8
3"	16	10 11/16	7 1/2	3/4
4"	20	12 3/16	9	15/16
6"	24 3/8	16 11/64	11	1
8"	32 1/8	20 7/8	13 1/2	1 1/8
10"	33 3/8	24 3/4	16	1 3/16
12"	34 3/8	29 5/16	19	1 1/4



All designs subject to change. Certified dimensions and specifications available upon request.

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Rev: 94407-3H

94407 Ordering Guide

Model Number Selection

The model number will consist of a base number **94407** followed by 6 digit letters. These digits will represent 3 option tables.

94407 - AB - CD - EF

Ordering Information

Specify:

1. Model 94407 Horizontal Inline Deflagration Flame Arrester
2. Size and Body Material
3. Tube Bank and Hardware Material
4. Type of Flange and Auxillary Connection
5. Special Materials or Coatings, If Required

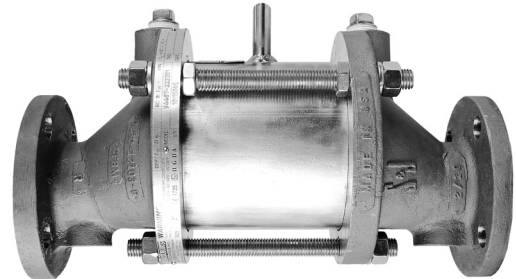


Table AB - Size & Body* Table CD - Tube Bank and Hardware*

Option AB	Size (Inches)	Body Material	Option CD	Tube Bank Material Shell/Element	Hardware Material	Shipping Wt. (lbs)
23	2	Cast Aluminum	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	15
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
33	3	Cast Aluminum	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	25
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
43	4	Cast Aluminum	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	45
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
63	6	Cast Aluminum	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	69
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
83	8	Cast Aluminum	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	93
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
03	10	Cast Aluminum	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	165
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
13	12	Cast Aluminum	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	200
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	

***NOTE:** Aluminum and Cast Iron bodies are only ISO 16852 approved.

Raised Face not available in Aluminum bodies. Other material combinations available.

94407**Horizontal Inline Deflagration Flame Arrester****SHAND & JURS****Table AB - Size & Body* Table CD - Tube Bank and Hardware***

Option AB	Size (Inches)	Body Material	Option CD	Tube Bank Material Shell/Element	Hardware Material	Shipping Wt. (lbs)
20	2	Cast Steel	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	43
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
22	2	Cast 316 SS	22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
30	3	Cast Steel	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	65
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
32	3	Cast 316 SS	22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
40	4	Cast Steel	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	120
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
42	4	Cast 316 SS	22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
60	6	Cast Steel	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	207
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
62	6	Cast 316 SS	22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
80	8	Cast Steel	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	280
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
82	8	Cast 316 SS	22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	

*NOTE: Aluminum and Cast Iron bodies are only ISO 16852 approved.

Raised Face not available in Aluminum bodies. Other material combinations available.

L&J Technologies or any of its subsidiaries assume no responsibility and shall not be liable for any damage, injury or death caused by the mis-application or improper installation of the products that it provides. Installation shall be per manufacturer's instructions in accordance to any applicable local, state or federal regulations. It is the responsibility of the purchaser to ensure these guidelines are followed and that the products are applied properly.

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Rev: 94407-3H

Table AB - Size & Body* Table CD - Tube Bank and Hardware*

Option AB	Size (Inches)	Body Material	Option CD	Tube Bank Material Shell/Element	Hardware Material	Shipping Wt. (lbs)
00	10	Cast Steel	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	480
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
02	10	Cast 316 SS	22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
10	12	Cast Steel	00	Carbon Steel / 304 Stainless Steel	18-8 Stainless Steel	610
			01	Carbon Steel / 304 Stainless Steel	304 Stainless Steel	
			22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	
12	12	Cast 316 SS	22	316 Stainless Steel / 316 Stainless Steel	316 Stainless Steel	

***NOTE:** Aluminum and Cast Iron bodies are only ISO 16852 approved.
Raised Face not available in Aluminum bodies. Other material combinations available.

Table E - Flange Type

Option E	Description
0	ANSI 125/150lb. FF
1	ANSI 125/150lb. RF*
2	EN1092-1 PN10 FF
3	EN1092-1 PN10 RF*
4	EN1092-1 PN16 FF
5	EN1092-1 PN16 RF*
6	FF JIS 10K FF
7	JIS 10K RF*
8	Special Drilling

*Raised Face Flanges Not Available in Aluminum.
Note: Raised faced flanges with smooth finish are standard.
For optional serrated raised faced flanges, consult factory.

Table F - Auxillary Connections

Option F	Description	Quantity
0 ⁽¹⁾	3/4" NPT (Uni-Directional)	1
1	3/4" NPT (Bi-Directional)	2
2	3/4" NPT (Bi-Directional)	4
4	3/4" NPT (U) ⁽²⁾ w/1/2" NPT (Drain)	3 ⁽³⁾
5	3/4" NPT (B) ⁽²⁾ w/1/2" NPT (Drain)	4 ⁽³⁾
6	3/4" NPT (B) ⁽²⁾ w/1/2" NPT (Drain)	6 ⁽³⁾

1. Flow Direction Label Affixed to Cast Body.
2. U = Uni-Directional; B = Bi-Directional.
3. Includes two drain connections - one per side.

Tube Banks Only*

Part Number	Size (Inches)	Tube Bank Material Shell/Element	Shipping Wt. (lbs.)
9440-10000	2	Steel / 304 SS	22
9440-10010		316 SS / 316 SS	25
9440-10001	3	Steel / 304 SS	30
9440-10011		316 SS / 316 SS	35
9440-10002	4	Steel / 304 SS	55
9440-10012		316 SS / 316 SS	60
9440-10003	6	Steel / 304 SS	90
9440-10013		316 SS / 316 SS	100
9440-10004	8	Steel / 304 SS	145
9440-10014		316 SS / 316 SS	160
9440-10005	10	Steel / 304 SS	225
9440-10015		316 SS / 316 SS	240
9440-10006	12	Steel / 304 SS	310
9440-10016		316 SS / 316 SS	325

*Other Material Combinations Available.

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Tab 6

(97120-061032103)

6" Sediment Trap

- Data Sheet

- Drawing



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L&J TECHNOLOGIES

A Cognesense Company

97120 Sediment Trap

The S&J 97120 Sediment Trap is designed to drop out sediment and condensate from "wet" gas immediately downstream from the digester. This protects downstream equipment from corrosion, blockage and water hammer to keep the gas train running for years with low pressure drop.

The inlet nozzle sets up a centrifugal force and a sharp drop in velocity. This combination of forces very efficiently removes the majority of sediment and condensate from the "wet" gas.

Available in large capacities (minimum of 6 gallons condensate and 6 gallons sediment) and with drip trap connection ports, the S&J 97120 provides easy clean out and minimized maintenance times.

Materials of construction include steel or stainless steel, 316 stainless steel hardware is standard. Units are designed for working pressures up to 25 psig.

An optional cooling coil is available to help supplement the removal of moisture in the gas. Cooling water (supplied by others) will be run through the coil inside of the vessel.

To monitor the level of condensate inside the tank, a sight glass is available. Two 1/2" NPT connections are supplied on either side of the 97120 Sediment Trap. Isolation Valves are available in Brass, 316 Stainless Steel or Bronze. A 5/8" diameter high pressure Pyrex sight glass tube is standard.



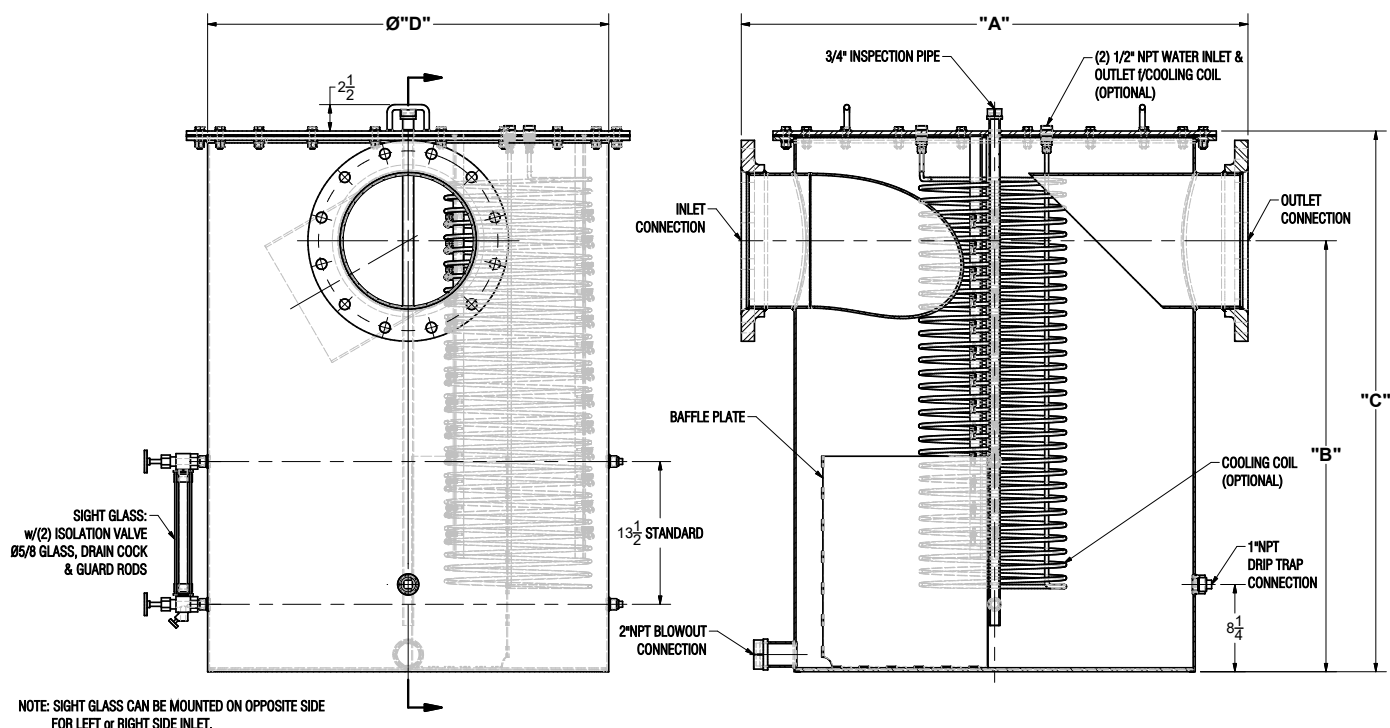
Features

- Removes Sediment and Condensate
- Low and High Pressure Units available up to 25 psi
- ~~Cooling Coil option available~~
- Large Capacity
- Easy Cleanout
- Inlet Sizes ~~2" to 18"~~ 6"
- Low Pressure Drop Design

97120 Sediment Trap

SHAND & JURS

Dimensions:



Line Diameter (Inches [mm])	Dimensions (Inches [mm])			
	"A"	"B"	"C"	"D"
2 [50]	22.5 [572]	22.25 [565]	27.13 [689]	16 [406]
3 [75]	23 [584]	21.5 [546]	27.13 [689]	16 [406]
4 [100]	23 [584]	20.75 [527]	27.13 [689]	16 [406]
6 [150]	27 [686]	24.75 [629]	32.13 [816]	20 [508]
8 [200]	39 [991]	31.25 [794]	38.13 [969]	30 [762]
10 [250]	39 [991]	30.63 [778]	38.13 [969]	30 [762]
12 [300]	48 [1219]	40.75 [1035]	50.5 [1283]	38 [965]
14 [350]	60 [1524]	46 [1168]	58 [1473]	48 [1219]
16 [400]	62 [1575]	50 [1270]	62 [1575]	48 [1219]
18 [450]	74 [1880]	58 [1473]	70 [1778]	62 [1575]

Air Flow Capacity in Standard Cubic Feet per Hour x 1000 @ 60°F

Pressure Drop Inches W.C.	Line Size, Volumetric Flow (SCFH)									
	2"	3"	4"	6"	8"	10"	12"	14"	16"	18"
0.25	1.2	3.1	5.6	13	23	36	51	65	85	108
0.5	1.9	4.4	8	18	32	50	72	92	120	152
0.75	2.3	5.5	10	22	40	62	89	113	147	187
1	2.7	6.6	11	26	46	72	103	130	170	216
1.5	3.3	8.0	14	32	56	87	125	160	209	264
2	3.9	9.3	16	36	64	100	145	185	241	305

All designs subject to change. Certified dimensions and specifications available upon request.

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Rev: 97120-3D

97120 Ordering Guide

Model Number Selection

The model number will have a base number **97120** followed by 8 digit numbers. These digits will represent 7 sets of option tables.

97120 - AB - CD - EF - GH - 3 = Special: AIS Compliant

Table AB - Line Size

Option AB	Size (Inches)
02	2"
03	3"
04	4"
06	6"
08	8"
10	10"
12	12"
14	14"
16	16"
18	18"

Table C - Connection / Pressure Rating

Option C	Connection
0	ANSI RF, 25 PSI
1	ANSI FF, 25 PSI
2	NPT 2" & 3" Only, 25 PSI
6	ANSI FF, 5 PSI
7	ANSI RF, 5 PSI
8	NPT 2" & 3" Only, 5 PSI

NPT Connection on 2" & 3" Sizes only, consult factory for NPT Connection for other sizes.

Table D - Finish

Option D	Finish
0	No Finish
1	Rust Inhibitive Primer Paint
2	Hot Dipped Galvanizing with Steel Only
3	Epoxy Internal & External
4	Epoxy Internal & External Rust Inhibitive Primer Finish

Table E - Tank Material

Option E	Material
1	Steel
2	304 Stainless Steel
3	316 Stainless Steel
4	316L Stainless Steel

Table F - Sight Glass Valves

Option F	Sight Glass
0	None
1	Brass
2	316 Stainless Steel
3	Bronze

Table G - Soft Goods

Option G	Soft Goods
0	NBR
1	CR

NBR (Nitrile-Butadiene)
CR = Chloroprene (Neoprene)

Table H - Cooling Coil

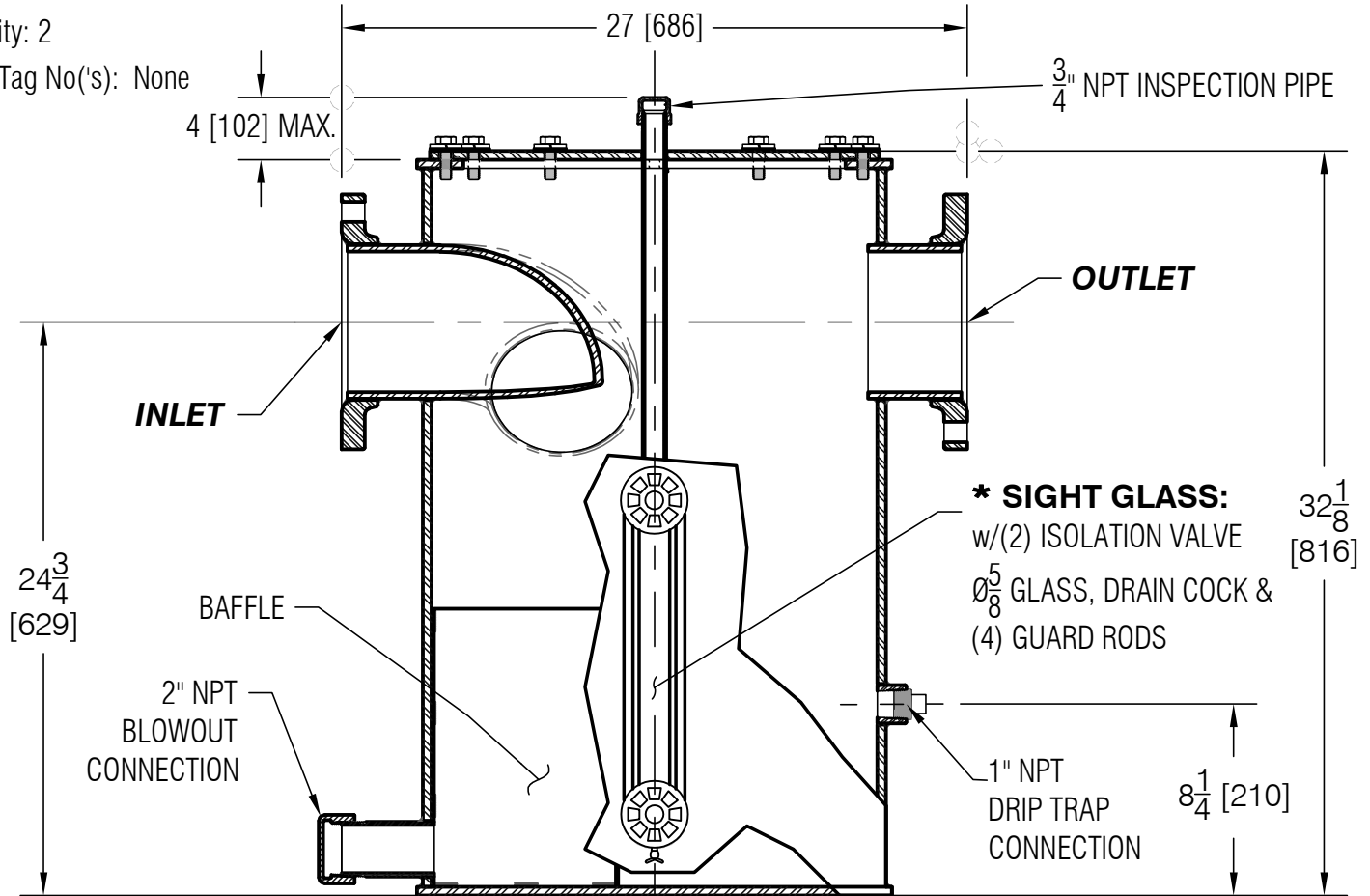
Option H	Cooling Coil
0	None
1	Cooling Coil

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***Note:** Sight Glass can be Mounted on Opposite Side for Left or Right Side Inlet.

Quantity: 2

Cust. Tag No('s): None



Dimensions Expressed In Inches [mm]

Flow Capacity (min.): 13,000 SCFH @ 0.25" ΔP W.C.

Maximum Pressure: 25 PSIG

Capacity (MIN.): 6 Gallons Sediment and 6 Gallons Condensate.

Connection Size Matches: 6"FF ANSI Class 150 Lb. Flange.

Inlet and Outlet Flanges Straddle Center Line.

Section View

AIS Compliant

Part Description: Material:

Body, Cover: ASTM A240T316SS

Isolation Valve, Drain Cock, Rods: 316 S.S.

Sight Glass (Ø5/8" O.D.): Pyrex

Cover Gasket: Neoprene (CR)

Hardware: 316 Stainless Steel

Finish: None

PROJECT: **HILO WWTP PROJECT**

CUSTOMER: **NAN, INC.**

CUSTOMER P.O. NO: **24077-00048**

DESCRIPTION: **Condensate & Sediment Trap**

MODEL NO: **97120-061032103**

SPECIFICATION: **11381** SECTION: **2.04**

S&J SALES ORDER NO: **!B2090** DATE: **June 13, 2025**

APPROVED ☐ APPROVED ☐ NOT APPR.-RESUB. ☐
w/CHANGES PER COMMENT

CUST. SIGNATURE: _____ DATE: _____



an **L&J TECHNOLOGIES** Company
HILLSIDE, ILLINOIS USA

97120 Sediment Trap
6"FF ANSI Dimensions

APPROVAL DATE DRAWN	SB	Type: Dim's	Size: A: 8.5 x 11
	Jun 13 25	Scale: None	Designed: Standard
	SB	T:\SO#\I\XXXXX\BXXXX\B2090\9712-00010-06-!B2090.dwg	
	Slaw	9712-00010-06-!B2090	

~	Approval Drawing	45	SB	Jun 13 25	Slaw
REV	DESCRIPTION	ECO	BY	DATE	APRV

Tab 7

(97570=160718AD2)

6" Combination Conservation Vent & Flame Arrester

- Data Sheet

- Drawing



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A Cognesense Company

97570 Combination Conservation Vent & Flame Arrester

The Shand and Jurs Model 97570 is a combination of our 94020 Conservation Vent and 94306 Flame Arrester in a single, easy to configure package providing for pressure and vacuum relief, as well as positive flame stop on low pressure tanks containing flammable liquids or solvents having a low flash point.

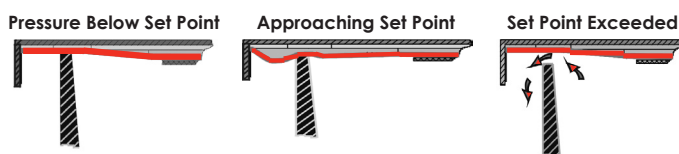
Continued design improvements provide the 94020 conservation vent with high efficiency, maximum flow capacity and minimum leakage. Standard materials of construction are low copper aluminum, cast iron, ductile iron, cast steel and 316 stainless steel for body materials. The conservation vent features air cushion seated diaphragms constructed of FEP for reliability, and extended service life. For high reliability, the pressure and vacuum pallets are both peripherally and center stem guided. Conservation Vents are available in a full range of sizes and configurations, such as open or closed vent (pipe-away). Standard pressure and vacuum settings are 1/2 oz./sq. in.

The 94306 not only provides exceptional protection against fire from external ignition, but also offers maximum flow capacity. The unitized tube bank provides maximum flow while minimizing pressure drop. Standard construction includes light weight cast aluminum, cast iron, cast steel, 304 stainless steel and 316 stainless steel body materials suitable for most environments. Tube bank is available in aluminum, 304 stainless steel and 316 stainless steel as standard. For highly corrosive and severe conditions, special materials and coatings are available. A complete range of sizes from 2" through 12" are available with flat face or raised face flanges to match ANSI 150 lb. connections.

Expanda-Seal

Shand & Jurs "Expanda-Seal" option is available on all pressure pallet assemblies. This feature significantly reduces leakage. The ballooning effect of the FEP diaphragm effectively seals the valve.

The "Expanda-Seal" feature ensures less than .5 SCFH of air at 95% of the set point.



Features

- Sizes ~~2" thru 12"~~ **6"** ANSI & ~~EN1092-1~~
- Cost effective, dual purpose
- Provides pressure & vacuum relief
- Serves as a barrier between external flame and internal vapors
- Provides protection against flame propagation
- ~~Optional~~ "all weather" coating and insulation jackets ~~available~~
- Open ~~or closed~~ vent configurations

94020 Specifications

Sizes:

~~2", 3", 4", 6", 8", 10" & 12"~~

Settings*:

Standard Pressure & Vacuum: 1/2 oz./in.² (.865 in. W.C.)

Expanda-Seal Pressure Setting:

1.5 oz./in.² Minimum (Consult Factory for lower settings)

Maximum Setting W/O Modification:

	(Pressure)	(Vacuum)**
2":	18 oz./in.²	8 oz./in.²
3":	18 oz./in.²	9 oz./in.²
4":	18 oz./in.²	11 oz./in.²
6":	12 oz./in.²	12 oz./in.²
8":	10 oz./in.²	14 oz./in.²
10":	8 oz./in.²	17 oz./in.²
12":	6 oz./in.²	21 oz./in.²

Temperature Range:

Body and Seal -40°F to 220°F. Options for Process Temperature Ranges of -300°F to 500°F

Diaphragm:

FEP (Fluorinated Ethylene Propylene), ~~EKM (Fluoroclastomer), NBR (Nitrile Butadiene)~~

Service and Body Material:

Normal: Cast Low Copper Aluminum

~~Low Temperature: Cast Low Copper Aluminum~~

~~Severe: Cast Iron, Ductile Iron, Cast Steel, Cast 316~~

~~Stainless Steel~~

~~Integral Seats: Same as body; AL, 316 SS, CS with 316~~

~~Stainless Steel seat overlay~~

Replaceable Seats:

~~Aluminum, 316 Stainless Steel, 316 SS Teflon Coated~~

Type of Flange Connection:

~~Screwed or flanged for: 2" & 3" sizes~~

~~Flanged for: 4", 6", 8", 10" & 12" sizes~~

~~Raised face flange available, except for aluminum body material.~~

Options Available:

~~Flame Snuffer for all sizes (open vent) and material, except low temperature service. Closed vent for all sizes and materials.~~

~~Material substitutions as required. Cleaning for LOX/LIN service.~~

94020 Standard Materials of Construction

Component	Normal Aluminum	Low Temperature	Severe		
Body	Cast Aluminum	Cast Aluminum	Cast Iron/Ductile Iron	Cast Steel	316 Stainless Steel
Cover	Aluminum	Aluminum	Steel	Steel	316 Stainless Steel
Hood	18-8 Stainless Steel ¹	Aluminum ²	18-8 Stainless Steel ¹	18-8 Stainless Steel ¹	316 Stainless Steel
Seats ³	Aluminum	Aluminum	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel
Stem Guide	Galvanized Iron	316 Stainless Steel	Galvanized Iron	Galvanized Iron	316 Stainless Steel
Pallets	Aluminum	Aluminum	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel
Pressure Stem	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel
Vacuum Stem	Aluminum ⁴	Aluminum ⁴	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel
Screens	Galvanized Steel	304 Stainless Steel	304 Stainless Steel	304 Stainless Steel	316 Stainless Steel
Retainer	Aluminum	Aluminum	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel
Hardware	Zinc Plated Steel	316 Stainless Steel	316 Stainless Steel	Zinc Plated Steel	316 Stainless Steel
Diaphragms	FEP	FEP	FEP	FEP	FEP

NOTES: 1. 18-8SS for 2", 3", 4", 6" & 8" sizes; spun alum. for 10" & 12" sizes.
2. 2", 3", 4", 6", 8", 10" & 12" sizes; alum. enclosure w/flapper.

3. Material same as body except CI and DI.
4. 316SS for elevated settings.

94306 Flame Arrester Specifications:

*Sizes:

~~2", 3", 4", 6", 8", 10" and 12"~~ Consult Factory for larger sizes

Max. Static Pressure:

10 PSI Standard. ~~For higher pressures, Consult Factory.~~

Flange Rating:

To match drilling of ANSI 125/150 lb. flat face ~~or 150 lb. raised face~~ (Cast Aluminum body option available with flat face only).

94306 Materials of Construction

Body:

~~Cast Iron, Cast Steel, Cast Aluminum, 304 Stainless Steel or 316 Stainless Steel~~

Hardware:

~~18-8 Stainless Steel Standard, 316 Stainless Steel~~

Tube Bank:

~~Aluminum with Aluminum Shell;~~

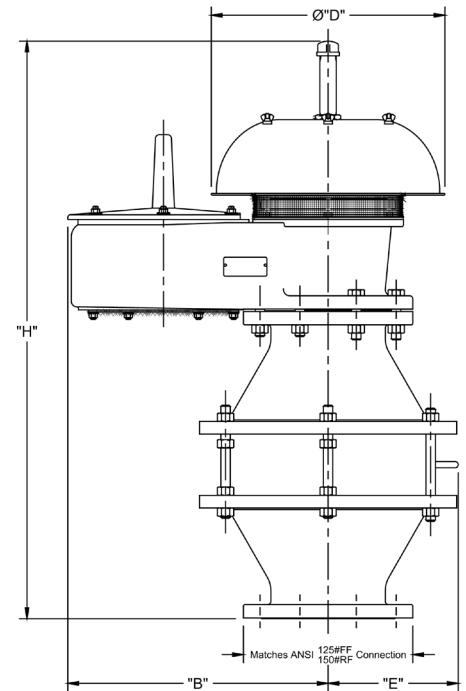
~~304 Stainless Steel with Steel Shell;~~

~~304 Stainless Steel with 304 Stainless Steel Shell;~~

~~316 Stainless Steel with 316 Stainless Steel Shell~~

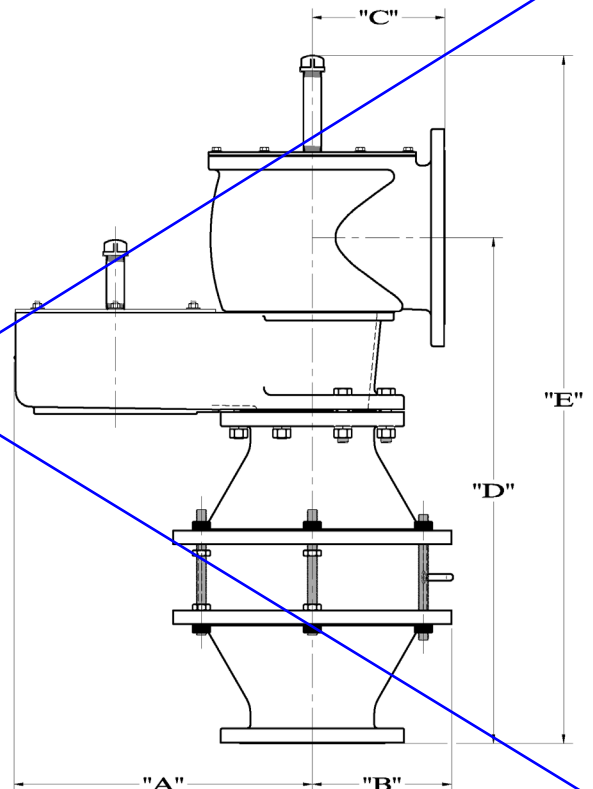
Open Vent Outline Dimensions

Dimensions (Inches [mm])				
Valve Size	"B"	Diameter "D"	"E"	"H"
2 [50]	8 13/16 [224]	12 7/8 [327]	7 5/8 [194]	26 9/16 [675]
3 [75]	11 3/16 [284]	13 9/16 [345]	8 5/16 [211]	29 7/16 [748]
4 [100]	13 9/16 [345]	15 15/16 [405]	6 13/16 [173]	33 11/16 [856]
6 [150]	16 11/16 [424]	18 5/8 [473]	8 1/4 [210]	38 11/16 [983]
8 [200]	20 1/2 [521]	18 5/8 [473]	10 1/4 [260]	45 13/16 [1164]
10 [250]	24 11/16 [627]	25 9/16 [649]	12 1/4 [311]	57 1/16 [1449]
12 [300]	29 1/16 [738]	29 7/8 [759]	14 3/8 [365]	65 3/16 [1656]



Closed Vent Outline Dimensions

Dimensions (Inches [mm])					
Valve Size Inlet x Outlet	"A"	"B"	"C"	"D"	"E"
2" x 2"	8 29/32 [226]	4 1/4 [108]	4 1/4 [108]	18 15/32 [468]	24 9/16 [624]
2" x 3"			4 1/4 [108]	18 7/16 [468]	24 9/16 [624]
3" x 3"	11 1/8 [283]	5 1/4 [133]	5 1/4 [133]	21 9/32 [540]	29 7/16 [748]
3" x 4"			5 1/4 [133]	21 5/8 [549]	29 7/16 [748]
4" x 4"	12 15/16 [328]	6 3/16 [157]	6 1/2 [165]	24 31/32 [633]	38 5/16 [973]
4" x 6"			5 15/16 [151]	24 15/16 [633]	34 1/4 [870]
6" x 6"	17 13/16 [452]	8 3/16 [208]	8 3/8 [213]	29 9/32 [743]	41 3/16 [1046]
6" x 8"			8 1/4 [210]	30 7/32 [767]	41 1/4 [1048]
8" x 8"	21 31/32 [558]	10 1/4 [260]	9 3/4 [248]	36 9/32 [920]	50 3/4 [1289]
8" x 10"			9 3/4 [248]	37 1/4 [946]	50 3/4 [1289]
10" x 10"	26 3/8 [670]	12 1/4 [311]	11 19/32 [294]	42 27/32 [1087]	59 3/16 [1503]
10" x 12"			11 19/32 [294]	43 23/32 [1110]	59 3/16 [1503]
12" x 12"	30 11/16 [780]	14 3/8 [365]	12 13/16 [325]	49 7/32 [1249]	67 [1702]
12" x 14"			12 13/16 [325]	49 3/4 [1264]	66 3/4 [1696]



All designs subject to change. Certified dimensions and specifications available upon request.

97570 Ordering Guide

Model Number Selection

The model number will have a base number **97570** followed by 9 digit numbers. These digits will represent 8 sets of option tables.

97570 - AB - CD - EF - GH - I

Table AB - Size and Body Material

Option AB	Size	Type of Connection	Body Material
11	2"	NPT	Aluminum
12	2"	Flanged	Aluminum
21	2"	NPT	Alum Cryo Hood
22	2"	Flanged	Alum Cryo Hood
*31/41	2"	NPT	Cast Iron/Ductile Iron
*32/42	2"	Flanged	Cast Iron/Ductile Iron
51	2"	NPT	Cast Steel
52	2"	Flanged	Cast Steel
71	2"	NPT	316 Stainless Steel
72	2"	Flanged	316 Stainless Steel
13	3"	NPT	Aluminum
14	3"	Flanged	Aluminum
23	3"	NPT	Alum Cryo Hood
24	3"	Flanged	Alum Cryo Hood
*33/43	3"	NPT	Cast Iron/Ductile Iron
*34/44	3"	Flanged	Cast Iron/Ductile Iron
53	3"	NPT	Cast Steel
54	3"	Flanged	Cast Steel
73	3"	NPT	316 Stainless Steel
74	3"	Flanged	316 Stainless Steel
15	4"	Flanged	Aluminum
25	4"	Flanged	Alum Cryo Hood

* 3X = Cast Iron

4X = Ductile Iron

Option AB	Size	Type of Connection	Body Material
*35/45	4"	Flanged	Cast Iron/Ductile Iron
55	4"	Flanged	Cast Steel
75	4"	Flanged	316 Stainless Steel
16	6"	Flanged	Aluminum
26	6"	Flanged	Alum Cryo Hood
*36/46	6"	Flanged	Cast Iron/Ductile Iron
56	6"	Flanged	Cast Steel
76	6"	Flanged	316 Stainless Steel
17	8"	Flanged	Aluminum
27	8"	Flanged	Alum Cryo Hood
*37/47	8"	Flanged	Cast Iron/Ductile Iron
57	8"	Flanged	Cast Steel
77	8"	Flanged	316 Stainless Steel
18	10"	Flanged	Aluminum
28	10"	Flanged	Alum Cryo Hood
*38/48	10"	Flanged	Cast Iron/Ductile Iron
58	10"	Flanged	Cast Steel
78	10"	Flanged	316 Stainless Steel
19	12"	Flanged	Aluminum
29	12"	Flanged	Alum Cryo Hood
*39/49	12"	Flanged	Cast Iron/Ductile Iron
59	12"	Flanged	Cast Steel
79	12"	Flanged	316 Stainless Steel

Table C - Flange Type

Option C	Description
0	FF ANSI 150 lb
1	*RF ANSI 150 lb
2	DIN PN 16 FF
3	DIN PN 16 RF*
4	JIS 10K FF
5	JIS 10K RF*

* RF not available in Aluminum

Table D - Vent Type

Option D	Description
1	Open Vent
2	Open Vent w/ Flame Snuffer
3	Closed, Standard Outlet > Inlet*
4	Pipe Away, P&V
6	Pipe Away, Same Size Inlet, Outlet, Seat
7	Open Vent with "All Weather" Coating
8	Open Vent with Flame Snuffer with "All Weather" Coating
9	Closed, Standard Outlet > Inlet with "All Weather" Coating
A	Pipe Away, Same Size Inlet, Outlet, Seat with "All Weather Coating
B	Open Vent with "All Weather" Coating with Insulation Jacket
C	Open Vent with Flame Snuffer with "All Weather" Coating & Insulation Jacket
D	Closed, Standard Outlet > Inlet with "All Weather" Coating & Insulation Jacket
E	Pipe Away, Same Size Inlet, Outlet, Seat with "All Weather Coating & Insulation Jacket

* Standard closed vent (outlet is one size larger than the inlet).
Ex. 2" Inlet X 3" Outlet, 6" Inlet X 8" Outlet.

NOTE: Option 6 replaced Option 5. Dimensions are not identical. Consult Factory if replacement valve is required.

Table F - Cleaning & Trim

Option F	Description	Material
1	**Standard Pressure and Vacuum Setting = 0.5 oz/in ²	Lead
2	Over 2.9 oz/sq. in. to Maximum Setting	Lead
3	> Standard to 2.9 oz/in ²	Lead
4	**Standard Setting	316 SS
5	Over 2.9 oz/in. ² to Maximum Setting	316 SS
6	> Standard to 2.9 oz/in. ²	316 SS
7	Incremental Weights Pressure Only 2.9 Oz. - 14 Oz. (5 in. W.C. - 24 in. W.C.)	Lead
8	Standard Incremental Weight Set Pressure Only 6" -16" W.C. in 1" Increments Including (1)-1/2 In. and (2)-1/4 In. W.C. Weights	Lead

**Expand-Seal min. pressure setting: 1.5 oz/in.²

Table G - Seat & Pallet Material

Option G	Seat	Pallet
A	Standard*	Standard*
E	Stainless Steel	Standard*
F	Teflon Coated SS	Standard*
G	Standard*	316 Stainless Steel
K	Stainless Steel	316 Stainless Steel
L	Teflon Coated SS	316 Stainless Steel

*See Materials of Construction

Options G-L are applicable for Aluminum bodies only. 316 SS Pallet is standard on all other body options.

Table E - Pallet / Seal Type

Option E	Description	Diaphragm
0	Standard	FEP
1	Expand-Seal	FEP
4	Standard	FKM
5	Expand-Seal	FKM
8	Normal Buna	NBR
9	Expand-Seal	NBR

Temperature Limits

Diaphragm Material	Temperature Range
FEP	-65 F to 400 F
FKM	-15 F to 400 F
NBR	-40 F to 250 F

Table H - Cleaning & Trim

Option H	Cleaning	Trim
A	Normal	Standard
B	LOX	Standard
C	LIN	Standard
D	Normal	316 Stainless Steel
E	LOX	316 Stainless Steel
F	LIN	316 Stainless Steel

LOX = Liquid Oxygen

LIN = Liquid Nitrogen

Trim Includes Stem, Stem Guides, Side Guides, Nuts, Bolts and Screen.

NOTE: LOX/LIN cleaning for Cryogenic Breathers of aluminum construction includes degreasing before assembly only!

Table I - Flame Arrester, Tube Bank & Shell*

Option I	Tube Bank Material	Shell Material
1	Aluminum	Aluminum
6	304 Stainless Steel	Cast Steel
2	316 Stainless Steel	316 Stainless Steel
4	304 Stainless Steel	304 Stainless Steel

* Tie Rods and Nuts available in 18-8 SS Standard. If 316 SS required, please refer to Option H.

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Notes:

- 1) Connection matches 6"FF ANSI $\frac{125}{150}$ Class Flange.
- 2) Dimensions expressed in INCHES [mm].
- 3) Max. Pressure 10 PSIG for 94306 Flame Arrester.
- 4) Minimum Net Free Area Of Flame Arrester = 3X Pipe Size Area.

97570 Combo Breather & Flame Arrester

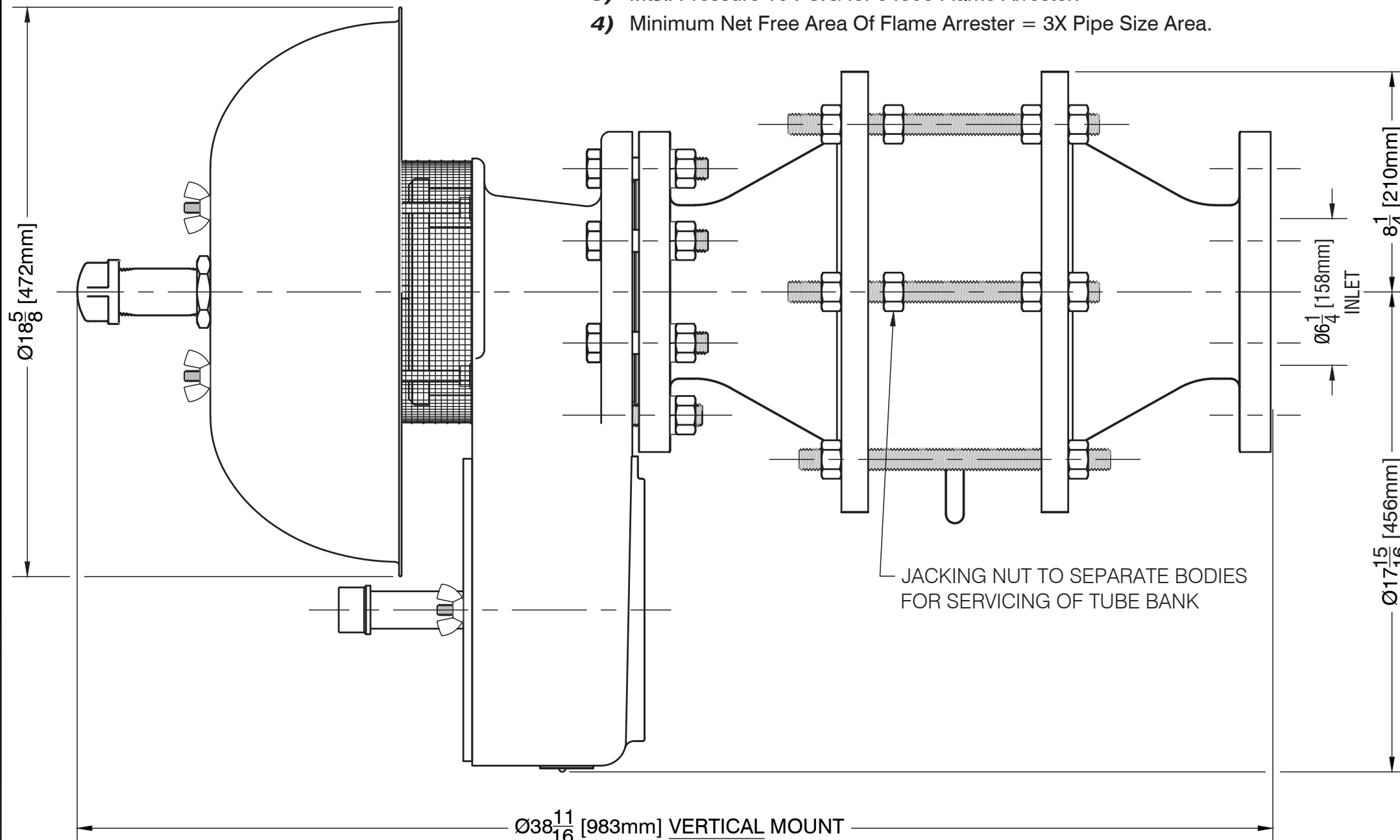
Part Description: Material:

Body: B26-356 Cast Aluminum
Seat (Part of Body): Aluminum* (Teflon Coated)
Expanda-Seal Pallet: Aluminum* (Teflon Coated)
Guide Posts + Stem: 316 Stainless Steel* (Teflon Coated)
Screen: HDPE
External Hardware: 316 Stainless Steel
Hood: 18-8 Stainless Steel
Weights: Lead; Incremental: 6"-16" W.C.
9" W.C. in 1" W.C. Increments
 $\frac{1}{2}$ " W.C. in (2) $\frac{1}{4}$ " W.C. Increments
 $\frac{1}{2}$ " W.C. in (1) $\frac{1}{2}$ " W.C. Increments

Diaphragm (Seat Insert): FEP Film (Teflon)
Gaskets Fiber Composite
Cover O-Ring: NBR

Flame Arrester Body: B26-356 Cast Aluminum
Bank & Element: 316 Stainless Steel
Hardware: 316 Stainless Steel

*All Weather Service: -30°F to 200°F
(Special Teflon Coating
Provides Antifreeze Protection)



JACKING NUT TO SEPARATE BODIES
FOR SERVICING OF TUBE BANK

Ø38 $\frac{11}{16}$ [983mm] VERTICAL MOUNT

Max. Pressure Leakage Rate: 0.5 SCFH @ 95% of Set Pressure

Pressure Setting: 15.0" W.C.
Vacuum Setting: -2.0" W.C.

Flow Capacity (Min):

Pressure: 36,230 SCFH @ 16.5" W.C. (Set Point Plus 10% Over Pressure)
Vacuum: 10,760 SCFH @ 2.2" W.C. (Set Point Plus 10% Under Vacuum)

Qty: 4

Cust. Tag No(s): N/A

PROJECT: **HILO WWTP PROJECT**
CUSTOMER: **NAN, INC.**
CUSTOMER P.O. NO: **24077-00048**
DESCRIPTION: **Combo Breather + Flame Arrester**
MODEL NO: **97570=160718AD2**
SPECIFICATION: **11381** SECTION: **2.11**
S&J SALES ORDER NO: **!B2090** DATE: **June 13, 2025**

APPROVED ☐ APPROVED ☐ NOT APPR.-RESUB.
CUST. SIGNATURE: _____ w/CHANGES PER COMMENT
DATE: _____

A	Approval Drawing	45	TKL	Jun 10 25	SB
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND&JURS an L&J TECHNOLOGIES Company HILLSIDE, ILLINOIS USA 97570 Combo Breather + Flame Arrester 6" Open Vent Dimensions					
APRV/CHKD/DATE	TKL	Type: Dim's	Size: B: 17 x 11		
	Jun 10 25	Scale: None	Designed: Standard		
	TKL	T:\SO#\1\XXXXXX\BXXXX\B2090\9757-00010-06-!B2090.dwg			
	SB	9757-00010-06-!B2090			

Tab 8

(97190-64331111)

6" 3-Way Safety Selector Valve

- Data Sheet

- Drawing



BIOGAS
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L&J TECHNOLOGIES

A Cognesense Company

97190 3-Way Safety Selector Valve

The S&J Model 97190 3-Way Selector Valve permits the safe and efficient maintenance of redundant pressure relief systems without any process interruptions. This is achieved through the smooth operation of the valve, allowing full flow from the primary system, to be diverted to the secondary system while providing positive shut off for the isolated system. The switching of the systems through a single valve only requires one connection into the pressure vessel, allowing the quick change over and avoiding the costly depressurizing of the system.

The 97190 3-Way Selector Valve, is available in sizes from 2" to 12" with ANSI B16.5 150 lb. and 300 lb. raised face or ANSI B16.5 150 lb. flat face flanged connections. The high flow internal configuration is designed to provide maximum capacity under emergency conditions, while providing positive shutoff for the isolated branch. Operation during "switchover" is performed effortlessly with the low friction design of bearings, seats and seals.

A wide selection of body, rotor and stem materials are available to suit the needs of your specific application. Available materials of construction include 304 Stainless Steel, 316 Stainless Steel, Cast Steel and Aluminum. Consult factory for other material selections. The standard seat seal material is Teflon or Buna-N. Consult the factory for other material selections.

Operators are provided for all sizes. The standard operator for the 2" through 12" valves is the nut drive.

Consult Factory for Remote Limit Switch option.



Features

- Permits "switchover" function
- High flow capacity with minimal pressure drop
- Positive shutoff of isolated branch
- Low cost option to multiple valve and piping configurations
- Low torque operation during switchover
- Provided with red position indicator and lockout protection
- ~~Remote limit switches (optional)~~

97190

3-Way Safety Selector Valve

SHAND & JURS

Standard Materials of Construction

Body:

Aluminum
~~304 Stainless Steel~~
~~316 Stainless Steel~~
~~Carbon Steel~~

Rotor:

~~Aluminum~~
~~304 Stainless Steel~~
316 Stainless Steel

Seal:

Teflon
~~Buna-N~~

Trim:

316 Stainless Steel

Isolation Disk, Shaft, Retraction Bush:

316 Stainless Steel
~~17-4 Stainless Steel~~

Specifications

Flanged Connections:

~~150 lb. ANSI RF Flange~~
~~300 lb. ANSI RF Flange~~
150 lb. ANSI FF Flange

Operator:

Nut Drive
~~Nut Drive/Insulation Jacket~~

Pressure/Temperature Rating:

25 PSIG Standard at 100°F (38°C).
~~Consult Factory for higher pressures.~~

Flow Arrangement:

Inlet to Outlet A/B
~~Inlet to Outlet A/B or A&B~~

Padlocking:

The foolproof provision for dual padlocking will prevent unwanted access to the units.

Bleed Valve:

By incorporating a bleed valve on both process connections, the SSV provides an effective way of venting trapped gas before servicing. It also enables field testing on the PRV vent when set pressure adjustments are made.

Active/Inactive Process Indication:

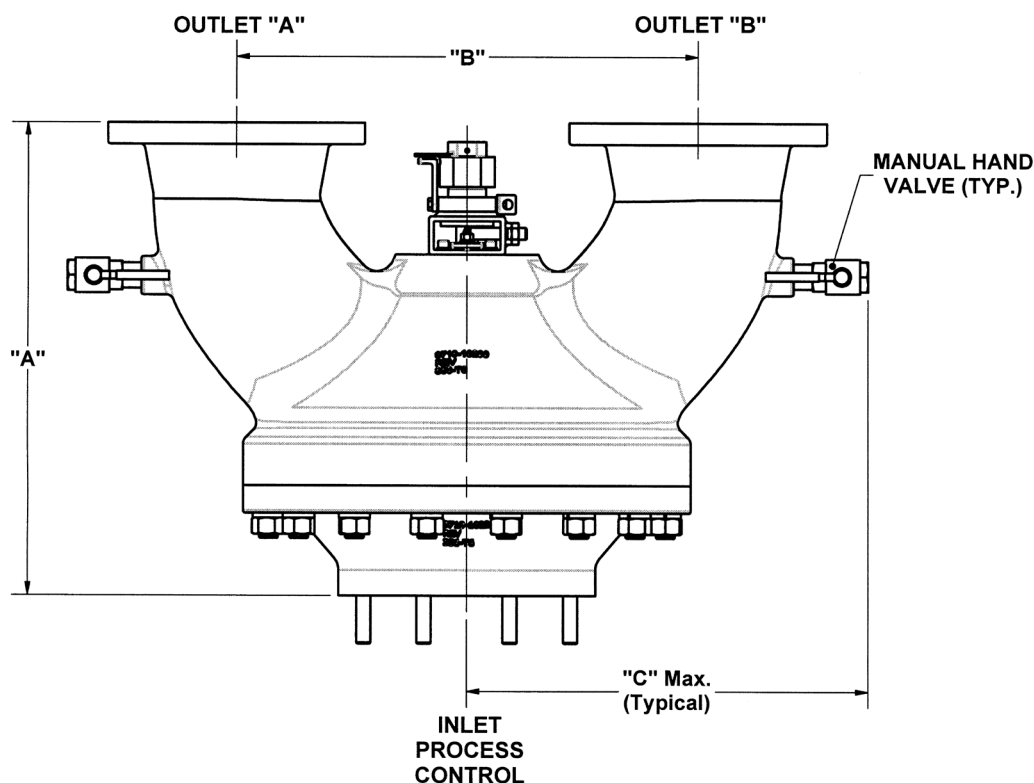
A bright red indicator for positive indication of the active process connection is available so there is never any question which process connection is active.

SHAND & JURS

5911 Butterfield Road Hillside, IL 60162 • tel: (708) 236-6000 • fax: (708) 236-6006
sales@ljtechnologies.com • www.ljtechnologies.com • ISO 9001:2015 Certified

Rev: 97190-3E

Outline Dimensions



Flange Connection	Dimension	97190						
		2"	3"	4"	6"	8"	10"	12"
ANSI B16.5 (150 lb. RF)	"A"	11.75" (298 mm)	13.26" (337 mm)	16.13" (410 mm)	20.17" (512 mm)	24.69" (627 mm)	30.25" (768 mm)	36.00" (914 mm)
	"B"	12.13" (308 mm)	14.00" (356 mm)	16.50" (419 mm)	20.75" (527 mm)	24.25" (616 mm)	28.00" (711 mm)	33.00" (838 mm)
	"C" (Max.)	11.00" (279 mm)	13.00" (330 mm)	15.00" (381 mm)	19.00" (483 mm)	22.00" (559 mm)	26.00" (660 mm)	30.00" (762 mm)
ANSI B16.5 (300 lb. RF)	"A"	11.95" (304 mm)	13.70" (348 mm)	16.63" (422 mm)	21.17" (538 mm)	25.69" (653 mm)	31.25" (794 mm)	37.00" (940 mm)
	"B"	12.13" (308 mm)	14.00" (356 mm)	16.50" (419 mm)	20.75" (527 mm)	24.25" (616 mm)	28.00" (711 mm)	33.00" (838 mm)
	"C" (Max.)	11.00" (279 mm)	13.00" (330 mm)	15.00" (381 mm)	19.00" (483 mm)	22.00" (559 mm)	26.00" (660 mm)	30.00" (762 mm)
ANSI B16.5 (150 lb. FF)	"A"	11.75" (298 mm)	13.26" (337 mm)	16.13" (410 mm)	20.17" (512 mm)	24.69" (627 mm)	30.25" (768 mm)	36.00" (914 mm)
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	"C" (Max.)	11.00" (279 mm)	13.00" (330 mm)	15.00" (381 mm)	19.00" (483 mm)	22.00" (559 mm)	26.00" (660 mm)	30.00" (762 mm)

All designs subject to change. Certified dimensions and specifications available upon request.

97190

3-Way Safety Selector Valve



97190 Ordering Guide

Model Number Selection

The model number will have a base number **97190** followed by 8 digit numbers. These digits will represent 8 sets of option tables.

97190 - AB - CD - EF - GH

Table A - Size

Option A	Description
2	2"
3	3"
4	4"
6	6"
8	8"
9	10"
0	12"

Table B - Body Material

Option B	Body Material
1	304 Stainless Steel
2	Carbon Steel
3	316 Stainless Steel
4	Aluminum

Table C - Rotor Material

Option C	Rotor Material
1	304 Stainless Steel
3	316 Stainless Steel

Table D - Flange Connection

Option D	Inlet	Outlet
1	ANSI 150 lb. RF**	ANSI 150 lb. RF**
2	ANSI 300 lb. RF**	ANSI 300 lb. RF**
3	ANSI 150 lb. FF	ANSI 150 lb. FF

**Raised face flange not available for Aluminum Bodies.
Note: Flange connection matches ANSI 150# or 300# Dimensions.

Table E - Flow Arrangement

Option E	Description
1	Inlet to Outlet A/B
2	Inlet to Outlet A/B or A&B

Table F - Seat Seal Material

Option F	Description
1	Teflon
2	Buna

Table G - Trim Material

Option G	Description
1	316 Stainless Steel
2	316 Stainless Steel / 17-4 Stainless Steel***

*** Isolation disk, shaft, retraction bushing

Table H - Operator

Option H	Description
1	Nut Drive
3	Nut Drive / Insulation Jacket



9719-00020-06-!B2090

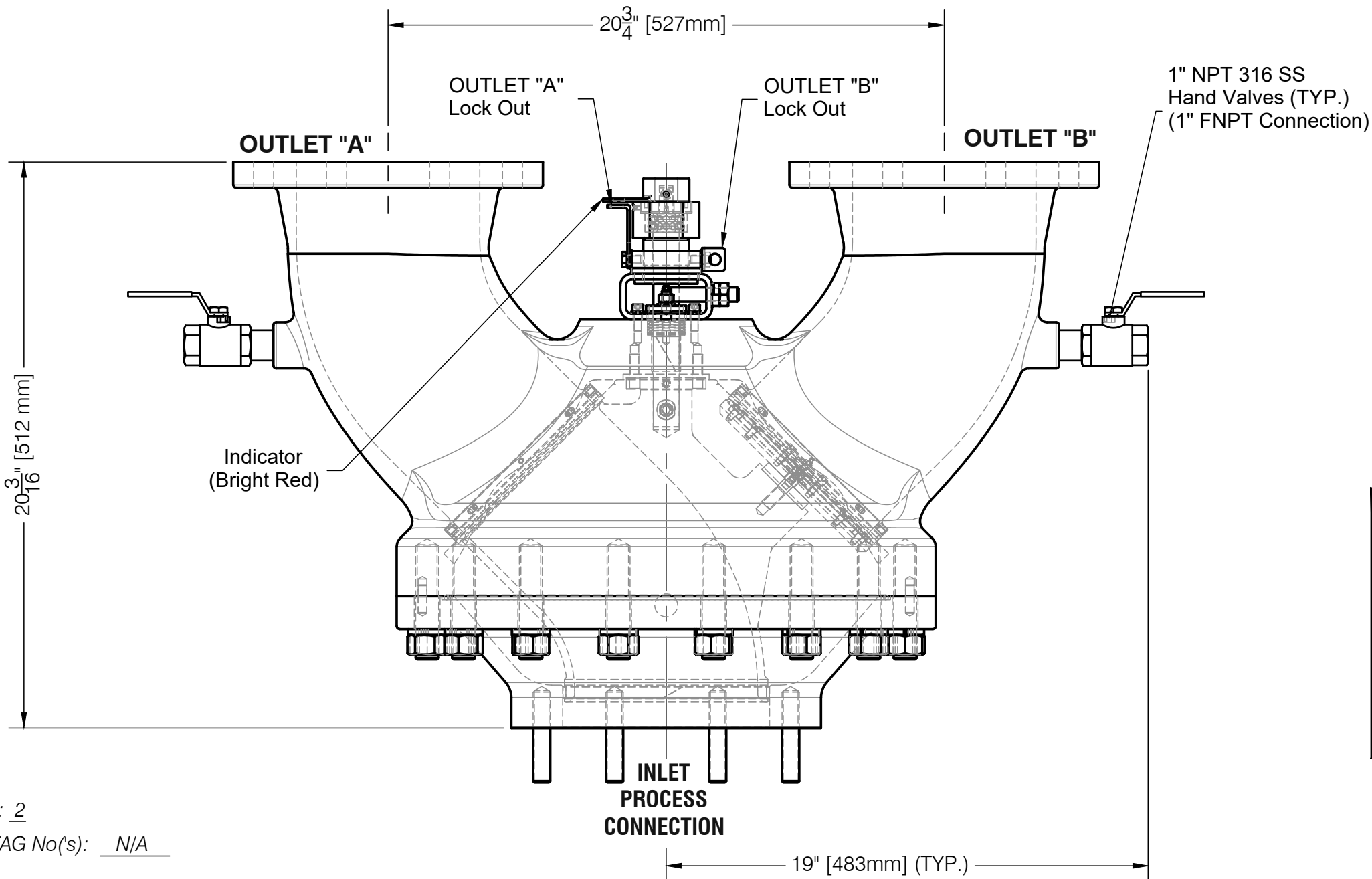
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97190 Safety Selector Valve

Part Description: Material:

Body / Base: B26-356 Cast Aluminum
Seat: 316 Stainless Steel
Seat Seal: Chloroprene
Rotor (Diverter): ASTM A351 Gr. CF8M (316 S.S.)
Isolation Disc: 316 Stainless Steel
Shaft Seals: Teflon
Index Shaft: 316 Stainless Steel
Bonnet-Guide: 316 Stainless Steel
Retraction Bushing: 316 Stainless Steel
Indicator: 316 Stainless Steel
Process Conn. Nut: 316 Stainless Steel
Process Conn. Stud: 316 Stainless Steel
Washers: 316 Stainless Steel



Quantity: 2

CUST. TAG No(s): N/A

Notes:

- 1) Inlet / Outlet Connections Match 6"FF ANSI 1²⁵/₁₅₀ lb Class.
- 2) All Bolt Patterns Straddle Center Line.
- 3) Flow Coefficient Cv = 4512.

PROJECT: **HILD WWTP PROJECT**
CUSTOMER: **NAN, INC.**
CUSTOMER P.O. NO: **24077-00048**
DESCRIPTION: **Safety Selector Valve**
MODEL NO: **97190-64331111**
SPECIFICATION: 11381 SECTION: **2.05**
S&J SALES ORDER NO: **!B2090** DATE: June 13, 2025

APPROVED ☐ APPROVED ☐ NOT APPR.-RESUB. ☐
w/CHANGES PER COMMENT
CUST. SIGNATURE: _____ DATE: _____

A	Approval Drawing	45	CWS	Jun 13 25	SB
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND&JURS an L&J TECHNOLOGIES Company HILLSIDE, ILLINOIS USA					
97190 Safety Selector Valve 6"FF Dimensions					
DRWN	CWS	Type:	Dim's	Size:	B: 17 x 11
DATE	Jun 13 25	Scale:	None	Designed:	Standard
CHKD	CWS	T:\SO#\XXXXX\BXXXX\B2090\9719-00020-06-!B2090.DWG			
APRV	SB	9719-00020-06-!B2090			

Tab 9

(97401-15112)

15" U-Tube Type Manometer

- Data Sheet

- Drawing



BIOGAS
SHAND & JURS



L&J TECHNOLOGIES

A Cognesense Company

97401 U-Tube Type Manometer

The S&J 97401 U-Tube Type Manometer is a direct reading single tube type gauge, that provides highly accurate, easy to read pressure measurements. Standard Green Fluid with specific gravity of 1.0 is used in the manometer.

The S&J 97401 U-Tube Type Manometer is used to gauge pressures at critical points in gas-train systems. Pressure, vacuum or differential pressure can be gauged to provide complete monitoring of the gas-train system.

Typical applications that the 97401 can be used in include anaerobic digester tanks/gas trains, boiler lines, engines, various plant utility lines and flares/waste gas burners.

Easy panel mounting of the manometers provides a convenient single point for monitoring various pressures in the system. The manometer have simple in-place scale adjustment.

Specifications:

Duplex Scale:

Graduated in English Units - Inches and Tenths W.C.
~~or Metric Units - Millimeters and Centimeters~~

Fluid:

Green Fluid (Standard)

Indoor/Outdoor Wall Panel Mounting:

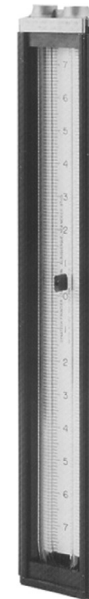
Aluminum, ~~Steel, Stainless Steel~~

Float Check Valve:

A Float Check Valve should be installed on the atmospheric and / or pressure side of each manometer. The valve closes as pressure surges beyond the manometer range, and opens when normal pressure is restored.

Moisture and Sediment Trap or Line Traps:

Line Traps are an inexpensive means to keep manometers and indicating fluids clean are recommended for installations. When installed in the line ahead of the manometer they permit entrained solids to settle out before they reach the manometer. When used with gases or vapors they trap moisture.



Features

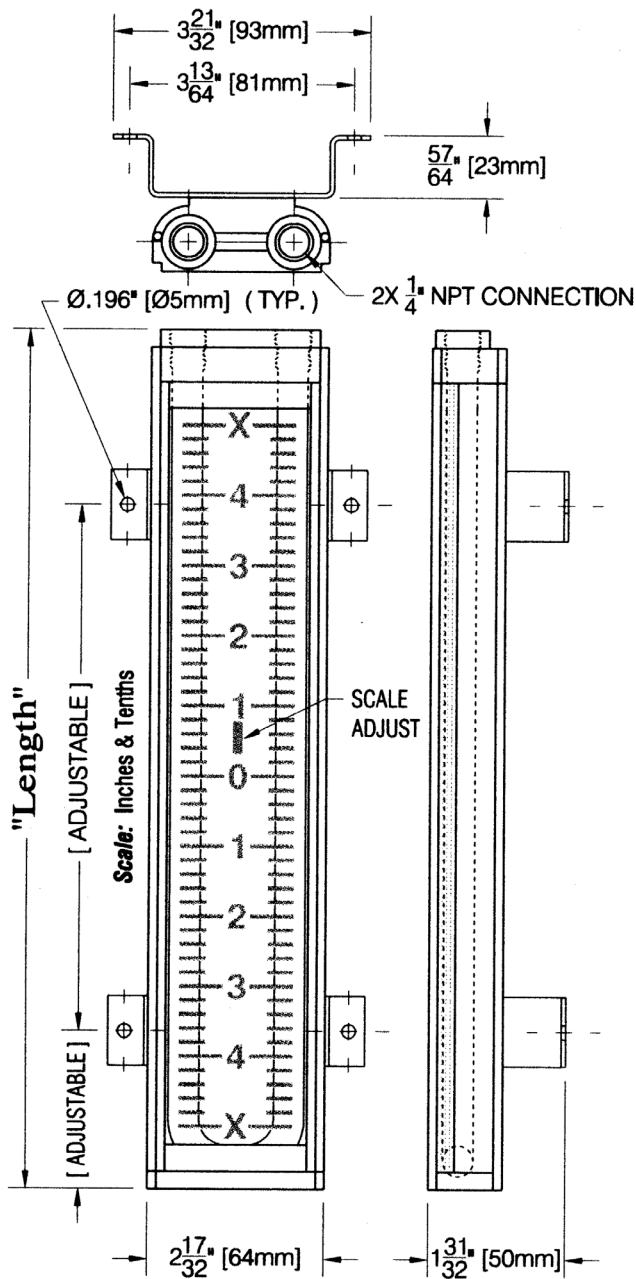
- Pressure or Vacuum Measurement
- Measures Differential or Point Pressure Readings
- Indoor or Outdoor Use
- Sizes ~~6", 10", 15", 20", 24", 30", 36", 50"~~
- Easily Panel Mounted
- Extruded Aluminum, Anodized Finish

97401

U-Tube Type Manometer

SHAND & JURS

Outline Dimensions



Range	Length in Inches (mm)
6"	9 $\frac{5}{8}$ (244)
10"	14 $\frac{1}{4}$ (362)
15"	17 $\frac{15}{16}$ (455)
20"	23 $\frac{3}{16}$ (589)
24"	27 $\frac{1}{2}$ (698)
30"	33 $\frac{11}{16}$ (855)
36"	39 $\frac{15}{16}$ (1014)
50"	54 $\frac{5}{16}$ (1379)

All designs subject to change. Certified dimensions and specifications available upon request.

SHAND & JURS

5911 Butterfield Road Hillside, IL 60162 • tel: (708) 236-6000 • fax: (708) 236-6006
 sales@ljtechnologies.com • www.ljtechnologies.com • ISO 9001:2015 Certified

Rev: 97401-3E

97401 Ordering Guide

Model Number Selection

The model number will have a base number **97401** followed by 5 digit numbers. These digits will represent 4 sets of option tables.

97401 - AB - CD - E

Table AB - Range

Option AB	Range
06	+/- 3" W.C.
10	+/- 5" W.C.
15	+/- 7.5" W.C.
20	+/- 10" W.C.
24	+/- 12" W.C.
30	+/- 15" W.C.
36	+/- 18" W.C.
50	+/- 25" W.C.

Table C - Scale Type

Option C	Description
1	English
2	Metric

Table D - Mounting Panel

Option D	Description
0	No Mounting Panel
1	Aluminum Mounting Panel
2	Stainless Steel Mounting Panel
3	Steel Mounting Panel

Consult Factory for Multiple Unit Mounting Panels

Table E - Accessories

Option E	Accessories
0	None
2	Float Check Valve, Stainless Steel with Teflon Float
5	Moisture and Sediment Trap
6	Line Trap

Accessories

Part Number	Description
9740-10054	Green Indicating Fluid S.G. = 1.0; 30°F - 100°F (standard)
9740-10061	Float Check Valve, 1/4" NPT, Stainless Steel with Teflon Float
9740-10080	Isolation Valve, 1/4" NPT, Bronze
9740-10282	20"/24" Mounting Panel Kit, Aluminum
9740-10283	20"/24" Mounting Panel Kit, Stainless Steel
9740-10280	12" Mounting Panel Kit, Aluminum
9740-10281	12" Mounting Panel Kit, Stainless Steel
9740-10284	30" Mounting Panel Kit, Aluminum
9740-10285	30" Mounting Panel Kit, Stainless Steel

Consult Factory for additional Indicating Fluids and temperature ranges. Scale must match S.G. of fluid. Fluid capacity 4oz.

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Materials of Construction

Housing Body: Black Anodized Extruded Alum.

Window: Acrylic

Connection Head: Type 316SS (Cleanout Head Style)

Indicating U-Tube: Pyrex™ Glass

O-Rings: Viton

Fluid: Green Fluid (S/G=1.0), 30°F-100°F

Float Check Valve: 316SS w/Teflon Ball

Venting Petcock:

2X 1/4" NPT CONNECTION
(Cleanout Head Style)

Scale: English

Range: 15" [±7.5"]

Quantity: 2

Cust. Tag No'(s): None

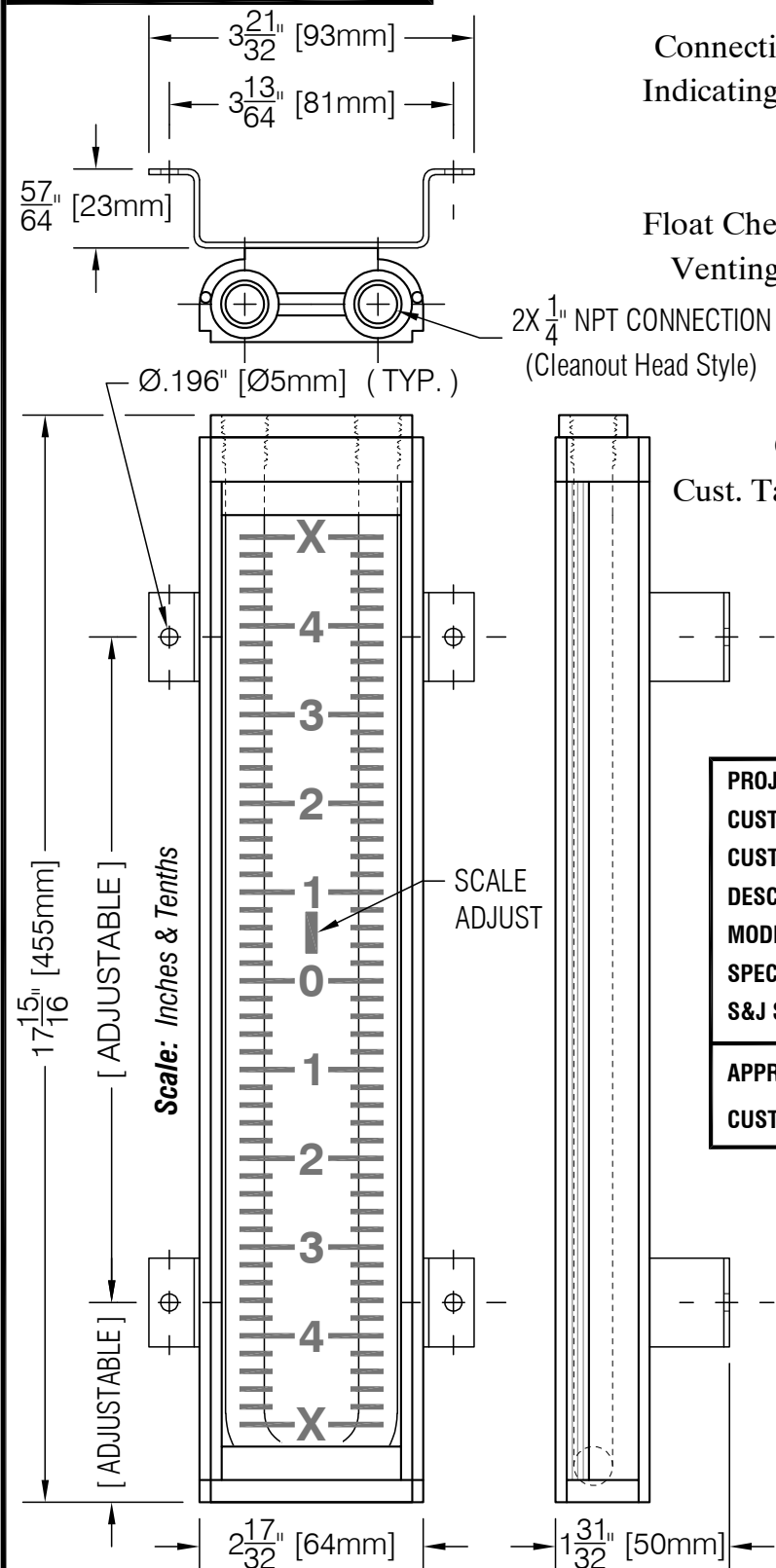
PROJECT: **HILO WWTP PROJECT**
CUSTOMER: **NAN, INC.**
CUSTOMER P.O. NO: **24077-00048**
DESCRIPTION: **U-Tube Manometer**
MODEL NO: **97401-15112**
SPECIFICATION: 11381 SECTION: 2.09
S&J SALES ORDER NO: **!B2090** DATE: June 13, 2025

APPROVED ☐ APPROVED ☐ NOT APPR.-RESUB.
w/CHANGES PER COMMENT
CUST. SIGNATURE: _____ DATE: _____

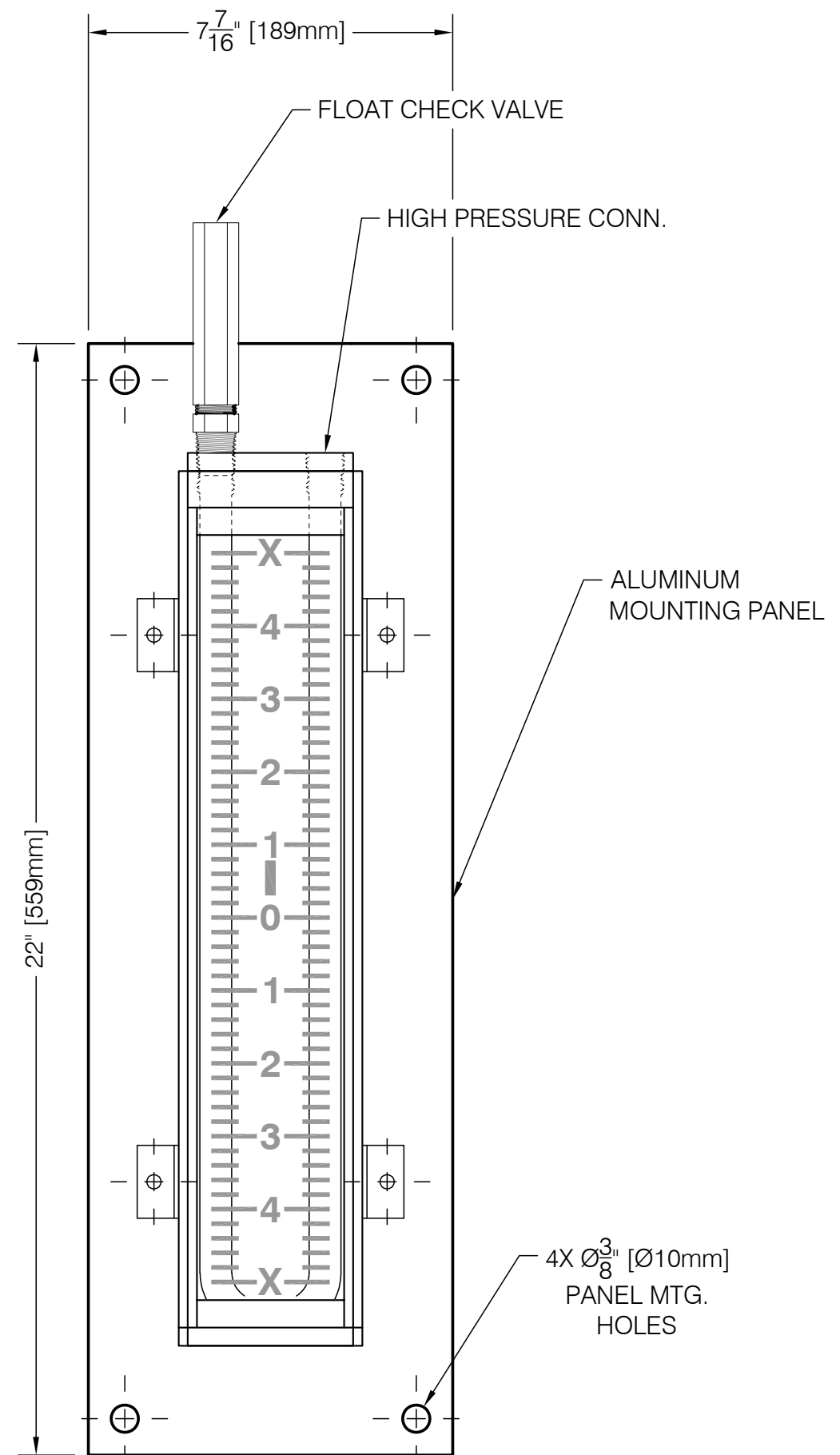
SHAND & JURS
an **L&J TECHNOLOGIES** Company
HILLSIDE, ILLINOIS USA

97401 U-Tube Manometer
50" Mounting Dimensions

SB	Type: Dim's	Size: A: 8.5 x 11
Jun 13 25	Scale: None	Designed: Standard
SB	T:\SO#\I\XXXXX\B\XXXX\B2090\9740-00041-15-!B2090.dwg	
Slaw	9740-00041-15-!B2090	



~	Approval Drawings	45	SB	Jun 13 25	Slaw
REV	DESCRIPTION	ECO	BY	DATE	APRV



Tab 10

(97150-042300003)

4" Single Port Regulator

- Data Sheet

- Drawing



BIOGAS
SHAND & JURS



L&J TECHNOLOGIES

A Cognesense Company

97150 Single Port Regulator

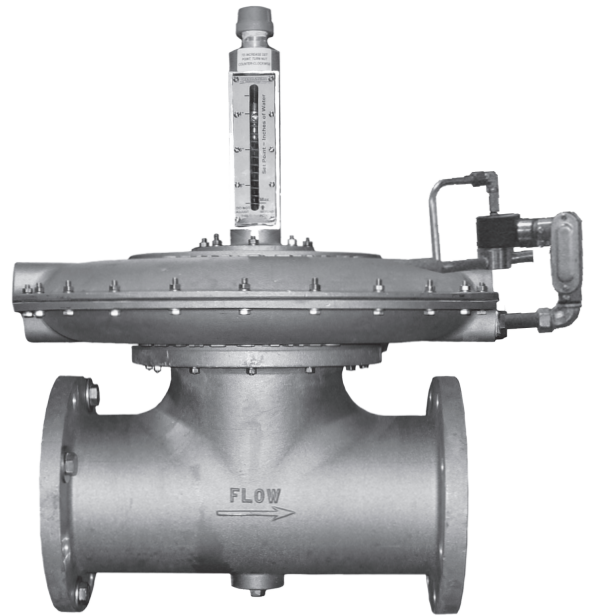
The S&J 97150 Single Port Back Pressure Regulator is a back pressure regulator with solenoid or thermally operated shut-off capabilities. The back pressure regulator function maintains a preset upstream pressure in low pressure (typically 2" - 8" w.c.) biogas lines such as those leading to flares and waste gas burners. Its normal operating position is closed until the set point is exceeded.

A large spring-loaded diameter diaphragm reduces line bounce for smoother gas flow. Field adjustment of set point is easily done by changing position of the spring with visual pointer indicator. The components are easy to access providing for simplified cleaning and maintenance. When external sense line is used, the 1/2" NPT connection on the lower diaphragm housing should be connected a minimum of 10 feet upstream from the regulator. The diaphragm-direct regulator can be ordered in two setting ranges: standard and high, which are defined for a given line size. The back pressure regulator's second function is to positively shutoff gas flow in emergency flame conditions when combined with a solenoid controlled sense line or fusible element.

This unit can be used alone or combined with a Flame Trap Assembly. If installed indoors, the vent connection should be piped outdoors. A 97200 Flame Check should be installed by the end of the vent line for flashback protection.

Standard materials of construction include an aluminum body with an aluminum pallet and stainless steel internals. The diaphragm is Nylon Reinforce Buna-N.

Its aluminum and stainless steel components withstand the severest of process environments. The S&J 97150 is especially designed for hydrogen sulfide and hot, wet methane which are the main components of digester gas streams in municipal waste water treatment facilities.



Features

- External Sense ~~or Internal Sense~~
- Large Diaphragm Operated/Spring Loaded for Sensitivity
- Maintain Upstream Pressure
- Simplified Cleaning and Maintenance
- Sizes ~~2" Through 12"~~ 4"
- 3-Way Solenoid ~~Option Available~~
- ~~3-Way Solenoid with Manual Bypass~~
~~Option Available~~
- Horizontal Configuration Only

97150

Single Port Regulator



Standard Materials of Construction:

Body:

Cast Aluminum

Diaphragm:

Nylon Reinforced Buna-N

Pallet:

Low Copper Aluminum, ~~316 Stainless Steel~~

Spring:

Stainless Steel

Specifications:

Flanged Connections:

125 lb. ANSI FF Flange

Maximum Operating Pressure:

20" W.C. (Sizes 2"-8")

~~15" W.C. (Sizes 10" and 12")~~

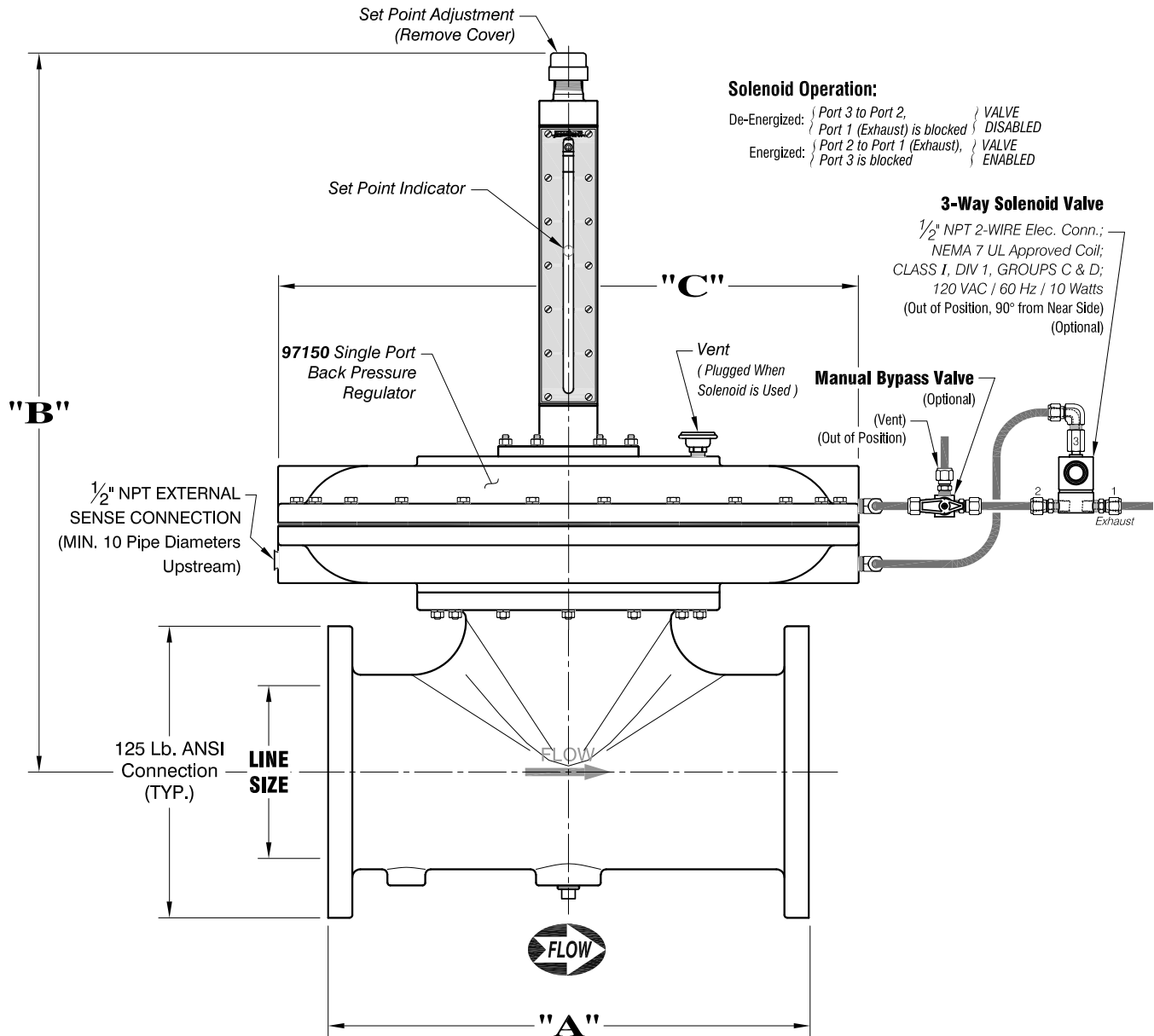
~~Consult Factory for other options.~~

Air Flow Capacity in Standard Cubic Feet per Hour x 1000 @ 60°F

Over Pressure Inches W.C.	Line Diameter						
	2"	3"	4"	6"	8"	10"	12"
1	1.3	2.3	5.9	13.5	25.0	35.0	55.0
2	1.9	4.5	8.9	19.5	37.1	54.0	88.0
3	2.3	5.4	10.8	24.2	46.5	73.0	115
4	2.7	6.3	12.6	28.5	54.2	86.0	137
5	3.1	7.2	14.3	32.0	60.3	97.0	153
6	3.4	8.2	15.8	35.0	66.8	110	170
7	3.8	8.8	17.1	38.0	73.0	119	183
8	4.0	9.5	18.6	40.7	78.0	128	193
9	4.3	10.2	19.6	43.3	82.6	137	208
10	4.6	10.7	20.9	45.5	87.0	146	218
11	4.8	11.2	21.8	47.8	90.9	153	227
12	5.1	11.7	22.7	49.8	95.1	160	237
13	5.3	12.2	23.8	51.6	99.0	165	245
14	5.5	12.7	24.7	53.7	103	170	252
15	5.8	13.2	25.4	55.5	106	175	258
16	6.0	13.8	26.1	57.4	109	180	265
17	6.2	14.3	26.8	58.9	112	184	271
18	6.4	14.7	27.4	60.2	115	188	276
19	6.5	15.0	28.0	62.0	118	191	281
20	6.6	15.3	28.6	63.1	120	194	285



Dimensions



Line Diameter (Inches [mm])	A (Inches [mm])	B (Inches [mm])	C (Inches [mm])
2 [50]	8 ¾ [222]	27 ⅞ [708]	21 [533]
3 [75]	10 [250]	28 ½ [724]	21 [533]
4 [100]	11 ¾ [289]	29 [736]	21 [533]
6 [150]	15 [381]	31 ⅞ [797]	26 ⅞ [683]
8 [200]	22 ¼ [565]	33 ⅞ [847]	26 ⅞ [683]
10 [250]	27 ⅜ [706]	37 ⅞ [962]	36 ½ [927]
12 [300]	33 [838]	41 ¼ [1047]	36 ½ [927]

All designs subject to change. Certified dimensions and specifications available upon request.

97150 Ordering Guide

Model Number Selection

The model number will have a base number **97150** followed by 6 digit numbers. These digits will represent 5 sets of option tables.

97150 - AB - CD - EF - 003 = Special: Set Point: 10" W.C.
Range: 7"-13" W.C.

Table AB - Line Size

Table C - Setting Range*

Option AB	Line Size	Option C	Setting Range
02	2"	1 - Standard	2" - 8" W.C.
		2 - High	>8" - 20" W.C.
03	3"	1 - Standard	2" - 8" W.C.
		2 - High	>8" - 20" W.C.
04	4"	1 - Standard	2" - 8" W.C.
		2 - High	7" - 13" W.C.
06	6"	1 - Standard	2" - 8" W.C.
		2 - High	>8" - 20" W.C.
08	8"	1 - Standard	2" - 8" W.C.
		2 - High	>8" - 20" W.C.
10	10"	1 - Standard	2" - 8" W.C.
		2 - High	>8" - 15" W.C.
12	12"	1 - Standard	2" - 8" W.C.
		2 - High	>8" - 15" W.C.

Specify operating set point. Valve range will be +/- 3" W.C. of set point.

Range will not exceed listed setting range (i.e. set point of 3" will have a range of 2"-8", set point of 19" will have a range of 14"-20").

For other ranges, please Consult Factory.

Table D - Sense Configuration/Solenoid*

Option D	Configuration
0	Internal
1	External
2	Internal with Solenoid
3	External with Solenoid
4	Internal with Solenoid and Manual Bypass
5	External with Solenoid and Manual Bypass

* Solenoid - 120VAC, NEMA 7

A three-way solenoid valve allows optimal control of digester gas flow to the waste gas burner, adding additional safety to the waste gas system. The three-way solenoid valve provides a regulator that remains closed until a pilot flame is proven, until then the unit stays in a de-energized state and maintains equal pressure on the regulator diaphragm.

Manual bypass overrides solenoid function. Pressure regulator will open when set point is reached.

Once a pilot flame is proven, an alarm contact or interposing relay in the waste gas burner control panel energizes the three-way solenoid valve, which then releases pressure from above the diaphragm and allows the regulator to open. When the three-way solenoid valve is de-energized, gas is applied to the top of the diaphragm, closing the regulator.

Table E - Hardware Material

Option E	Hardware
0	Stainless Steel
1	316 Stainless Steel

Table F - Accessories

Option F	Accessories
0	None
1	Insulation Jacket

9715-00014-04-!B2090

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Set Point: 10" W.C.

Set Point Range: 7-13" W.C.

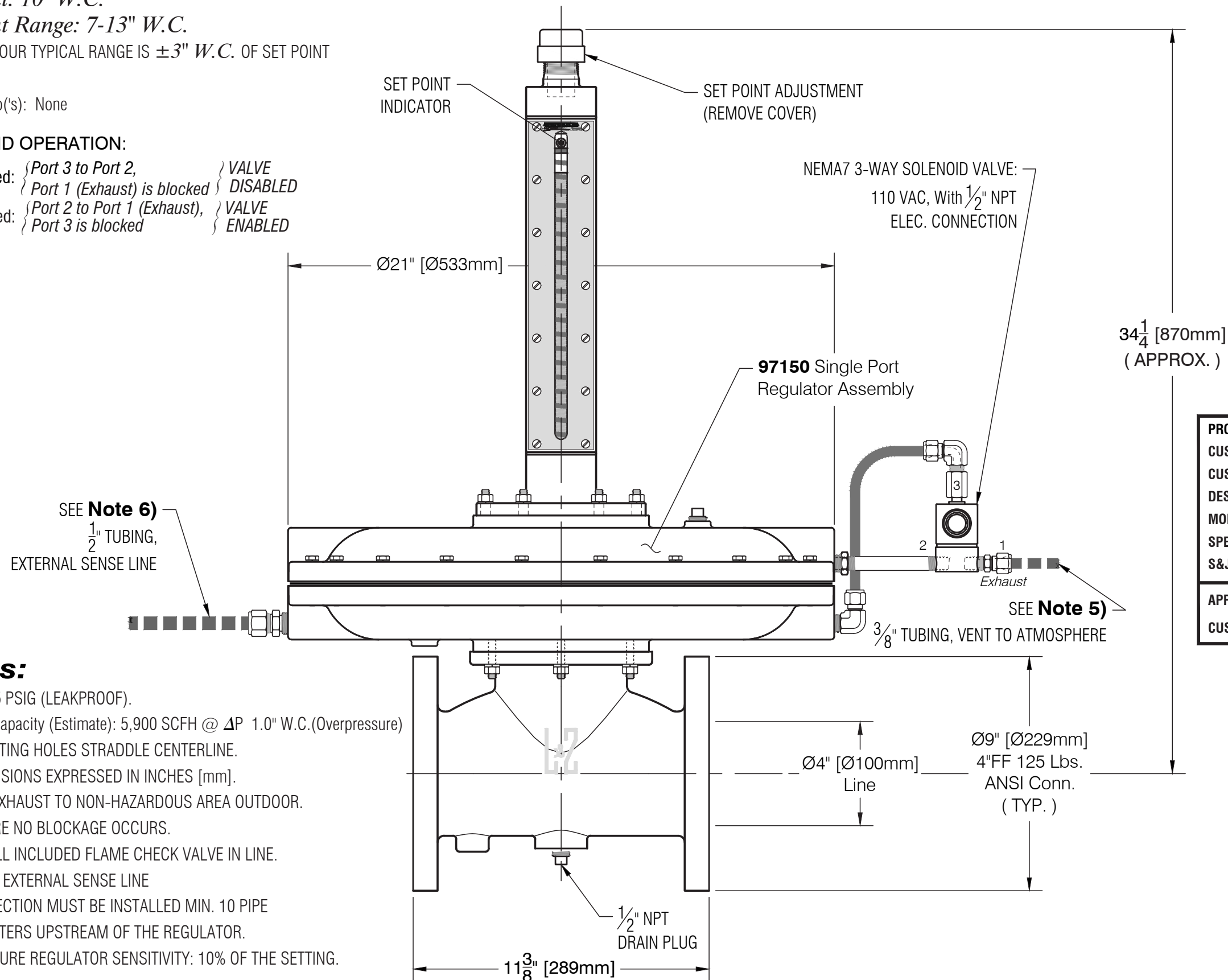
NOTE: OUR TYPICAL RANGE IS $\pm 3"$ W.C. OF SET POINT

Quantity: 1

Cust. Tag No('s): None

SOLENOID OPERATION:

De-Energized: { Port 3 to Port 2, } VALVE
 { Port 1 (Exhaust) is blocked } DISABLED
Energized: { Port 2 to Port 1 (Exhaust), } VALVE
 { Port 3 is blocked } ENABLED



97150 Single Port Regulator

Maximum Pressure: 5 PSIG

Part Description: Material

Valve Body, Housings: 356-T6 Cast Aluminum

Pallet: Aluminum

Diaphragm Plate: Coated Steel

O-Ring(s) / Gasket(s): NBR / Fiber Composite

Diaphragm: NBR

Spring: 17-7 PH S.S. (Electroless Nickel Plated)

Hardware: 316 Stainless Steel

Internal Parts: Types 304/316 S.S. + Aluminum

PROJECT: **HILO WWTP PROJECT**
CUSTOMER: **NAN, INC.**
CUSTOMER P.O. NO: **24077-00048**
DESCRIPTION: **Back Pressure Regulator**
MODEL NO: **97150-042300003**
SPECIFICATION: **11381** SECTION: **2.02**
S&J SALES ORDER NO: **!B2090** DATE: **June 13, 2025**

APPROVED ☐ APPROVED ☐ NOT APPR.-RESUB. ☐
w/CHANGES PER COMMENT
CUST. SIGNATURE: _____ DATE: _____

Notes:

- 1) MAX. 5 PSIG (LEAKPROOF).
- 2) Flow Capacity (Estimate): 5,900 SCFH @ ΔP 1.0" W.C.(Overpressure)
- 3) MOUNTING HOLES STRADDLE CENTERLINE.
- 4) DIMENSIONS EXPRESSED IN INCHES [mm].
- 5) PIPE EXHAUST TO NON-HAZARDOUS AREA OUTDOOR. ENSURE NO BLOCKAGE OCCURS. INSTALL INCLUDED FLAME CHECK VALVE IN LINE.
- 6) 1/2" NPT EXTERNAL SENSE LINE CONNECTION MUST BE INSTALLED MIN. 10 PIPE DIAMETERS UPSTREAM OF THE REGULATOR.
- 7) PRESSURE REGULATOR SENSITIVITY: 10% OF THE SETTING.

~	Approval Drawing	45	SB	Jun 13 25	Slav
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND & JURs an LEJ TECHNOLOGIES Company HILLSIDE, ILLINOIS USA					
97150 Single Port Pressure Regulator 4"FF Dim's w/Solenoid					
DRWN	SB	Type:	Dim's	Size:	B: 17 x 11
DATE	Jun 13 25	Scale:	None	Designed:	Standard
CHKD	SB	T:\SO#\XXXXX\B2090\9715-00014-04-!B2090.dwg			
APRV	Slav	9715-00014-04-!B2090			

Tab 11

(97100E631203)

6 Qt. Electric Drip Trap

- Data Sheet

- Drawing



BIOGAS
SHAND & JURS



L&J TECHNOLOGIES

A Cognesense Company

97100E Electric Drip Trap

The S&J 97100E Electric Drip Trap provides for the continuous, safe removal of condensate from low points in low pressure gas piping systems. Available in either 3 or 6 quart sizes and with 1" NPT inlets and outlets, the S&J 97100E adapts to any condensation or drainage collection requirement. The 97100E can be mounted in Class 1, Division 1 & 2, Gas Group D hazardous locations and is rated for both NEMA 4 and 7.

The unit collects liquid until the ported rotating disc valve is electrically opened. The actuator seals the gas inlet port prior to opening the drain port allowing for removal of condensate without gas escaping. A visual indicator is located on the front to identify the "open" or "closed" condition. An optional timer controlled electrical actuator is available which will automatically drain the trap at a user programmable time interval.

An optional Local Control Panel, the 97101P, is available and can control up to 8 units maximum. The panel is available in NEMA 4X or NEMA 7 enclosures.

For higher pressure systems, see the Shand & Jurs model 97101E.

Specifications:

Available Sizes/Capacity

~~3 Quart~~, 6 Quart

Inlet/Outlet Connections

1" NPT

Materials of Construction

Low Copper Aluminum ~~or Stainless Steel (SS) body,~~
cover plate and (Drip Trap) disc; Stainless Steel
internals

Maximum Working Pressure

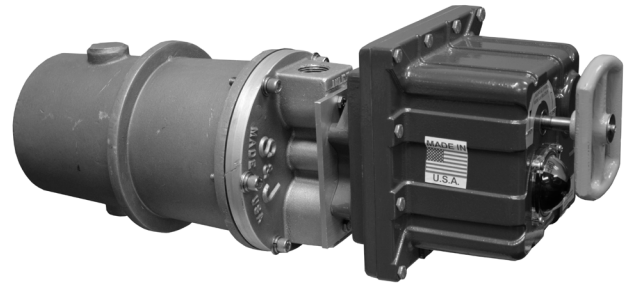
5 PSIG Standard, ~~25 PSIG Optional~~

Mounting Bracket

~~Available in Aluminum or~~ Stainless Steel

Electronic Actuator:

120 VAC 60 Hz, 1 Phase (220 VAC Model available)
Temp. Rating: -40°F to +150°F (with heater)
NEMA 4/7, Aluminum Enclosure Class 1, Div 1 & 2,
Gas Group D CSA Approved
Permanent Lubrication; High Torque Capability
Visual Indication: Open/Close
Indication Contacts: Open/Close SPDT;
Auxiliary Limit Switches (Optional): Open/Close SPDT



97100E Drip Trap Standard Model with Handwheel

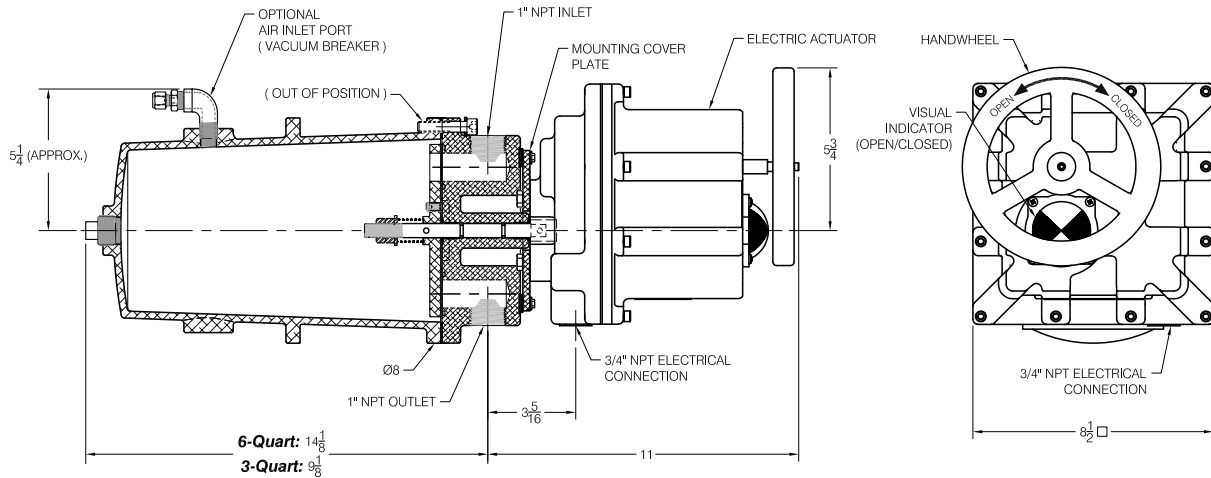
Features

- Positive Gas Shut-off While Draining
- Automatic Operation with an Electronic Timer
- Local Control Panel Available
- ~~5 PSI or 25 PSI~~ Operating Pressures Available
- ~~3 or 6~~ Quart Capacities
- Air Inlet ~~and Insulation Jackets~~ available
- Manual Handwheel Standard
- New - Easy to Read Visual Indicator
- NEMA 4/7 Approved
- Isolation Valve Available

97100E Electric Drip Trap

SHAND & JURS

Dimensions:



All designs subject to change. Certified dimensions and specifications available upon request.

97100E Ordering Guide

Model Number Selection

The model number will have a base number **97100E** followed by 6 digit numbers. These digits will represent 6 sets of option tables.

97100E - AB - CD - EF

Table A - Capacity

Option A	Capacity	MAWP
3	3 Quart	5 PSIG
6	6 Quart	
4	3 Quart	25 PSIG
7	6 Quart	

Table D - Options

Option D	Options
0	None
1	Insulation Jacket
2	Air Inlet
3	Air Inlet with Insulation Jacket

Isolation valve available upon special request.

Table B - Body Material

Option B	Material
2	Aluminum
3	Aluminum w/ Anodized Disc/Cover
4	Aluminum w/ Anodized Disc/Cover/Body
5	304 Stainless Steel
6	316 Stainless Steel

Table E - Hardware

Option E	Options
0	Stainless Steel Bracket / Stainless Steel Hardware
2	Aluminum Bracket / Stainless Steel Hardware
3	316 Stainless Steel Bracket / Stainless Steel Hardware

Table C - Softgoods

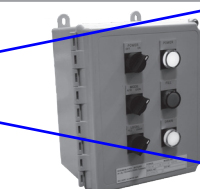
Option C	Softgoods
0	NBR (Nitrile-Butadiene)
1	CR = Chloroprene (Neoprene)

Table F - Actuator

Option F	Actuator
2	No Timer, NEMA 4/7 w/ Manual Handwheel
3	With Timer, NEMA 4/7 w/ Manual Handwheel

97100P Local Control Panel

The 97100P Local Control Panel can control up to eight 97100E units maximum. Selectors include Auto/Local Mode and Local Fill/Drain. Indicators show when the drip trap is filling or draining. An optional fail indicator is also available. Optional status and fail dry contacts are available for user control. The Local Control Panel is available for manual control only or with automatic timers. Typical enclosures available are NEMA 4X Fiberglass, 304 Stainless Steel, 316 Stainless Steel or NEMA 7 Aluminum.



NEMA 4X Fiberglass Panel



NEMA 7 Panel

SHAND & JURS

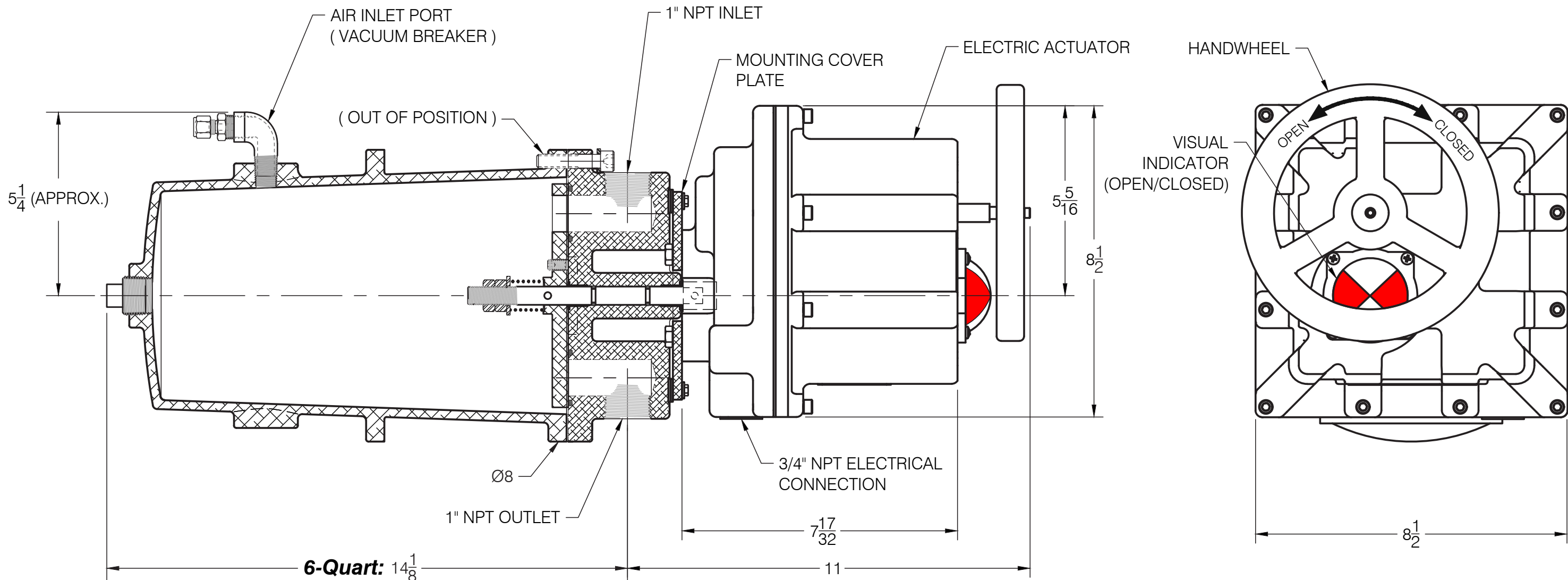
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sales@ljtechnologies.com • www.ljtechnologies.com • ISO 9001:2015 Certified

Rev: 97100E-3G

9710-00046-!B2090

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ACTUATOR SPECIFICATION:

- 1) 120VAC 60HZ, 1 PHASE, 1 AMP.
- 2) TEMP.RATING -40° TO +150°F.
- 3) NEMA 4X/7 ENCLOSURE,
Class *I* Groups *C & D*;
Class *II* Groups *E, F, & G*. DIV 1 & 2.
- 4) PERMANENT LUBRICATION.
- 5) VISUAL INDICATION - OPEN/CLOSE.
- 6) HIGH TORQUE CAPABILITY, REVERSIBLE.
BUILT IN OVERLOAD PROTECTION.
- 7) MANUAL OVERRIDE DECLUTCHING HANDWHEEL.
- 8) DIMENSIONS ARE SHOWN IN INCHES.
- 9) INTERNAL PROGRAMMABLE TIMER.
5 MINUTE TO 10 HOURS (BUILT IN).

PROJECT: <i>HILD WWTP PROJECT</i>		
CUSTOMER: <i>NAN, INC.</i>		
CUSTOMER P.O. NO: <i>24077-00048</i>		
DESCRIPTION: <i>Electric Drip Trap</i>		
MODEL NO: <i>97100E631203</i>		
SPECIFICATION: <i>11381</i>		SECTION: <i>2.11</i>
S&J SALES ORDER NO: <i>!B2090</i> DATE: <i>June 13, 2025</i>		
APPROVED ☐	APPROVED ☐ w/CHANGES	NOT APPR.-RESUB. PER COMMENT ☐
CUST. SIGNATURE: _____		DATE: _____

Max. Working Pressure: 5 PSIG

Capacity: 6 Quart

Quantity: 6

Cust. Tag No(s): N/A

97100 Electric Drip Trap

Part Description: Material:

Body: *Low Copper Aluminum Alloy 356*
Cover: *Low Copper Aluminum Alloy 356, Anodized*
Disc: *Low Copper Aluminum Alloy 6061, Anodized*
Internals: *Stainless Steel*
Shaft: *Stainless Steel*
O-Rings: *Chloroprene*
Coupling: *Stainless Steel*
Hardware: *Stainless Steel*
Bracket/ Handle: *Stainless Steel*

A	Approval Drawing	45	CWS	Jun 13 25	SB
REV	DESCRIPTION	ECO	BY	DATE	APRV
SHAND&JURS an L&J TECHNOLOGIES Company HILLSIDE, ILLINOIS USA					
97100E Drip Trap Dimensions 6 Qt. Dim's w/ Vacuum Breaker; Actuator With Timer					
DRWN	CWS	Type:	Dim's	Size:	B: 17 x 11
DATE	Jun 13 25	Scale:	None	Designed:	Standard
CHKD	CWS	T:\SO#\XXXXXX\B2090\9710-00046-!B2090.dwg			
APRV	SB	9710-00046-!B2090			